

Anomalies as Structural Obstructions

Gauge Inconsistency, Global Response, Inflow, Matching, and
Normal-Form Closure in the Standard Model

A fixed-target anomaly reference for the Standard Model symmetry arc

Principal result

Let D_λ be a declared Reference-complete anomaly target. A non-degenerate determinate physical claim on D_λ necessarily instantiates Admissibility, Standing, Reference, and Irreversibility. Conventional anomaly theory fixes the target-specific currents, measures, determinant phases, descent classes, global obstructions, inflow relations, matching data, and renormalized operator identities. Independently derived target obligations and role-blind physical anomaly objects then determine a lawful normal-form groupoid $\mathcal{N}(D_\lambda)$. For each of the ten closed targets treated in this volume,

$$|\pi_0 \mathcal{N}(D_\lambda)| = 1.$$

The result retains nontrivial presentation morphisms, automorphisms, and stabilizers within the unique isomorphism class. Global-form, line-operator, monopole, higher-group, and non-invertible inventories remain explicit interfaces to SYM-9.

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Combined archival reference

Abstract

Quantum field theory uses the word *anomaly* for physically different phenomena: failure of a dynamical gauge symmetry at the quantum level, anomalous nonconservation of a meaningful global current, obstruction to gauging a global symmetry, topology-sensitive phases invisible to local anomaly polynomials, mixed-background noninvariance, bulk–boundary inflow, ultraviolet–infrared matching, and renormalized trace identities. These phenomena do not share one physical status. Some make a gauge theory inconsistent; others preserve a physical global action while constraining currents, backgrounds, defects, or possible infrared realizations.

This volume develops a fixed-target classification of those distinctions for the Standard Model symmetry program. The logical starting point is that any non-degenerate determinate physical claim already requires fixed identity, lawful comparison and transformation, consequence-bearing use, and irreversible fixation; these instantiate Admissibility, Standing, Reference, and Irreversibility. Conventional anomaly theory then supplies the target-specific current, measure, determinant, descent, index, inflow, matching, and renormalization facts. The standing obligations forced by each target are derived independently of the physical anomaly objects, and the two systems first meet in the eligibility calculation.

The closed family contains ten targets: the one-generation Standard Model local anomaly ledger; vector/axial descent with Bardeen structure; a symmetric vectorlike six-Weyl theory; the ordinary and newer $SU(2)$ global anomalies; massless three-flavor QCD $U(1)_A$; electroweak $B + L$; a four-dimensional Weyl boundary with five-dimensional inflow; three-flavor QCD gauged-Wess–Zumino–Witten matching; and the massless-QCD trace anomaly in a fixed $\overline{MS}$ operator target. Across these targets, denial analysis yields 63 role positions and the independent physical construction yields 590 candidates. All 4412 same-fiber pairs are evaluated. The resulting assignment spaces contain 91695 complete maps; 91597 are rejected by physical compatibility and 98 survive. Each closed fiber has one lawful normal-form isomorphism class.

The construction distinguishes gauge inconsistency from anomalous global response, local from global obstruction, counterterm redistribution from physical cancellation, inflow from retrospective repair, and matching constraints from uniqueness of infrared realization. The target-relative anomaly records are exported to SYM-4 through SYM-10 under explicit target-preserving transport rules. The theorem does not claim that anomaly cancellation uniquely derives the Standard Model, that matching selects one infrared phase, or that all generalized anomaly sectors are already inventoried. Global-form, line-operator, monopole, higher-group, and non-invertible inventories remain interfaces to SYM-9, while whole-arc candidate exhaustion remains the burden of SYM-11. Detailed construction ledgers and reproducibility evidence appear in the Technical Audit Companion.

Read This First: Dependency and Proof Spine

Dependency lock

AASC is not an optional interpretation applied after anomaly physics is independently complete. A determinate anomaly claim already requires a fixed target, a fixed acting or background structure, a lawful comparison class, admissible transformations, consequence-bearing use, and a completed reporting act. These instantiate Admissibility, Standing, Reference, and Irreversibility by necessity.

The proof order is

$$\begin{aligned}
 &\text{NonDegDet}(D_\lambda^{\text{phys}}) \implies K_{D_\lambda}, \\
 &K_{D_\lambda} + \text{fixed QFT data} \implies C_\lambda^{\text{phys}}, \\
 &D_\lambda \implies B_\lambda \implies R_\lambda = B_\lambda / \sim_{\text{gov}}, \\
 &D_\lambda + C_\lambda^{\text{phys}} \implies O_\lambda \implies F_\lambda \implies X_\lambda = F_\lambda / \sim_{\text{pres}}, \\
 &(R_\lambda, X_\lambda) \implies E_\lambda, J_\lambda, \Gamma_\lambda^{\text{lawful}}, \mathcal{N}(D_\lambda), \\
 &|\pi_0 \mathcal{N}(D_\lambda)| = 1 \quad \text{for each closed target.}
 \end{aligned}$$

Kernel necessity is logically prior. QFT is evidentially prior to final role reporting: the transformation, current, measure, determinant, partition function, topology, counterterm, inflow relation, symmetry map, and operator identity are fixed before the structural role is reported. The no-second-classifier consequence enters only after the local normal-form theorem.

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Part I

Scope, Targets, and Structural Distinctions

Dependency Order, Necessary Kernel Activation, and Proof Class

1.1 AASC necessity, not interpretive addition

A determinate anomaly claim must remain the same theorem object across formation, comparison, use, and report. Its target, symmetry or background, acted-upon carrier, transformation, quotient, anomaly locus, consequence, and endpoint must be fixed. Its comparison class must distinguish redescription from physical change, same-target repair from successor target, and admissible continuation from hidden selector. A completed classification must remain the completed act rather than being silently rewritten by later evidence. These are the functions of Reference, Admissibility, Standing, and Irreversibility.

The denial costs are structural rather than terminological. Without Reference, a local polynomial in one background may be substituted for a determinant phase in another. Without Standing, a Ward identity remains syntax without fixed consequence-bearing use. Without Admissibility, counterterm redistribution, aggregate cancellation, inflow, added matter, and target change are indistinguishable. Without Irreversibility, a later bulk, background map, or operator category can be projected backward into an already completed act. A critic retaining all four functions retains the kernel under another vocabulary; a critic removing one changes or destroys the determinate theorem object.

1.2 Pre-role physical establishment of the antecedent

The antecedent is established before the standing-role construction, physical-object construction, eligibility relation, or assignment analysis is introduced. Each closed target has one fixed physical input and two completed classification acts evaluated under one fixed discriminator: a standing-positive report and a standing-negative report. The acts may differ in asserted disposition, relation, or consequence, but not in target-defining physical data. For the Standard Model ledger, for example, the same species and hypercharges are reported once with the exact aggregate-neutrality result and once with a false gauge-inconsistency result; exact rational arithmetic distinguishes the acts. In the inflow fiber the same oriented bulk-boundary target is reported once with the computed relative-class cancellation and once with a false boundary-only or noncancelling disposition; the class sum distinguishes them. In the Wess–Zumino–Witten fiber the fixed level and fixed symmetry map are held constant while the positive act reports the computed pullback equality and the negative act denies it. Hypercharge controlled variations, changed doublet counts, reversed orientations, altered WZW levels, and changed beta-function data are retained as destructive tests or target updates, never as same-target T01 evidence.

SYM3-T01 – Pre-role non-degenerate determinacy. For every physical fiber reported as closed, a target-fixed physical comparison is completed before role formation. Each act reconstructs its own target fingerprint from its complete inputs; acts that change a target-defining coordinate are routed to TargetUpdate and excluded from the witness pair. The remaining same-target comparison contains at least one standing-positive and at least one standing-negative completed construction under one fixed comparison rule; the gate is neither empty nor universal; the exclusion changes a physical or theorem-bearing consequence; and a failed act is not repairable retrospectively as the same act. Therefore $\text{NonDegDet}(D_\lambda^{\text{phys}})$ is locally established without using later assignment counts.

Claim status: Local physical theorem, one physical witness per closed target. **Supporting record:** The target-fixed witness constructions are summarized in the scientific appendices; complete derivations appear in the Technical Audit Companion.

SYM3-T02 – Necessary kernel activation. For every closed SYM-3 fiber, $\text{NonDegDet}(D_\lambda^{\text{phys}})$ necessarily instantiates $K_{D_\lambda} = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$. The local witness supplies the antecedent; the imported AASC theorem supplies the necessity implication. No anomaly equation, source result, role architecture, finite normal-form calculation, AMetric theorem, or apex theorem is an additional activation premise.

Claim status: Imported AASC necessity theorem with local physical instantiation.

1.3 Kernel-first in necessity; QFT-first in certification

The logical and evidentiary orders must not be conflated. The kernel is logically prior because no physical record can be fixed, compared, used, or completed without it. QFT is evidentially prior to final role reporting because the transformation, current, measure, determinant, partition function, topology, counterterm, inflow relation, symmetry map, and operator-locus facts must be established without consulting the desired AASC role. This ordering blocks role labels from manufacturing their own premises while preserving the correct necessity direction.

1.4 Five sequential proof layers

1. **Necessary kernel layer.** Establish the pre-role physical antecedent and necessary kernel activation.
2. **Reference-complete QFT layer.** Fix target coordinates and source-supported physical facts.
3. **Independent construction layer.** Derive standing burdens and physical objects in process-isolated pipelines.
4. **Local normal-form layer.** Execute presentation, eligibility, compatibility, automorphism, and orbit calculations.
5. **Downstream apex layer.** Only after local singleton closure, apply the separately imported no-second-classifier result.

No layer substitutes for another. A finite normal-form calculation cannot generate the kernel; a continuum theorem cannot establish candidate saturation; the apex cannot choose an occupant or repair a missing relation.

1.5 Proof classes and falsification routes

Every theorem is typed as Imported QFT, Imported AASC, Local physical theorem, Executable finite theorem, Grammar-relative coverage theorem, Conditional frontier interface, Transport theorem, Open burden, Rejected, or Superseded. A criticism has standing if it preserves the same target while breaking kernel necessity, invalidates an imported proposition or specialization, exposes a missing burden, object, constructor, presentation witness, relation, or automorphism, or supplies a second same-target assignment orbit. Vocabulary preference, premise deletion, target change, and the demand that software generate the kernel do not by themselves refute the theorem. An implementation defect has standing against any conclusion that depends on it and leaves that conclusion open until repaired; it does not render the physical domain kernel-free.

Indexed Anomaly Targets and Reference Completeness

2.1 Why one undifferentiated target is insufficient

Anomaly status can change when the faithful global form, matter content, background fields, boundary or bulk system, operator category, phase, scale convention, or ultraviolet–infrared map changes. A theorem that silently aggregates those coordinates does not preserve Reference. SYM-3 therefore uses one integrated indexed family,

$$D_{\text{SYM3}}^* = (D_0, \Lambda, \{D_\lambda\}_{\lambda \in \Lambda}, \mathcal{T}),$$

whose common grammar and admissible transports are fixed while every physical conclusion remains fiber-indexed.

2.2 Reference-complete closed targets

A closed target fixes every coordinate used by its imported results, physical constructions, presentation maps, relations, failure routes, and downstream anomaly interface. The analyzed family contains ten such targets. Their complete fixed scopes are summarized in Table 2.1.

Table 2.1: Reference-complete closed targets and explicit SYM-9 frontiers

Target	Physical scope	Selected fixed coordinates	Status
F-SM-LOCAL-1GEN	Standard Model local anomaly ledger	gauge group local=['SU3c', 'SU2L', 'U1Y']; generations=1; normalization=Q=T3+Y; quantum regime=perturbative-local; right handed neutrino=False	closed
F-VA-BARDEEN-DESCENT	$U(1)_V \times U(1)_A$ descent and Bardeen structure	background group=['U1V', 'U1A']; counterterm basis=['V A FV', 'V A FA']; fermion content=one massless dirac; gauged subgroup=U1V; normalization=integer charge trace; common overall anomaly factor suppressed	closed
F-VECTORLIKE-3PAIR-SYMMETRIC	Symmetric vectorlike six-Weyl target	flavor breaking data=False; gauge group local=['U1']; observable algebra=charge-sector symmetric; quantum regime=perturbative-local; spacetime dimension=4	closed
F-SU2-WITTEN-GLOBAL	Ordinary $SU(2)$ global anomaly	field content=one left weyl doublet; gauge group=SU2; large transformation class=pi4 generator; operator category=partition function determinant line; spacetime dimension=4	closed
F-SU2-NEW-SPIN-ANOMALY	New spin- $SU(2)$ anomaly	chosen spin structure=False; field content=one isospin 3 over 2 weyl; gauge group=SU2; operator category=partition function 5d bulk invariant; spacetime dimension=4	closed
F-QCD-U1A-NF3	Massless three-flavor QCD $U(1)_A$	Nc=3; Nf=3; axial charge normalization=qL=+1,qR=-1; gauge group=SU3c; phase=chirally broken vacuum	closed
F-EW-BPLUSL-3GEN	Electroweak $B + L$	gauge group=SU2LxU1Y; generations=3; operator category=currents selection rules partition function; phase=electroweak quantum target; spacetime dimension=4	closed

continued

Target	Physical scope	Selected fixed coordinates	Status
F-U1-WEYL-INFLOW-4D5D	Weyl anomaly with five-dimensional inflow	anomaly basis=['U1 cubic', 'U1 gravity squared']; closed boundary charges=[1]; boundary dimension=4; bulk dimension=5; gluing map=restriction	
F-QCD3-WZW-MATCH	Three-flavor QCD gauged-WZW matching	Nc=3; Nf=3; background map=identity; bardeen closed convention=vector-preserving; ir target=SU3 chiral field	
F-QCD3-TRACE-MSBAR	QCD trace anomaly in a fixed $\overline{\text{MS}}$ target	Nc=3; Nf=3; gauge group=SU3c; loop order=2; closed operator basis=['F2', 'EOM', 'total derivative']	
F-SM-GLOBALFORM-LINES	SYM-9 frontier target	completion requires SYM-9 target data	frontier
F-SM-HIGHER-FLAVOR	SYM-9 frontier target	completion requires SYM-9 target data	frontier
F-SM-NONINVERTIBLE-DEFECT	SYM-9 frontier target	completion requires SYM-9 target data	frontier

The symmetric vectorlike target is not an auxiliary combinatorial universe. It is a physical anomaly-neutral theory with three indistinguishable charge-+1 and three indistinguishable charge-−1 left-handed Weyl species. Its nontrivial species permutations demonstrate that one normal-form isomorphism class need not mean one stored assignment or a trivial automorphism group.

2.3 Exact SYM-9 frontier targets

Three targets remain open by design: faithful Standard Model global form with genuine line and monopole sectors; higher-flavor or higher-group completion; and non-invertible chiral-defect completion. Each identifies the missing global-form, line, monopole, extension, defect, or operator-category data and assigns completion to SYM-9. These records are exact interfaces, but they do not discharge closed-target burdens or enter the local normal-form quantifier.

SYM3-T03 – Reference-complete indexed-target fixation. Every SYM-3 claim is attached to exactly one fixed target, an explicitly declared transport between fixed targets, a target update, or an open frontier burden. A target is eligible for local closure only when every coordinate used by its QFT results, physical constructions, presentation maps, compatibility relations, continuations, and failure routes is fixed and mutually consistent. A change of field content, global form, background, boundary or bulk content, operator category, phase, scale convention, or ultraviolet–infrared map is a target update unless already parameterized by the fixed target.

Claim status: Local Reference-completeness theorem with target-coordinate and premise validation.

2.4 Transports and successors

A transport $\tau : D_\lambda \rightarrow D_\mu$ is not a same-target repair. It states which coordinates are preserved, which change, which physical results remain applicable, and which conclusions must be recomputed. Conservative evidentiary enrichment is separated from standing-changing target modification by explicit maps. A later SYM-9 result may close a frontier target, but it produces a successor target and successor anomaly record rather than silently rewriting a completed predecessor.

2.5 The cost of target ambiguity

Without Reference there is no fixed anomaly class. Without Admissibility there is no principled distinction among presentation change, physical cancellation, target enlargement, and illicit repair. Without Standing the output cannot carry a determinate theorem consequence. Without Irreversibility later global-form or operator information could rewrite an already completed classification

as though it were the same act. Exact target specification is therefore part of the theorem burden, not merely an expository convenience.

Atomic Anomaly Objects, Facts, and Independent Pipelines

3.1 Four controlled layers

The analysis separates:

1. **Primitive target data:** immutable fiber coordinates and comparison rules.
2. **Proposition-operative physical facts:** source-supported claims whose premises match the fiber.
3. **Generated physical objects:** constructor outputs with derivation paths and inherited limitations.
4. **Derived conclusions:** residues, eligibility edges, relation dispositions, assignments, orbits, records, and theorem statuses.

Derived conclusions are never accepted as primitive premises.

3.2 Two isolated construction processes

The burden process reads target obligations, denial losses, loci, continuation, compatibility, and failure profiles. It has no candidate or source-object input. The physical process reads target coordinates, source-supported facts, physical seeds, and the typed constructor grammar. It has no burden, role, capability predicate, intended assignment, or stored compatibility verdict. The outputs meet for the first time in the all-pairs eligibility calculation.

The independence requirement is substantive: altering role terminology cannot change the physical object universe, while omitting a physically applicable generation rule leaves the target incomplete.

3.3 Top-level object kinds

The SYM-0 sum type remains operative:

$$\text{Action} \sqcup \text{Structure} \sqcup \text{Diagnostic} \sqcup \text{Transport}.$$

An anomaly polynomial is not a group action; a current is not its divergence; a determinant phase is not a local cocycle; an inflow pair is not a counterterm; a matching map is not an ordinary symmetry on one Hilbert space. Candidate decomposition is therefore target- and locus-sensitive.

SYM3-T04 – Anomaly-kind noncollapse. Gauge inconsistency, anomalous global-current response, local descent class, global determinant/topology obstruction, mixed anomaly, obstruction to gauging, inflow relation, ultraviolet-infrared matching obligation, trace/RG diagnostic, generalized-operator anomaly, neutral aggregate ledger, and target update are pairwise non-identical statuses. No status follows from the word “anomaly” alone; the target, dynamical/background status, locus, quotient, operator category, topology, and consequence must be fixed.

Claim status: Central target-indexed physical classification theorem.

3.4 Primitive data prohibition

Primitive candidate records may not contain `eligible_role`, expected occupant, intended assignment, canonical/defective label, stored pass/fail verdict, or a completeness flag that decides the theorem. Any derived object at constructor depth greater than zero must have an executed derivation. An unsupported manual insertion has no standing in the normal-form proof.

Independent Burdens and the Governance Quotient

4.1 The burden process

A standing burden is derived when denial of a fixed target interface destroys a declared anomaly construction and the loss is neither a presentation change nor a downstream witness. The burden process reads target coordinates, theorem queries, denial operations, continuation requirements, and failure routes. It does not read candidate identities, physical constructors, eligibility records, or assignments.

For each target, the baseline theorem queries are evaluated first. Each admissible denial then removes one interface and recomputes the available consequences. The resulting transition records which theorem queries disappear, which remain, and whether target identity is preserved. Burdens are therefore extensional: they are fixed by the physical consequences of denial rather than by a preferred role vocabulary.

4.2 Governance equivalence

For raw burdens b and b' , define

$$b \sim_{\text{gov}} b'$$

when their executed denial transitions have the same lost consequences, continuation behavior, target relation, and failure profile. The role architecture is

$$R_\lambda = B_\lambda / \sim_{\text{gov}},$$

with explicit multiplicity only when the fixed target itself contains lawfully interchangeable standing positions. Across the ten closed fibers, this construction yields 63 role positions. The symmetric vectorlike target contains genuine multiplicity because its three positive-charge and three negative-charge carrier positions are physically indistinguishable before flavor-breaking data are introduced.

SYM3-T05 – Extensional occupant-independent role necessity. Relative to each fixed target and its declared burden grammar, realization requires some independently constructed physical object to restore the theorem queries lost under the corresponding denial. Role positions and lawful multiplicities are derived before candidate inspection from the recorded losses, continuation changes, and failure behavior. A complete same-target denial behavior not represented in the governance quotient is a role-completeness counterexample and reopens the local theorem.

Claim status: Local finite theorem derived from executed denial transitions.

Table 4.1: Extensional burden quotient and lawful role multiplicity

Fiber	Raw burdens	Governance classes	Role positions	Representative denial losses
F-SM-LOCAL-1GEN	6	5	5	aggregate local class; ledger complete; quantum descent; species contributions; su2 global parity
F-VA-BARDEEN-DESCENT	7	5	5	axial ward identity; consistent ward polynomial; left charge ledger; presentation relation; six form charge tensor; vector gauge resolution
F-VECTORLIKE-3PAIR-SYMMETRIC	12	10	10	aggregate abelian anomaly; carrier response::charge +1; carrier response::charge -1; lawful species automorphisms; physical presentation quotient
F-SU2-WITTEN-GLOBAL	5	5	5	determinant phase; global consistency; local polynomial; ordinary mod2; representation ledger
F-SU2-NEW-SPIN-ANOMALY	6	6	6	global consistency; new mod2; ordinary mod2; representation ledger; tangential packet; w2w3 bulk
F-QCD-U1A-NF3	6	6	6	axial ward identity; classical axial action; discrete remnant; massless content; measure index anomaly; vacuum realization
F-EW-BPLUSL-3GEN	6	6	6	baryon rule; combined rule; electroweak index; generation content; lepton rule; nonperturbative status
F-U1-WEYL-INFLOW-4D5D	6	6	6	boundary class; boundary content; bulk class; inflow cancellation; integral quantization; relative gluing
F-QCD3-WZW-MATCH	7	7	7	background map; counterterm; gauged variation; ir target; uv anomaly; uv ir matching
F-QCD3-TRACE-MSBAR	7	7	7	beta gamma; finite scheme map; operator basis; operator quotient; qcd parameters; scheme invariant trace

4.3 Refinement discipline

A proposed additional role has four possible dispositions: governance-equivalent to an existing role; a genuine same-target refinement; a target update; or a downstream witness without independent standing function. No vocabulary rule forbids refinement. A nonisomorphic complete same-target denial structure is an explicit falsifier.

Conventions, Literature Basis, and Imported Results

5.1 Conventions

The anomaly conventions follow standard differential-form and background-field notation. Local consistent anomalies are treated through BRST cohomology and descent; global anomalies are treated through determinant phases, index or eta-invariant data, and their target-specific topological refinements. The distinction between dynamical gauge transformations and background transformations is explicit throughout. Currents, charges, Ward identities, anomaly polynomials, counterterms, and symmetry actions are recorded as different objects even when a theorem relates them.

5.2 Literature basis

The local cohomological treatment uses the Wess–Zumino consistency condition and the relative BRST cohomology framework [1–3]. Consistent and covariant anomalies, counterterm redistribution, and Bardeen–Zumino relations are treated with the standard primary sources and open technical derivations [4–7]. Global anomalies use the ordinary and newer $SU(2)$ analyses together with determinant-line, eta-invariant, and bordism formulations [8–12]. Inflow and anomaly matching are treated through bulk–boundary relations, ’t Hooft matching, and gauged Wess–Zumino–Witten realizations [6, 13–15]. The trace-anomaly treatment is tied to renormalized composite operators and operator mixing rather than to a beta-function coefficient alone [16–20].

5.3 Imported result versus local specialization

An imported theorem is used only at the strength of its stated hypotheses. The local SYM-3 burden is to match those hypotheses to a fixed target, state the limitation, and derive the target-specific anomaly record. A change of field content, background, global form, boundary condition, operator category, phase, or scale is a target update unless it is already part of the declared target family.

All load-bearing QFT claims are supported by freely accessible primary sources or explicit open composite derivations. Full source reviews, exact locators, identity records, and the evidence graph are collected in the Technical Audit Companion rather than repeated in the scientific narrative.

Part II

Quantum-Field-Theoretic Anomaly Architecture

Quantum Descent and Gauge Inconsistency

6.1 Classical invariance is not quantum descent

A classical gauge transformation may preserve the action while the regulated quantum measure or effective action fails to be invariant. The distinction is structural. Classical invariance is a proposition about an action functional on field representatives. Quantum descent is a proposition about whether the measure, renormalized effective action, partition function and physical state/operator construction are well-defined on gauge-equivalence classes. The latter cannot be inferred from the former.

Let $W[A]$ be a renormalized effective action for a background or dynamical gauge field. For an infinitesimal gauge parameter α , write

$$\delta_\alpha W[A] = 2\pi i \int_{M_d} I_d^{(1)}(\alpha, A).$$

When the symmetry is dynamically gauged, a nontrivial variation that cannot be removed within the fixed target obstructs the quantum quotient. In a BRST formulation the same statement is represented by a nontrivial ghost-number-one relative cohomology class. The formulas are coordinates of the obstruction; the physical endpoint is failure of descent.

SYM3-T06 – Gauge-anomaly quantum-descent failure. For a dynamically gauged symmetry in the fixed target, a nontrivial uncanceled obstruction to descent of the measure, effective action, partition function or BRST structure makes the proposed quantum gauge quotient undefined or inconsistent. It is not permitted explicit breaking, not a spontaneous vacuum effect and not a physically acceptable anomalous gauge symmetry.

Imported interface: The perturbative and measure statements are imported from Wess and Zumino [1], Bardeen [4], Adler [21], and Fujikawa [22]; the fixed-domain standing classification imports Maley [23, 24].

6.2 Individual contributions and aggregate quantum standing

A chiral matter representation can contribute a nonzero anomaly coefficient while remaining a lawful representation of the classical gauge architecture. Quantum admissibility is a property of the complete declared chiral ledger. For generators T^a in left-handed Weyl representations, the local cubic coefficient is represented schematically by

$$\mathcal{A}^{abc} = \sum_i \text{tr}_{R_i} \left(T^a \{T^b, T^c\} \right).$$

Mixed Abelian and non-Abelian coefficients, Abelian cubic sums and mixed gravitational-Abelian sums are separate coordinates. The aggregate target requires every relevant gauge obstruction to cancel. Termwise vanishing is neither necessary nor generally true.

The distinction blocks two opposite mistakes. A nonzero coefficient for one multiplet does not make that multiplet an inconsistent object by itself. Conversely, a list of individually meaningful representations does not guarantee a consistent quantum gauge theory. The aggregate representation, charge, multiplicity and global-group data must be tested as one ledger.

6.3 The quantum quotient as a conditional bridge

The manuscript distinguishes the classical proper-gauge quotient, BRST state cohomology, local operator cohomology and quantum effective-action descent. Under regularity and anomaly-neutrality assumptions these objects can be related by certified bridges. If nilpotency fails or a global obstruction remains, the bridge is blocked. No notation such as $H^0(s)$ is used to hide the difference between state, operator and relative complexes.

In particular, a local anomaly candidate belongs schematically to

$$H^{1,d}(s \mid d),$$

where the domain is the selected local field/ghost/antifield complex and equivalence is modulo BRST-exact and total-derivative terms. A nontrivial class is a candidate local obstruction. Its coefficient and target specialization must still be fixed. Global anomalies can remain even when this local group yields no obstruction.

6.4 Counterterms and cancellation

A local counterterm $S_{\text{ct}}[A]$ changes the representative by

$$\mathcal{A}(\alpha; A) \longmapsto \mathcal{A}(\alpha; A) + \delta_\alpha S_{\text{ct}}[A].$$

This can redistribute anomaly between currents, select a consistent representative, or trivialize a cohomologically exact term. It cannot remove a nontrivial obstruction class without satisfying the exact cohomological and quantization conditions. If the proposed neutralization adds new fields, a Green–Schwarz sector, a bulk invertible theory or new boundary modes, the target relation must be declared.

Gauge-anomaly falsifiers

The gauge-descent theorem is reopened by a correct demonstration that the registered coefficient, representation multiplicity, hypercharge, regulator, BRST complex, global form or counterterm class is wrong; by a missing global obstruction; or by a same-target neutralization omitted from the aggregate ledger. It is not refuted by assuming the anomalous quantum quotient exists and then demanding factorization through it.

6.5 AASC translation

On the role axis, the successful anomaly-neutral aggregate occupies the quantum-descent architecture. Individual coefficients, triangle diagrams, Ward identities and cancellation equations are derived witnesses. A nonzero aggregate is an obstruction. A regulator or current representative is skin only

when a complete witness preserves the invariant obstruction and consequences. Added matter or inflow is a target update unless already part of the declared target fiber.

This translation preserves the physics. Calling an anomaly coefficient a witness does not make it unphysical. Calling the successful aggregate a carrier constraint does not calculate the coefficient. The role language records dependency and standing, while the QFT calculation records the actual quantum result.

Local Anomaly Polynomials, Wess-Zumino Consistency, and BRST Descent

7.1 The anomaly polynomial as a coordinate source

For chiral fermions in even spacetime dimension $d = 2n$, perturbative gauge and gravitational anomalies can be encoded by a closed invariant polynomial in $d + 2$ dimensions. In index-theoretic notation the fermionic contribution is represented by the degree- $(d + 2)$ part of

$$I_{d+2} = \left[\widehat{A}(TM) \text{ch}_R(F) \right]_{d+2}.$$

The normalization and reality conditions depend on the fermion type and conventions. The polynomial is not itself an action on physical states and is not a complete global anomaly proof record. It is a local coordinate source whose descent generates consistent local anomaly representatives. The operative open derivations are the self-contained treatments of characteristic classes, Chern–Simons forms, descent, Wess–Zumino consistency and relative BRST cohomology in Barnich, Brandt, and Henneaux [2] and Bilal [3]; the historically original papers remain attribution only where their full text is not freely available.

Locally one writes

$$I_{d+2} = dI_{d+1}^{(0)}, \quad \delta_\alpha I_{d+1}^{(0)} = dI_d^{(1)}(\alpha, A).$$

The anomaly of the effective action is then represented by the integral of $I_d^{(1)}$. Different choices of Chern–Simons form or local counterterm can change the representative while preserving the invariant polynomial and cohomology class. Global information can obstruct a globally defined $I_{d+1}^{(0)}$ or introduce torsion data invisible to de Rham forms.

SYM3-T09 – Local polynomial and descent coordinate theorem. For the declared local perturbative target, the imported BRST/descent theorem identifies the consistent local anomaly class represented by the specified polynomial and descent complex. The finite SYM-3 calculation verifies the target-level coefficient and consistency data used by that specialization; it does not rederive the continuum cohomology theorem. The polynomial and descent forms are faithful coordinates only at the local/cohomological locus and do not by themselves determine global anomaly freedom, physical quotient consistency, or infrared realization.

Claim status: Open operative QFT structure from Barnich, Brandt, and Henneaux [2], Bilal [3], and Minasian, Papadimitriou, and Yi [7]; Wess and Zumino [1], Zumino, Wu, and Zee [25], Manes, Stora, and Zumino [26], and Alvarez-Gaume and Witten [27] are retained as historical attribution.

7.2 Wess-Zumino consistency

Gauge transformations form an algebra. Therefore the anomaly functional must satisfy an integrability relation. With $\mathcal{A}(\alpha; A) = \delta_\alpha W[A]$, the consistent anomaly obeys

$$\delta_\alpha \mathcal{A}(\beta; A) - \delta_\beta \mathcal{A}(\alpha; A) = \mathcal{A}([\alpha, \beta]; A).$$

This relation is necessary for an anomaly to arise as the variation of one effective action and is derived openly in Bilal [3, Sec. 9] and in the local BRST analysis of Barnich, Brandt, and Henneaux [2]. It restricts the allowed local form but does not say that the anomaly vanishes. It also does not make every solution physically admissible: coefficients, representations, target, counterterms and global conditions remain to be fixed.

The consistency equation has an exact role in the proof record. It is a local-coordinate compatibility condition. A candidate expression that fails it cannot occupy the local anomaly coordinate role. An expression that satisfies it is not automatically an obstruction to a dynamical gauge target; the class may be trivial, the aggregate coefficient may cancel, or the symmetry may be global rather than gauged.

7.3 Relative BRST cohomology

The BRST form packages the Wess-Zumino condition as a cocycle equation. If a_d^1 is a local d -form of ghost number one, consistency is

$$sa_d^1 + da_{d-1}^2 = 0,$$

with equivalence

$$a_d^1 \sim a_d^1 + sm_d^0 + dn_{d-1}^1.$$

The class lies in $H^{1,d}(s \mid d)$. This is the appropriate local complex for integrated local anomaly candidates. It is distinct from state cohomology and ordinary local operator cohomology. The target must state whether the gauge algebra is irreducible, open or reducible; which antifields are included; and whether the analysis is on-shell or off-shell.

The proof record records three separate propositions: the existence of a local cocycle, its nontriviality in the chosen complex, and the nonzero aggregate coefficient in the physical representation ledger. Omitting any one produces an incomplete obstruction record.

7.4 Covariant characteristic data and local limits

The polynomial method is powerful because it organizes gauge, mixed and gravitational terms uniformly and automatically satisfies consistency after descent. Its limits are equally important. Torsion, mod-two indices, large gauge transformations and global-form distinctions can evade local differential-form detection. A local cancellation can coexist with a global anomaly. The local chapter therefore exports an explicit `global_status=unresolved` field unless a separate global analysis has been completed.

7.5 Coordinate readiness

The AASC anomaly-algebraization source [28] supplies a readiness discipline. A trace, charge sum, mixed product or index is admissible as an anomaly coordinate only when the representation, chirality, multiplicity, normalization, probe and target are fixed. For example, $\sum_i Y_i^3$ is not meaningful until all fields are expressed in one left-handed convention with multiplicities. A mixed

coefficient requires both background factors and their joint target relation. An index or parity condition requires the global topology and operator domain.

Local-coordinate checklist

A complete local analysis states: spacetime dimension and orientation; spin or other tangential structure; gauge/global symmetry and representation; chirality convention; regulator and quantum object; invariant polynomial with normalization; descent complex; consistent representative; counterterm class; aggregate coefficient; boundary assumptions; scale/perturbative range; global-status handoff; and exact nonclaims.

Consistent and Covariant Anomalies, Currents, and Counterterm Quotients

8.1 Two current notions

The consistent current is obtained by differentiating one effective action with respect to the gauge field. Its divergence gives the consistent anomaly, which satisfies Wess-Zumino consistency. The covariant current transforms covariantly but generally is not the functional derivative of the same effective action. Its anomaly has a different coefficient or local form. The two currents and their anomalies are related by a local Bardeen–Zumino polynomial. The operative open evidence is the background-gauged current construction of Harvey, Hill, and Hill [6] and the systematic consistent-to-covariant analysis of Minasian, Papadimitriou, and Yi [7]; Bardeen and Zumino [5] is retained as historical attribution.

Schematically,

$$J_{\text{cov}} = J_{\text{cons}} + J_{\text{BZ}}, \quad \mathcal{A}_{\text{cov}} = \mathcal{A}_{\text{cons}} + D \cdot J_{\text{BZ}}.$$

The relation does not make the two currents identical. It states that they are linked by a specific local transport preserving the underlying obstruction and cancellation condition. A manuscript that lists both as independent anomalies without the relation double-counts one local class; a manuscript that uses one formula while claiming the properties of the other conflates loci.

SYM3-T10 – Consistent/covariant representative relation. A consistent anomaly, covariant anomaly, consistent current, covariant current, and Bardeen-Zumino polynomial are distinct atomic records. They belong to one relation class only where an imported source theorem and an explicit target-level transport or canonicalization witness preserve the target, invariant anomaly class, aggregate cancellation condition, and physical consequences. Mere agreement of a selected coefficient is reported as invariant agreement, not as a full presentation isomorphism.

Claim status: Open-composite QFT relation from Harvey, Hill, and Hill [6] and Minasian, Papadimitriou, and Yi [7] plus witnessed presentation-quotient theorem.

8.2 Bardeen counterterms and current allocation

When several currents are present, a local counterterm can shift which current carries the anomaly while preserving an invariant mixed obstruction. Bardeen’s vector/axial allocation is the historical canonical example. The operative freely accessible construction, including the modified counterterm required when background sources are present, is given by Harvey, Hill, and Hill [6, Secs. I–IV]; Bardeen [4] remains attribution. The choice can be physically forced: if one current is gauged,

gauge consistency requires a representative in which that gauge current is conserved, leaving the anomaly in a global current when the aggregate theory permits it.

The counterterm quotient therefore depends on the target. A redistribution acceptable for background global sources may be inadmissible when it spoils a dynamical gauge symmetry. A counterterm allowed on a closed manifold may require boundary terms or fail under large transformations. Quantization conditions can make a formally local shift globally unavailable.

8.3 Regulator and measure presentations

The regulated fermion-measure Jacobian gives a powerful coordinate for the axial anomaly. The operative open derivation used here is Bilal [3, Sec. 3]; Fujikawa [22] remains the historical attribution. Diagrammatic, effective-action and measure computations can agree on the invariant result when their regulators and normalization are matched. The measure is not an oracle: changing basis, phase or regulator cannot be called cancellation unless the complete quantum target remains fixed and the obstruction class becomes trivial or aggregate-neutral.

The presentation witness for two regulators includes equality of renormalized physical amplitudes, anomaly class, counterterm relation, boundary/global status and downstream consequences. If one regulator preserves one symmetry at the expense of another, the relation is a redistribution with explicit target consequences, not a neutral notation change.

SYM3-T13 – Counterterm-equivalence theorem. A local counterterm, current improvement, regulator change or Bardeen-Zumino transport is presentation-equivalent only when a validated witness preserves the invariant anomaly class, target, aggregate cancellation condition, global status, boundary data and reportable consequences. A nontrivial obstruction cannot be removed by a representative change alone.

8.4 Relation lattice

The proof record distinguishes:

1. identical representative;
2. same cohomology class, different local representative;
3. consistent/covariant transport;
4. anomaly redistribution between currents;
5. cohomologically trivial removal;
6. aggregate cancellation by other matter already in target;
7. cancellation by an integrated inflow sector;
8. target enlargement by new matter or bulk;
9. incompatible or unresolved relation.

No pair is assigned merely from similar equations. The complete relation-lattice table appears in the Technical Audit Companion.

Representative-quotient falsifier

A valid counterexample exhibits two records that the quotient identifies although they differ in invariant class, large-transformation phase, boundary variation, operator category, target relation or physical consequence; or two records kept separate despite a complete invertible witness preserving the full tuple. Display-name differences and source-record IDs are not physical distinctions.

Global Currents, ABJ Anomalies, and Physical Response

9.1 Global action, current, and anomaly are distinct

For a continuous global symmetry, a transformation may act nontrivially on physical fields, states, or operators. Under suitable assumptions Noether's theorem supplies a current, and boundary conditions may supply a charge. Quantization can alter the Ward identity. The action, current, integrated charge, Ward identity, measure variation, and anomaly class remain distinct objects, as already required for spacetime symmetry in SYM-2 [29].

A current anomaly does not necessarily erase the transformation from the classical theory, eliminate every useful selection rule, or make the gauge target inconsistent. It states that the quantum functional cannot preserve all the corresponding background or current relations. The physical status depends on whether the symmetry is exact, anomalous, explicitly broken, spontaneously realized, approximate, or replaced by a remnant in the fixed target.

SYM3-T07 – Global-current action/conservation separation. A global symmetry can be a quotient-nontrivial physical action while its associated classical current is quantum mechanically anomalous. The action record, current record, Ward-identity record, anomaly diagnostic, and surviving remnant receive separate statuses.

9.2 Axial $U(1)_A$ in massless QCD

Consider N_f massless quarks. Classically the axial transformation

$$q_L \mapsto e^{i\alpha} q_L, \quad q_R \mapsto e^{-i\alpha} q_R$$

leaves the massless action invariant. The singlet axial current satisfies the anomalous quantum relation

$$\partial_\mu J_5^\mu = \frac{g^2 N_f}{16\pi^2} \text{tr}(F_{\mu\nu} \tilde{F}^{\mu\nu}) + 2i \sum_f m_f \bar{q}_f \gamma_5 q_f,$$

up to normalization conventions. The anomaly term and explicit mass term have different causal ancestry. In the massless target, the local density is a coordinate of the ABJ anomaly; in topologically nontrivial sectors its integral is related to the index and topological charge [21, 22, 30].

The SYM-3 record therefore assigns:

- classical physical action: exact in the massless parameter-limit target;
- quantum current: anomalous;

- local anomaly: ABJ class;
- topology: nonperturbative sensitivity;
- vacuum: state-dependent and linked to chiral realization;
- mass deformation: explicit breaking route;
- possible discrete/non-invertible remnant: separate generalized target.

Calling the action “absent” would erase the object whose quantum nonconservation is being diagnosed. Calling it an exact quantum $U(1)$ would ignore the anomaly.

9.3 Baryon and lepton currents

At the renormalizable classical level the Standard Model admits accidental baryon and lepton phase symmetries. Electroweak topology violates $B + L$ nonperturbatively while preserving $B - L$ in the appropriate local gauge-anomaly sense. For topological charge Q_{top} , the selection rule is conventionally summarized by

$$\Delta B = \Delta L = N_g Q_{\text{top}}, \quad \Delta(B + L) = 2N_g Q_{\text{top}}, \quad \Delta(B - L) = 0,$$

for N_g generations in the minimal target [31–33].

The phrase “baryon number is conserved” is therefore incomplete without perturbative/nonperturbative, operator-dimension, neutrino, and EFT scope. Higher-dimension operators can violate B or L even when the renormalizable Lagrangian does not [34]. The anomaly record is not a primitive carrier constraint; it is an effective physical action plus an anomaly/selection diagnostic in a stated target.

9.4 Background-field interpretation

A global symmetry is probed by coupling to a nondynamical background field A_{bg} . If the generating functional transforms nontrivially under background gauge transformations, the theory has a ’t Hooft anomaly for that global symmetry or symmetry combination. The background field is a probe, not a new dynamical gauge field. This distinction prevents a background anomaly from being reported as inconsistency of the original ungauged theory.

The physical global action can remain meaningful even though one cannot promote the background to a fully dynamical gauge field without additional structure. The anomaly can enforce selection rules, protected contact terms, boundary states, or IR matching. Its structural role is obstruction/witness, while the transformation remains a physical action.

9.5 Current redistribution and Bardeen choice

When vector and axial currents participate in a mixed anomaly, a local Bardeen counterterm can shift which Ward identity is manifestly conserved. In a gauge theory the dynamical gauge current must be preserved, so the anomaly is assigned to the global current. The ability to redistribute local representative terms does not make the mixed incompatibility conventional. It reflects a choice of which symmetry is required to be exact in the target [4, 5].

The counterterm record therefore carries:

- source and target current assignments;

- local polynomial shift;
- preserved dynamical gauge invariance;
- invariant mixed anomaly class;
- boundary terms and global conditions;
- exact nonclaim that no physical obstruction is erased merely by relabeling currents.

9.6 Adler–Bardeen nonrenormalization

For the appropriate perturbative target, the axial anomaly coefficient is one-loop exact [35]. SYM-3 imports this as a coefficient-stability proposition, not as a general claim that all anomalies receive no higher corrections or that all global anomaly data are perturbative. The theorem has exact regulator, current, and renormalization assumptions. Its role is to stabilize a coordinate of the anomaly record across perturbative continuation.

9.7 Vacuum and spontaneous realization

A global symmetry can also be spontaneously realized. The action remains a symmetry of the quantum dynamics while the vacuum is not invariant. An anomaly and spontaneous symmetry breaking can coexist. In QCD, chiral symmetry, the axial anomaly, condensates, and Goldstone modes occupy distinct records. SYM-6 will analyze the complete chiral/vacuum sector, but it must import the anomaly record from SYM-3 unchanged.

An anomaly can be matched in a broken phase by a Wess–Zumino–Witten term for Goldstone fields [15]. That term is not a same-target cancellation of the UV anomaly; it is an IR realization of the same 't Hooft anomaly.

9.8 Global-current status table

Object	Candidate/status	Required distinction
Global transformation	transforma-Physical action	Acts on states/operators after quotient
Noether current	Derived local operator	Requires target and boundary assumptions
Integrated charge	Generator/witness	May fail to exist or conserve because of flux/anomaly
Ward identity	Diagnostic	Encodes local consequence, not the action itself
ABJ density	Local anomaly coordinate	Representative of invariant quantum obstruction
Topological charge/index	Global diagnostic	Connects integrated density to topology under hypotheses
Mass term	Explicit deformation	Distinct from anomaly
Non-invertible remnant	Generalized action	Requires defect-completed target

9.9 Failure routes

The chapter fails if it describes every anomalous global symmetry as meaningless, infers a global charge from a local current without boundary conditions, uses the ABJ anomaly as ordinary explicit mass breaking, or treats a counterterm assignment as a second physical anomaly. It also fails if it imports a massless or zero-Yukawa limit into the observed target without an explicit parameter-limit coordinate.

Global, Topological, Eta, and Bordism Anomalies

10.1 Why local cancellation is not enough

Perturbative anomaly polynomials detect infinitesimal noninvariance near the identity of the gauge or background group. A theory can have vanishing local polynomial and nevertheless fail under a large transformation or on a topologically nontrivial background. Global anomaly classification therefore requires target data absent from a local Lie-algebra ledger: faithful global group, tangential structure, bundle topology, representation, manifold class, and partition-function phase.

The public object is not merely a sign encountered in one calculation. It is the obstruction to defining the fermion determinant or Pfaffian, quantum partition function, or state consistently over the full configuration/background space. Modern formulations describe the partition function as a section of a determinant line; an anomaly is nontrivial holonomy or a failure to trivialize that line compatibly. Eta invariants and invertible field theories encode the corresponding global information. The operative open authority is Yonekura [10], Freed and Hopkins [11], and Freed [36]; Dai and Freed [37] is historical attribution.

SYM3-T11 – Global anomaly determinant/topology theorem. Vanishing of every target-relevant perturbative local anomaly coefficient does not imply global anomaly freedom. A complete gauge or background anomaly record must include the declared global group, tangential structure, topology, representation, and global invariant or a certified proof that no such obstruction exists.

10.2 The original Witten $SU(2)$ anomaly

For four-dimensional $SU(2)$ gauge theory, $\pi_4(SU(2)) \cong \mathbb{Z}_2$. The ordinary parity obstruction for an odd number of relevant left-handed Weyl multiplets is used through the open review and extension in Wang, Wen, and Witten [9] and the Standard Model global-form analysis of Davighi, Gripaios, and Lohitsiri [12]; Witten [8] is the historical source. The perturbative cubic $SU(2)^3$ anomaly vanishes because the representations are pseudoreal. The obstruction is global and nonperturbative.

The SYM-3 family separates:

- the $SU(2)$ gauge transformation and bundle path;
- the fermion Pfaffian/determinant line;
- the mod-two spectral flow or index;
- the representation parity condition;

- the quantum gauge inconsistency for an odd obstruction;
- the Standard Model even-doublet witness.

The sign is not treated as a common unphysical state-vector phase. It is a holonomy of the functional determinant over configuration space and prevents a globally consistent gauge-invariant definition.

10.3 The new $SU(2)$ anomaly

A different anomaly arises when the theory is formulated without a chosen spin structure using a spin- $SU(2)$ structure. An odd number of fermion multiplets of isospin $4r + \frac{3}{2}$ is inconsistent; the five-dimensional invariant can be expressed by $(-1)^{\int w_2 w_3}$ [9]. The selected closed fiber imports that theorem for one isospin-3/2 Weyl multiplet and executes its representation-congruence, tangential-structure, ordinary-index, newer-index, and phase-consistency coordinates. It does not claim a software derivation of the bordism invariant on arbitrary manifolds. This record has a different tangential structure and cannot be merged with the original anomaly merely because both involve $SU(2)$ and a mod-two condition.

The comparison illustrates target discipline. Choosing an ordinary spin structure can remove the new anomaly by changing the target assumptions. Embedding $SU(2)$ into $U(2)$ can convert aspects of the global anomaly into perturbative mixed constraints under additional charge data [38]. These are structured relations between targets, not contradictions.

10.4 Eta invariants and determinant lines

Let Y^{d+1} be a manifold bounding or interpolating the d -dimensional spacetime/background data. For suitable fermionic systems, the exponentiated eta invariant provides a global phase

$$\exp(-2\pi i \eta_Y/2)$$

that combines with the boundary determinant to form a well-defined quantity under the Dai–Freed theorem. The operative open derivation and determinant-line interpretation are taken from Yonekura [10, Secs. 2–4]; Dai and Freed [37] is historical attribution. The precise normalization, reduced eta invariant, zero modes, and tangential structure are source- and convention-dependent; the manuscript stores them in the proposition proof record rather than presenting one universal formula.

The determinant-line language clarifies several points:

- a local anomaly is curvature of the anomaly line/connection;
- a global anomaly can remain as nontrivial holonomy when local curvature vanishes;
- choosing a local phase convention is a presentation, not necessarily a global trivialization;
- a bulk invertible theory can provide a canonical inflow trivialization for the combined system.

10.5 Bordism and invertible anomaly theory

Modern anomaly classification often identifies deformation classes of anomalies with invertible $(d+1)$ -dimensional field theories, constrained by the symmetry and tangential structure [11, 36, 39]. Bordism groups can organize candidate global anomalies. The use is conditional: one must

specify whether the theory is bosonic or fermionic, spin, spin^c , pin, spin- G , oriented, or otherwise structured, and whether the symmetry is internal or includes spacetime action.

SYM-3 does not equate “bordism group is nonzero” with “this particular theory is anomalous.” The theory defines a homomorphism or anomaly invariant on the relevant bordism group; its value must be computed from the representation and target. Conversely, a vanishing group under the exact structure can certify absence of anomalies of that deformation type. The bordism interface is therefore a classification arena plus representation-specific evaluation, not a magic lookup table.

Corollary – Bordism readiness. A bordism or invertible-field-theory statement enters an atomic anomaly record only after the spacetime dimension, tangential structure, faithful symmetry group, background category, representation, and deformation/equivalence notion are fixed. Changing any of these coordinates is a target update or a distinct proof record.

10.6 Global gravitational anomalies

Gravitational anomalies can be local or global and depend strongly on dimension and chirality. The operative open perturbative and index-theoretic treatment is Bilal [3, Secs. 8–11]; determinant-line, eta and bordism interfaces are supplied by Yonekura [10] and Freed and Hopkins [11]. Alvarez-Gaume and Witten [27] and Witten [40] remain historical attribution. SYM-3 treats these as external/background anomalies unless gravity is dynamical in the target. A mixed gauge-gravitational anomaly of a dynamical gauge current is a gauge inconsistency; a mixed gravitational anomaly of a global current is a ’t Hooft/background obstruction.

The fixed Standard Model target has nondynamical Minkowski spacetime, but curved backgrounds are admitted as controlled anomaly probes. A statement true only on curved or topologically nontrivial backgrounds is not dismissed as irrelevant; it is target-indexed. A claim about dynamical quantum gravity is a target update.

10.7 Faithful global form and line sectors

The local Lie algebra does not determine the global gauge group or bundle sectors. Different quotients G_{SM}/Γ can share perturbative particle content while differing in allowed bundles, line operators, monopoles, theta periodicities, and global anomalies [41, 42]. Every global-sensitive anomaly theorem therefore carries a Γ field or quantifies over a declared family.

A global anomaly calculation performed for $SU(2)$ cannot be silently transported to $SO(3)$, $U(2)$, or a quotient product. The relation may be computable, but it is a target relation requiring an explicit bridge. Likewise, admitting line operators or monopoles can change the generalized-symmetry and anomaly record.

10.8 Global anomaly proof record

A complete record contains:

1. exact global group and tangential structure;
2. spacetime dimension and manifold class;
3. representation or field content;
4. large transformation, loop, or bordism cycle;

5. determinant/Pfaffian/partition-function object;
6. local anomaly status;
7. global invariant and normalization;
8. physical consequence;
9. inflow or trivialization relation;
10. source, limitations, and target-update edges.

10.9 Failure routes

The chapter fails if it infers global freedom from local cancellation, uses $\pi_d(G)$ alone as the anomaly without evaluating the fermion determinant, ignores the tangential structure, treats an eta phase as an arbitrary state-vector phase, or imports a result between different global forms without a bridge. It also fails if a bordism classification is presented as a computation of every theory's anomaly.

Mixed, Discrete, and Gauge–Gravitational Anomalies

11.1 Mixed anomalies are relational objects

A mixed anomaly involves two or more independently typed symmetries, backgrounds, or geometric structures. Typical local coordinates in four dimensions include

$$F_A \operatorname{tr}(F_G^2), \quad F_A F_B^2, \quad F_A p_1(T),$$

and their non-Abelian or higher-form analogues. The anomaly does not belong to either factor alone. It expresses an incompatibility among simultaneous background gauge invariance, dynamical gauging, current conservation, or global definition.

A counterterm can redistribute a local mixed anomaly between currents, but it cannot make all incompatible requirements simultaneously true unless the invariant mixed class is trivial. The target must declare which symmetries are dynamical gauge redundancies and which are global probes. Preserving a dynamical gauge symmetry can force the anomaly into a global Ward identity; changing that priority changes the target requirement, not the underlying calculation.

SYM3-T12 – Mixed-anomaly nonfactorization. A mixed anomaly is indexed by the complete ordered set of participating symmetry/background factors and by the counterterm convention. It cannot be assigned to one factor or inferred from separate pure-anomaly records.

11.2 Gauge–global and global–global cases

For a dynamical gauge group G and global $U(1)_X$, a $G^2 U(1)_X$ anomaly may be harmless for gauge consistency if the G current is preserved and the anomaly is interpreted as nonconservation or a 't Hooft anomaly of X . If one attempts to gauge X , the same coefficient becomes part of the enlarged gauge-consistency conditions. The status changes because the target changes.

For two global symmetries, a mixed anomaly obstructs gauging both simultaneously and constrains the joint IR realization. One may preserve one background invariance manifestly through a counterterm choice. The relation lattice records this redistribution without creating two independent anomaly classes.

11.3 Gauge–gravitational and global–gravitational cases

A mixed gauge–gravitational anomaly in a dynamical gauge current is inconsistent. For an Abelian gauge factor in four dimensions, the local coefficient is proportional to the sum of left-handed charges, subject to exact normalization and representation multiplicities. The Standard Model hypercharge sum vanishes generation by generation.

A mixed gravitational anomaly of a global symmetry instead means that the global current cannot be conserved in arbitrary gravitational backgrounds. For $B - L$, the answer depends on neutrino content. Adding a right-handed neutrino per generation changes the target and can render $B - L$ fully anomaly-free and gaugeable under the appropriate conditions. Without it, cubic and gravitational obstructions remain. Non-invertible remnants in gravitational backgrounds are a separate generalized-symmetry construction [43].

11.4 Discrete anomalies

Discrete symmetries cannot always be analyzed by naively replacing continuous charges with residues in triangle equations. Some discrete anomaly constraints descend from a continuous embedding; others require group cohomology, eta invariants, bordism, or topological actions [44, 45]. The exact result depends on whether the theory is bosonic or fermionic, the tangential structure, the faithful discrete group, and whether the symmetry is gauged or global.

A discrete anomaly record contains:

- the exact finite group and action;
- background cocycle or bundle data;
- spacetime/tangential structure;
- local embedding, if used;
- global invariant, congruence, or topological action;
- counterterm equivalence;
- physical consequence and matching status.

The word “mod N ” is not a sufficient proof record.

11.5 Discrete chiral symmetry and gapped phases

Certain anomalies of four-dimensional discrete chiral symmetries forbid a symmetry-preserving trivially gapped vacuum under stated assumptions [46]. The conclusion is stronger than generic matching but weaker than a unique IR prediction. The theory may be gapless, break the symmetry, or realize topological order compatible with the anomaly. The proposition proof record records unitarity, locality, symmetry, and gap assumptions.

11.6 Bardeen redistribution matrix

For a mixed anomaly among currents J_A and J_B , local counterterms can implement a relation of the form

$$(\nabla \cdot J_A, \nabla \cdot J_B) \mapsto (\nabla \cdot J_A + \Delta_A, \nabla \cdot J_B + \Delta_B),$$

with the invariant incompatibility preserved. The counterterm matrix in the Technical Audit Companion records which currents are required exact, which are global, the local shifts, and the unchanged anomaly class.

11.7 Higher mixed structures

A zero-form symmetry and one-form symmetry can combine into a 2-group, with background transformations that mix their gauge fields. The corresponding anomaly is not a sum of separate zero-form and one-form anomalies; the Postnikov class and background transformation law are target data [47, 48]. SYM-3 types the anomaly relation; SYM-9 owns the complete Standard Model generalized-symmetry inventory.

11.8 Mixed-ledger stability

The indexed proof record checks that product gauge factors, representations, charges, global symmetries, backgrounds, and global form share one target. A coefficient computed with one charge normalization or neutrino target cannot be combined with another row. Mixed-ledger stability requires:

1. common field content and chirality convention;
2. common faithful global group and bundle category;
3. common background normalization;
4. complete multiplicities;
5. declared counterterm convention;
6. local/global completion;
7. compatible vacuum, boundary, and scale scope.

11.9 Failure routes

The chapter fails if it treats a mixed anomaly as two pure anomalies, uses continuous embedding formulas as complete discrete tests, ignores gravitational/tangential assumptions, or reports a counterterm redistribution as simultaneous conservation. It also fails if a neutrino or global-form change is used to repair a source target without an update record.

't Hooft Anomalies and Obstruction to Gauging

12.1 Definition by background gauging

A global symmetry G has a 't Hooft anomaly when the theory cannot be coupled to arbitrary background G gauge fields with a partition function invariant under background gauge transformations, subject to the declared locality and counterterm equivalences. Equivalently, the anomaly is an obstruction to promoting the global symmetry to a dynamical gauge symmetry without additional degrees of freedom or a higher-dimensional bulk. It is not generally an inconsistency of the original ungauged theory.

The background-field construction fixes the symmetry acting faithfully on the theory, possible quotient by a trivially acting subgroup, the background bundles, tangential structure, operator category, and allowed local counterterms. Failure to specify these coordinates produces an ambiguous anomaly statement.

SYM3-T08 – 't Hooft gauging-obstruction theorem. A nontrivial 't Hooft anomaly obstructs gauging the declared global symmetry/background structure within the fixed target. It does not erase the global physical action, and it becomes a gauge inconsistency only in a target where the anomalous symmetry is already dynamical.

12.2 Anomaly as RG-invariant data

Anomalies of exact global symmetries are invariant under RG flow when the symmetry, background structure, and target relation are preserved. The UV and IR generating functionals can differ by local counterterms, but their anomaly classes must agree. This is 't Hooft anomaly matching [14, 49].

The word “matching” is often used too loosely. A complete statement fixes:

- the UV theory and exact symmetry;
- the background fields and anomaly class;
- the RG relation and preserved subgroup;
- the IR degrees of freedom or phase;
- the local counterterm equivalence;
- the equality or inflow relation between anomaly responses.

A similarity of group names is not a matching proof record.

12.3 No unique infrared phase follows in general

One anomaly can be matched by several physically inequivalent IR realizations:

1. massless composite or elementary degrees of freedom;
2. spontaneous symmetry breaking with Goldstone modes and Wess–Zumino terms;
3. a topological quantum field theory or topological order;
4. a symmetry-enriched phase with emergent gauge fields;
5. a boundary of a higher-dimensional invertible theory;
6. a non-invertible or categorical remnant in a separately declared target.

Therefore anomaly matching constrains but usually does not uniquely determine the IR. Additional dynamics, positivity, operator content, or minimality assumptions are required for uniqueness.

SYM3-T16 – IR nonuniqueness theorem boundary. Equality of a nontrivial UV and IR 't Hooft anomaly class is necessary for a symmetry-preserving RG relation, but by itself does not select one unique IR spectrum, phase, or topological order.

12.4 Obstruction to a trivially gapped symmetric phase

A nontrivial anomaly rules out a completely trivial, symmetry-preserving, gapped IR under appropriate assumptions because such a phase has no degrees of freedom or topological response capable of reproducing the anomaly. Stronger theorems may exclude broader classes of gapped phases for particular discrete anomalies [46]. The exact conclusion depends on the symmetry, spacetime dimension, tangential structure, and allowed topological order.

The atlas records a graded conclusion:

- no trivially gapped symmetric phase;
- no gapped symmetric phase under a cited theorem;
- gapped topological matching allowed;
- symmetry breaking allowed;
- gapless matching required or permitted;
- unresolved.

This prevents a generic anomaly from being overclaimed as proof of massless particles.

12.5 Spontaneous symmetry breaking and Wess–Zumino terms

If G breaks to H , the anomaly of the unbroken symmetry and the full background response must still be matched. Goldstone fields can carry a Wess–Zumino–Witten term whose variation reproduces the anomaly [15]. The symmetry-breaking pattern, target manifold, quantization, and global extension must be fixed. The Wess–Zumino term is an IR realization witness, not a cancellation of the UV anomaly.

12.6 Symmetry extension and emergent gauge structure

An anomalous symmetry can sometimes be extended by a larger group or combined with emergent gauge structure so that the extended symmetry is anomaly-free while the original anomaly is reproduced by topological degrees of freedom. This can produce symmetric gapped boundaries or interfaces [50]. Such constructions add target data. They are not proof that the original anomaly was absent.

The relation lattice distinguishes:

- same anomaly class under presentation change;
- pullback-trivialization under symmetry extension;
- IR topological matching;
- added-sector target update;
- genuine cancellation in a larger declared target fiber.

12.7 Anomaly matching across explicit deformations

If a deformation explicitly breaks the symmetry, the original matching obligation need not survive unchanged. The correct record transports to the preserved subgroup or treats the deformation as a target change. An approximate symmetry has an approximate/effective anomaly utility only in a stated regime; it is not assigned an exact 't Hooft class without qualification.

Mass deformations can integrate out fermions and generate local topological terms. The resulting low-energy anomaly record includes threshold contributions and possible invertible phases. Matching is performed with the complete effective action, not by deleting heavy fields without their induced terms.

12.8 Relation to gauge anomaly

The same polynomial may appear in a gauge-anomaly cancellation equation and a 't Hooft anomaly. The role depends on the target:

- dynamical gauge factor: nonzero aggregate coefficient is inconsistency;
- global symmetry background: nonzero class is obstruction to gauging and matching data;
- mixed dynamical/global factors: preserve gauge invariance and assign the anomaly to the global background, subject to counterterm convention;
- both factors gauged in an enlarged target: all gauge conditions must cancel.

Matching proof record. A complete record fixes the UV and IR targets, the symmetry/background map, invariant anomaly class, counterterm convention, preserved subgroup, phase or topological content, global and operator data, and literature basis. A claimed match fails if the symmetries differ without a map, the global form or tangential structure changes silently, heavy-field induced terms are omitted, only one local coefficient is compared while a global component remains, or the proposed IR phase has no object capable of carrying the anomaly.

Anomaly Matching and Infrared Realization

13.1 The matching problem

The matching role is not satisfied by repeating that anomalies are RG invariant. It requires one exact relation between a UV anomaly record and an IR physical realization. The target fixes the UV symmetry and backgrounds, the preserved IR symmetry or subgroup, the RG/EFT map, local counterterm equivalence, global form, tangential structure, and operator category. A change in any of these coordinates must be transported explicitly.

Let α_{UV} and α_{IR} denote anomaly classes in the same declared classification group or invertible anomaly theory. The general necessity and broken-phase alternatives are supported by the open analysis of Hason, Komargodski, and Thorngren [51]; generalized-symmetry matching is scoped by Gaiotto et al. [52]. A match has the schematic form

$$\alpha_{UV} = \varphi^* \alpha_{IR} + \delta S_{\text{local}},$$

where φ is the symmetry/background map and the local term belongs to the allowed counterterm equivalence. This formula is schematic; the exact category may be cohomological, differential, eta/bordism, or defect-theoretic.

SYM3-T15 – Anomaly matching transport theorem. When an exact global symmetry, its faithful background structure, and the relevant operator category survive an RG flow, the imported anomaly-matching theorem requires equality of the UV and IR anomaly classes under the declared symmetry/background map, up to allowed local counterterms. In the selected three-flavor QCD fiber, the finite calculation checks the corresponding coefficient pullback for levels $k = 1, 2, 3, 4, 5$, the identity or exchanged background map, and the declared counterterm conventions. A mismatch rules out that proposed symmetry-preserving IR relation; the finite check is not a fresh derivation of the full gauged Wess–Zumino–Witten functional.

13.2 Executed $N_f = 3$, $N_c = 3$ QCD/Wess–Zumino–Witten fiber

The closed matching fiber fixes massless $SU(3)_c$ QCD with three colors and three flavors, background $SU(3)_L \times SU(3)_R$, the identity background map, the vector-preserving counterterm convention, the ultraviolet coordinate

$$a_{UV} = (3, -3) \in \mathbb{Z} \oplus \mathbb{Z},$$

and an infrared chiral field $U \in SU(3)$. The declared finite candidate family contains the quantized levels $k = 1, 2, 3, 4, 5$, the identity and left–right-swapped background maps, and either the vector-

preserving counterterm convention or no such counterterm. Under the fixed coefficient convention, the infrared coordinate is $(k, -k)$ and the identity-map pullback test is

$$a_{UV} - f^* a_{IR} = (3 - k, -3 + k).$$

Only $k = N_c = 3$, together with the fixed identity map and vector-preserving convention, satisfies every target-level relation. The other levels or maps are complete finite candidates and fail one or more of the level, background-map, counterterm, or pullback checks. The operative open WZW/background-current authority is Harvey, Hill, and Hill [6]. This fiber implements the published coefficient and background-current relations at the stated target level; it does not claim to reconstruct the complete gauged WZW functional for every coset, global form, or background category.

13.3 Massless matching

A gapless IR can match an anomaly through massless fermions, interacting conformal degrees of freedom, or emergent gauge sectors. The proof record must compute the IR anomaly using the actual massless representations and global form; it may not merely count candidate composites. Classic anomaly-matching arguments use this constraint to test proposed spectra. The operative open authority for the present nonuniqueness and broken-phase claims is Hason, Komargodski, and Thorngren [51]; Hooft [14] and Frishman et al. [49] are historical attribution.

The matching condition is necessary, not generally sufficient for the existence of the proposed IR theory. Dynamics, unitarity, operator algebras, and other anomalies can still fail. SYM-3 records ANOMALYCOMPATIBLE separately from DYNAMICALLYESTABLISHED.

13.4 Spontaneously broken phases

If G is spontaneously broken to H , the IR contains Goldstone modes on a coset or related order-parameter space. A gauged Wess–Zumino–Witten term can reproduce the UV anomaly when its level, background map and counterterm convention are fixed. The operative open construction is Harvey, Hill, and Hill [6]; Witten [15] remains historical attribution. The matching record includes the symmetry-breaking pattern, topology of the target space, quantized coefficient, unbroken subgroup anomaly, and coupling to backgrounds.

The existence of a Wess–Zumino term does not mean the anomaly disappears. It is the mechanism by which the low-energy fields realize the same anomaly. A different vacuum or breaking pattern is a different IR target.

13.5 Topological matching

A gapped phase with topological order can match a 't Hooft anomaly through anyon/defect symmetry fractionalization, emergent gauge fields, or a TQFT response. Such a phase is not trivially gapped. The operator category and emergent lines/surfaces must be admitted in the target. A point-local ledger is insufficient.

Some anomalies impose stronger constraints that forbid a symmetry-preserving gapped phase under specified assumptions [46]. The project does not generalize those conclusions beyond their theorem scope. Each benchmark records whether topological matching is allowed, required, forbidden, or unresolved.

13.6 Symmetry extension

A group extension

$$1 \longrightarrow K \longrightarrow \tilde{G} \longrightarrow G \longrightarrow 1$$

can pull back an anomaly to a trivial class on $\tilde{G}$. An IR theory with emergent K gauge structure can then realize the original G anomaly. The construction changes the physical content and operator category. It is not a counterterm cancellation in the source theory.

The relation lattice distinguishes pullback trivialization, emergent gauge realization, and complete physical equivalence. A symmetry extension may solve the anomaly-matching problem while producing multiple possible IR phases.

13.7 Boundary and interface matching

An anomalous d -dimensional theory can appear at a boundary or interface of a $(d+1)$ -dimensional phase. Matching requires the boundary anomaly equal the inflow of the bulk. Different boundary conditions can realize the same bulk anomaly through gapless, broken, or topologically ordered states. The bulk therefore constrains but does not uniquely select the boundary.

13.8 Mass deformations and threshold transport

Integrating out massive chiral fermions can induce Chern–Simons, theta, Wess–Zumino, or invertible topological terms. A low-energy anomaly calculation that simply deletes heavy fields can fail. The transport record includes threshold terms and the sign/phase of masses. Across a mass sign change, the invertible phase can jump and a boundary mode may appear.

For EFT matching, anomaly coefficients associated with exact symmetries are transported along with Wilson coefficients and local terms. If a higher-dimension operator explicitly violates the symmetry, the target and matching obligation change. The operator dimension and matching scale are mandatory fields.

13.9 Vacuum, global form, and operator-category synchronization

Matching is sensitive to:

- whether the symmetry is unbroken, broken, or emergent;
- faithful global group and quotient;
- admitted line and defect operators;
- spin or generalized tangential structure;
- boundary and topology;
- whether background fields are ordinary, higher-form, or 2-group coupled.

A UV local-field anomaly cannot be compared to an IR defect-completed anomaly without an inclusion or completion map.

13.10 Matching alternatives table

IR route	Anomaly carrier	Additional certification
Massless fields	Chiral representations and currents	Spectrum, interactions, global form, all anomaly classes
Spontaneous breaking	Goldstone fields and term	Breaking pattern, quantization, unbroken subgroup
TQFT/topological order	Lines, surfaces, anyons, topological action	Defect category, symmetry action, fractionalization
Symmetry extension	Emergent gauge sector and extended group	Exact extension, pullback trivialization, quotient
Boundary realization	Bulk inflow plus boundary state	Bulk target, orientation, gluing, boundary conditions
Non-invertible nant	rem-Topological defect/fusion category	Defect construction, fusion, action, target limit

13.11 No unique IR theorem

Corollary – No universal IR uniqueness. An anomaly record determines a constraint class on admissible infrared realizations. It does not, without additional premises, determine a unique massless spectrum, symmetry-breaking pattern, TQFT, boundary condition, or non-invertible completion.

This theorem is important for the AASC arc. Structural necessity must not be laundered into unsupported dynamical uniqueness. SYM-3 can prove that an IR proposal is anomaly-incompatible or anomaly-compatible. A stronger uniqueness claim belongs to a separate theorem with explicit dynamics and rival exhaustion.

13.12 Matching falsifiers

A proposed match is rejected when:

- UV and IR target IDs or symmetry maps are absent;
- one compares different global forms or tangential structures;
- heavy-field induced terms are omitted;
- local polynomials match but a global anomaly does not;
- the IR phase has no object capable of carrying the anomaly;
- a broken symmetry is treated as unbroken without a WZW/background analysis;
- a TQFT is invoked without its symmetry/defect data;
- matching is presented as proof of unique dynamics.

Anomaly Inflow, Bulk–Boundary Systems, and Repair Discipline

14.1 Inflow is a relation, not a slogan

Anomaly inflow describes a combined bulk–boundary system in which the variation of a bulk topological or invertible action cancels the anomalous variation of the boundary effective action. The invariant object is the combined partition function. The boundary theory alone remains anomalous as a stand-alone target. Therefore the phrase “inflow cancels the anomaly” is complete only when the bulk, boundary, orientation, background extension, quantization, and cancellation map are part of one declared target.

Let $M_d = \partial Y_{d+1}$ with compatible orientation. Schematically,

$$\delta W_\partial[A] = -\delta S_{\text{bulk}}[A], \quad Z_{\text{combined}}[A] = Z_\partial[A] e^{iS_{\text{bulk}}[A]}$$

is invariant. The local descent construction realizes this with a Chern–Simons-type bulk term; global inflow can be expressed by eta invariants or an invertible anomaly theory. The operative open authority for determinant-line/eta inflow and gluing is Yonekura [10], Freed and Hopkins [11], and Freed [36, 53]; Callan and Harvey [13] and Dai and Freed [37] are historical attribution.

SYM3-T14 – Inflow integrated-target theorem. An anomaly is inflow-resolved only for a target that already contains the boundary theory, bulk theory, common symmetry/background data, extension or gluing conditions, quantized inflow functional, and cancellation relation. The selected Weyl fiber imports the quantization theorem and verifies cancellation in the declared two-coordinate cubic/mixed gauge-gravitational lattice. Adjoining the bulk after classification is a target update, not same-act repair.

14.2 Executed charge-one Weyl boundary/5d bulk fiber

The closed inflow fiber fixes one left-handed four-dimensional Weyl fermion of background $U(1)$ charge one on $X = \partial Y$, an oriented five-dimensional bulk Y , the restriction gluing map, and the two-coordinate anomaly basis

$$(U(1)^3, U(1)\text{--gravity}^2).$$

The boundary class is evaluated in scaled integral coordinates with lattice scales (6, 24). The declared finite bulk candidates are

$$(-1, 1), \quad (0, 0), \quad (1, -1), \quad (-1, 0), \quad (0, 1).$$

For each candidate the finite relation engine evaluates the complete boundary coordinate pair, the declared lattice-quantization test, orientation, restriction gluing, and vanishing of the relative class. Exactly the target-compatible opposite class cancels the selected boundary class; zero, same-sign, and one-coordinate candidates fail the class or completeness checks. The continuum theorem that licenses the determinant-line and invertible-field-theory interpretation is imported from the cited literature. The finite engine certifies this selected coordinate model; it does not derive a general spin/bordism quantization theorem from the candidate list.

14.3 Callan–Harvey prototype

The defect-zero-mode/current-inflow prototype is used through the open determinant-line and gluing formulation of Yonekura [10] and Freed [53]; Callan and Harvey [13] is retained as historical attribution. The physical record contains:

- a spatially varying mass or defect background;
- localized chiral modes;
- bulk massive fermions and induced topological term;
- boundary/defect current nonconservation;
- bulk current and total conservation;
- orientation and normalization.

The localized theory is not rendered anomaly-free in isolation. The combined system is consistent.

14.4 Descent inflow

For a local anomaly polynomial $I_{d+2} = dI_{d+1}^{(0)}$, a bulk action built from $I_{d+1}^{(0)}$ varies by a boundary term $I_d^{(1)}$. The descent relation explains local inflow, but the global definition of the Chern–Simons functional requires quantization and extension independence. A local formula that cancels an infinitesimal variation can fail globally if the coefficient or global form is wrong. The inflow proof record therefore contains both local descent and global quantization fields.

14.5 Global inflow and anomaly field theory

In the determinant-line/invertible-field-theory picture, the anomaly of the d -dimensional theory is a $(d+1)$ -dimensional invertible theory. The boundary partition function is a vector or section in the state space/line assigned by the anomaly theory, and the combined object can be canonically defined. Eta invariants give a fermionic realization under the Dai–Freed theorem, with the operative open derivation in Yonekura [10]; Dai and Freed [37] is historical attribution.

This framework clarifies that inflow need not be a dynamical bulk in the phenomenological sense. It can be a bookkeeping of the anomalous theory’s relative nature. Nevertheless, whether the bulk is a physical sector, external anomaly theory, UV completion, or mathematical extension is a target field. SYM-3 does not erase this distinction.

14.6 Boundary conditions, charges, and flux

A local current identity does not automatically give a conserved integrated charge. Boundary flux, edge modes, and nonintegrable surface terms can modify the global statement. SYM-2 established this for Noether charges; SYM-3 applies the same product-status discipline to anomalous currents and inflow.

The boundary record contains

$$BStat = (B_{\text{bulk}}, B_{\partial}, Q_{\partial}, \Phi_{\partial}, E_{\partial}, I_{\text{inflow}}).$$

Bulk gauge redundancy, boundary physical action, surface charge, flux, edge data, and inflow may coexist. A surface current or charge is not itself the anomaly-canceling action.

14.7 Repair relation lattice

Every proposed repair receives one of the following relations.

1. **Presentation-equivalent shift:** a local counterterm, basis, or convention changes the representative while preserving the invariant class.
2. **Same-domain aggregate neutralization:** contributions already contained in the target sum to a trivial anomaly class.
3. **Integrated inflow:** the frozen target already contains the bulk–boundary system and the combined anomaly vanishes.
4. **Added-sector neutralization:** new fields or topological sectors are introduced; this is a target update relative to the source theory.
5. **Scope enlargement:** a boundary theory becomes part of a higher-dimensional system or UV completion.
6. **Hidden selector/illicit repair:** a regulator, branch, counterterm, or bulk is inserted after the result to force a preferred outcome without changing the target ID.

Corollary – Inflow repair classification. Relative to the frozen target and allowed transformation class, every proposed anomaly repair is classified as representative shift, same-domain neutralization, integrated inflow, added-sector neutralization, scope enlargement, or illicit hidden repair. No repair mechanism is admitted without one relation-lattice disposition.

14.8 Counterterms versus inflow

A boundary local counterterm changes the boundary representative. A bulk Chern–Simons or eta theory changes the combined target and can cancel a nontrivial relative anomaly. The two can interact: different boundary counterterms correspond to different trivializations or polarizations of the anomaly line. They are not interchangeable. The source classification table and counterterm lattice identify which part is local, global, quantized, and target-changing.

14.9 Green–Schwarz and Wess–Zumino-type mechanisms

In some theories an axion-like or higher-form field transforms so that its variation cancels a gauge anomaly. The mechanism can be consistent when the field and couplings are part of the target and all

factorization/quantization conditions hold. SYM-3 classifies it as same-domain aggregate/mediated neutralization for that enlarged target. Adding the field to a previously anomalous target is a target update. The structural analysis does not claim the Standard Model uses such a mechanism for its ordinary gauge anomalies.

14.10 Inflow and 't Hooft anomaly

A 't Hooft anomaly can be represented as the boundary of an invertible bulk theory. This does not make the d -dimensional global symmetry gaugeable on its own. It states that the anomalous theory is a relative boundary theory or that a higher-dimensional bulk supplies inflow. The matching constraint persists under RG as the same anomaly theory.

14.11 Failure routes

The inflow chapter fails if it:

- calls the boundary theory anomaly-free without the bulk;
- omits coefficient quantization or global extension independence;
- treats added bulk fields as same-target repair;
- confuses boundary charge with anomaly functional;
- assumes every local descent term defines a global inflow action;
- uses an unphysical mathematical extension without declaring its target status;
- claims inflow uniquely determines the boundary dynamics.

Inflow export

Every inflow record exports two linked but nonidentical records: the stand-alone boundary anomaly status and the combined bulk–boundary consistency status. Downstream papers may not cite the second as if it were the first.

Generalized, Higher-Group, and Non-Invertible Frontier Anomaly Sectors

15.1 Operator-category dependence

Generalized symmetries act on extended operators and defects rather than only on point-local fields. A p -form symmetry in d dimensions has charged p -dimensional operators and topological codimension- $(p + 1)$ symmetry operators. Coupling to a background $(p + 1)$ -form gauge field can expose anomalies invisible to a point-particle ledger [52].

The target must therefore declare the operator category, faithful global group, screening matter, admitted monopoles, line/surface spectrum, defect fusion data, and parameter limit. A candidate one-form symmetry may disappear when charged matter screens its lines or when monopoles are admitted. A non-invertible defect construction may exist only in a massless or background-probe target.

SYM3-T17 – Generalized-operator anomaly theorem. A generalized anomaly is defined only relative to a fixed extended-operator/background category. No zero-form point-local record is promoted to generalized closure without line, surface, or defect data, faithful global form, screening/monopole assumptions, and a source-supported anomaly or matching relation.

Claim status: Target-indexed conditional theorem. Full Standard Model generalized-sector closure is owned by SYM-9.

15.2 Three explicit frontier fibers

SYM-3 retains three physically meaningful but non-closed fibers:

1. **Faithful $\mathbb{Z}_6$ global-form/line completion.** The local Lie algebra and point-particle representations are insufficient to fix genuine lines, monopoles, theta periodicities, or one-form symmetry.
2. **Higher-flavor two-group background.** Zero-form flavor backgrounds and magnetic hypercharge one-form data require fixed global form, Postnikov data, background transformations, and monopole assumptions [54].
3. **Massless non-invertible chiral defect target.** The topological defect, fusion law, charged local/line operators, massless limit, and absence or treatment of dynamical monopoles must be specified [55].

These fibers are interface-complete: they have target IDs, source-supported propositions, limitations,

no-promotion rules, and explicit SYM-9 successor interfaces. They are reported as **OPEN-FRONTIER**, remain outside the closure quantifier, and do not satisfy closed-fiber role obligations.

15.3 Higher-form and global-form sensitivity

Electric and magnetic one-form symmetries depend on the faithful gauge group, matter screening, and monopole sectors. Different quotients of a common local gauge product can share local perturbative content while differing in genuine Wilson/'t Hooft lines and generalized anomalies [41, 42]. A mixed anomaly between a discrete zero-form symmetry and a one-form center symmetry can constrain theta dependence and infrared phases only after these data are fixed.

15.4 Higher-group structure

Zero-form and one-form symmetries can form a 2-group whose background transformations mix through an action and Postnikov class [47, 48].

The structural extension class, physical defect actions, and anomaly/matching record are distinct atomic objects. A pair of independent zero-form and one-form tables cannot reconstruct the higher-group law after the fact.

15.5 Non-invertible defect noncollapse

A non-invertible symmetry is represented by topological defects with non-group fusion. It is not a generic synonym for an anomalous current. A classical continuous action, its anomalous quantum current, a surviving discrete subgroup, and a conditional non-invertible defect remnant occupy different targets and loci.

SYM3-T18 – Non-invertible defect noncollapse. A non-invertible symmetry record must specify the defect, fusion law, operator action, target limit, global-form and monopole assumptions, and anomaly or matching relation. It is neither an ordinary zero-form group action nor a mere representative of the anomalous current from which the construction may descend.

Claim status: Conditional target-indexed theorem; no role-completeness burden before SYM-9 closes the required global and operator data.

15.6 No-promotion and handoff rule

A frontier record may support a conditional physics discussion, source-supported example, or downstream construction request. It may not occupy the global-completion, generalized, or synchronization burden of a closed fiber. SYM-9 may later create a new closed fiber or successor record. That action creates an explicit successor target and transport relation; it is not a silent retroactive correction.

Falsifier: A generalized anomaly claim fails if it omits charged extended operators, background degree, faithful global form, screening or monopole assumptions, fusion data, parameter limit, or operator category. A frontier record promoted into a closed theorem without those data reopens the target and occupation results.

Trace Anomaly, Scale Symmetry, and RG Transport Handoff

16.1 Trace anomaly is not a gauge anomaly

Scale and conformal anomalies concern the response of the renormalized theory to dilatations or Weyl transformations. They do not generally signal inconsistency of a gauge quotient. The object can be a quantum nonconservation law or local operator identity for the stress tensor. It is therefore classified as a diagnostic/obstruction to exact scale symmetry, with a separate physical scale-action record.

For a renormalized theory, the trace identity has the schematic form

$$T^\mu_\mu = \sum_i \beta_i(g) [\mathcal{O}_i] + \sum_a (1 + \gamma_{m_a}) m_a [\bar{\psi}_a \psi_a] + \text{improvement, curvature, and contact terms,}$$

with exact operator normalization and scheme conventions specified. The operative open trace/operator package is Hatta, Rajan, and Tanaka [17], Ahmed, Chen, and Czakon [18], Panagopoulos, Panagopoulos, and Spanoudes [19], and Costa et al. [20]; Callan, Coleman, and Jackiw [16] and Collins, Duncan, and Joglekar [56] are historical attribution. A classical massless action can be scale invariant while the quantum theory has running couplings and dimensional transmutation.

16.2 Scale action versus RG transport

A dilatation acting on spacetime and fields is a candidate physical action. RG evolution is a transport on theory/effective-description space. A change of renormalization scale with compensating running can be representative redescription; beta functions, scaling violations, anomalous dimensions, and dimensional transmutation carry physical content. SYM-0 already calibrates these distinctions; SYM-10 will complete the Standard Model scale/RG analysis [57].

Trace/RG handoff rule

Classical scale action, quantum trace anomaly, renormalization-scale coordinate choice, scheme reparameterization, physical RG flow, fixed-point scale symmetry, and EFT matching are distinct records. The trace anomaly is not typed as gauge inconsistency.

16.3 Executed massless $N_f = 3$ QCD trace fiber

The closed trace fiber fixes massless $SU(3)_c$ QCD with $N_c = N_f = 3$, dimensional regularization, the $\overline{\text{MS}}$ scheme, a two-loop beta-function coordinate packet, and the declared renormalized operator

basis $\{[F^2], \text{EOM}, \partial J\}$ over an interval with no threshold crossing. Its leading coefficient is

$$\beta_0 = \frac{11}{3}N_c - \frac{2}{3}N_f = 11 - 2 = 9.$$

The number 9 is one coordinate of the fixed perturbative target, not the full trace identity. The imported literature supplies the renormalized quark/gluon energy-momentum-tensor and operator-mixing framework at the stated perturbative scope [17–20]. The finite target record tests the basis subsets $\{F^2\}$, $\{F^2, \text{EOM}\}$, and $\{F^2, \text{EOM}, \partial J\}$ together with the finite scheme parameters $0, \pm 1/5$. Only the complete quotient-aware basis and invertible finite scheme transports that preserve the selected trace-class coordinate survive compatibility. The engine therefore certifies the declared finite operator-coordinate and scheme-transport record; it does not independently rederive the full continuum composite-operator renormalization theorem.

SYM3-T22 – QCD axial/remnant and trace-target alignment. The massless three-flavor QCD axial fiber and the massless three-flavor $\overline{\text{MS}}$ trace fiber retain distinct targets, currents, anomaly groups, topological data, vacuum status, operator bases, and scale relations. The axial fiber fixes the ABJ class and $\mathbb{Z}_6$ remnant. The trace fiber imports the renormalized operator theorem at the stated perturbative scope and executes the declared β_0, β_1 , basis-completeness, quotient, and finite scheme-transport checks; it does not identify the number 9 with the complete trace identity. Neither fiber is inferred from the other, and Wess–Zumino–Witten matching belongs to its separate non-Abelian chiral target.

Claim status: Target-indexed theorem combining imported QFT results with finite target-level calculations.

16.4 Scheme dependence and invariant content

Beta-function coefficients and operator bases can change under coupling redefinitions or scheme transformations, while physical predictions and fixed-point critical data transform covariantly. The anomaly record stores both representative coordinates and scheme-independent consequences. A term proportional to equations of motion or an improvement can alter the local trace representative without changing physical amplitudes.

A claim that “the scale anomaly is scheme dependent” is incomplete. One must specify which coefficient, operator identity, or observable is changed. Conversely, no scheme choice can make physical running disappear when invariant scale dependence is present.

16.5 Weyl anomaly on curved backgrounds

In a curved background, even a conformal field theory can have a Weyl anomaly involving curvature invariants. The coefficients and total-derivative terms have different scheme status. The target must distinguish flat-space trace anomaly, curved-background Weyl anomaly, boundary Weyl anomaly, and dynamical gravity. SYM-3 records the anomaly grammar; curved/dynamical spacetime remains outside the core Standard Model target unless explicitly admitted.

16.6 Anomaly transport under RG and EFT matching

Exact ’t Hooft anomaly classes are invariant under RG when the symmetry and background persist. Local trace-anomaly coordinates evolve with couplings and operator mixing. Across thresholds, heavy-field effects are encoded in matching terms. The downstream anomaly record distinguishes:

- RG-invariant global anomaly class;
- running local coordinate;
- scheme convention;
- threshold contribution;
- fixed-point status;
- explicit symmetry-breaking operator.

16.7 Fixed points and emergent symmetry

At a fixed point with vanishing beta functions, scale or conformal symmetry may be exact under stated assumptions. The fixed-point action is a physical symmetry record; the RG trajectory approaching it is transport. An emergent IR symmetry can carry a 't Hooft anomaly inherited from the UV or derived relative to an enlarged emergent structure. Matching requires an explicit symmetry map.

16.8 SYM-10 export

SYM-3 exports to SYM-10:

1. trace-anomaly type and source propositions;
2. operator identity and scheme fields;
3. 't Hooft anomaly records that must persist under RG;
4. threshold/matching relation fields;
5. exact distinction between scale action and theory-space transport;
6. open global-form or defect dependencies.

SYM-10 must add the complete scale interval, beta-function status, fixed-point assumptions, and physical observables. It may not reinterpret RG covariance as an ordinary internal symmetry.

16.9 Failure routes

The chapter fails if it calls every running coupling an anomaly, treats a scheme coordinate as physical action, infers exact conformal symmetry from a classically massless Lagrangian, or uses trace anomaly to claim gauge inconsistency. It also fails if a curved-background Weyl anomaly is imported into the flat target without scope qualification.

Part III

Standard Model Anomaly Analysis

Exact Standard Model Perturbative Gauge-Anomaly Ledger

17.1 Target and convention

The closed local fiber uses one generation of left-handed Weyl fermions in the minimal renormalizable Standard Model. Conventional right-handed particle fields are represented as left-handed charge-conjugate fields:

$$\begin{aligned} Q : (\mathbf{3}, \mathbf{2})_{1/6}, \quad u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-2/3}, \quad d^c : (\bar{\mathbf{3}}, \mathbf{1})_{1/3}, \\ L : (\mathbf{1}, \mathbf{2})_{-1/2}, \quad e^c : (\mathbf{1}, \mathbf{1})_1. \end{aligned}$$

No sterile-neutrino field appears in this minimal fiber. A right-handed-neutrino extension is a different target record. Color and weak multiplicities are included in Abelian sums, and hypercharge normalization is fixed by $Q_{\text{em}} = T_3 + Y$.

17.2 Exact computed coordinates

The canonical ledger stores multiplicity, $SU(3)$ and $SU(2)$ representation data, hypercharge, $B - L$, dimensions, and indices. The primary implementation computes exact fractions for

$$[SU(3)]^3, \quad [SU(3)]^2 U(1)_Y, \quad [SU(2)]^2 U(1)_Y, \quad [U(1)_Y]^3, \quad \text{grav}^2 U(1)_Y,$$

and the mod-two number of weak doublets. A second implementation independently reconstructs the unmutated results.

For one generation,

$$A_{SU(3)^3} = 0, \quad A_{SU(3)^2 Y} = 0, \quad A_{SU(2)^2 Y} = 0, \quad A_{Y^3} = 0, \quad A_{\text{grav}^2 Y} = 0,$$

and the weak-doublet count is four, hence even. Individual multiplets need not have vanishing contributions; quantum consistency is an aggregate property of the complete chiral ledger.

SYM3-T19 – Standard Model aggregate gauge-neutrality theorem. In the minimal renormalizable local Standard Model fiber, the complete one-generation left-handed representation and hypercharge ledger cancels the perturbative $[SU(3)]^3$, $[SU(3)]^2 U(1)$, $[SU(2)]^2 U(1)$, $[U(1)]^3$, and mixed gravitational- $U(1)$ gauge obstructions by exact rational computation. The statement is aggregate, convention-controlled, and local; it neither fixes the faithful global group nor excludes every global anomaly.

Claim status: Exact-rational computed theorem with independent implementation and source-supported convention.

17.3 Negative controls

Three destructive controls verify sensitivity rather than stored zero values.

1. Changing $Y(e^c)$ from 1 to $4/5$ produces $A_{Y^3} = -61/125$ and $A_{\text{grav}^2 Y} = -1/5$.
2. Removing the lepton doublet produces nonzero mixed and Abelian coefficients and an odd weak-doublet count.
3. Adding one color-singlet weak doublet with $Y = 0$ leaves local Abelian coefficients unchanged but flips the global weak parity.

These controlled variations are derived from the primitive field ledger and are part of the scientific completion criterion.

17.4 Minimal and right-handed-neutrino $B - L$

In the same left-handed convention, minimal $B - L$ has cubic and mixed-gravitational coefficients -1 per generation. Adding one charge-conjugated right-handed neutrino per generation cancels both. This is an explicit target bifurcation, not a repair of the minimal fiber.

17.5 What anomaly cancellation does not determine

Gauge anomaly cancellation is necessary for the quantum gauge architecture, but it does not determine gauge couplings, Yukawa eigenvalues, masses, mixing angles, generation number, unification scale, or a unique faithful global group. Vectorlike extensions can preserve local cancellation while changing thresholds, global form, or flavor backgrounds. Every such extension receives a target update.

Local Standard Model falsifier

A wrong representation, chirality convention, multiplicity, hypercharge, index normalization, omitted coefficient, arithmetic discrepancy between independent implementations, or undeclared additional chiral matter reopens the result. A consistently transported normalization convention is not a counterexample; a changed global form is a different target coordinate.

Standard Model Global Anomalies and Faithful Global Form

18.1 The mod-two weak anomaly

Although the perturbative $[SU(2)]^3$ coefficient vanishes, a four-dimensional $SU(2)$ gauge theory with an odd number of left-handed doublets has Witten's global anomaly [8]. One Standard Model generation contains three color copies of the quark doublet and one lepton doublet, for a total of four. The mod-two obstruction therefore cancels generation by generation:

$$N_{\text{doublets}} = 3 + 1 = 4 \equiv 0 \pmod{2}.$$

This parity test is independent of the local pseudoreality argument. It uses the global topology of the gauge transformation and the representation count.

SYM3-T20 – Standard Model global $SU(2)$ parity theorem. For the conventional one-generation Standard Model weak representation, the number of left-handed $SU(2)$ doublets is even, so the Witten mod-two obstruction is absent. The conclusion is target-relative and does not replace the complete global-form/bordism analysis.

18.2 The family of faithful gauge groups

The local Lie algebra and perturbative matter representation are compatible with faithful global groups

$$G_\Gamma = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\Gamma}, \quad \Gamma \subseteq \mathbb{Z}_6,$$

with commonly considered $\Gamma \cong 1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$. Different choices admit different bundles, line operators, monopoles and theta periodicities. A local anomaly calculation does not choose Γ .

The global-anomaly record is therefore indexed by Γ . Modern bordism computations for these groups show a subtle pattern: some forms have no global anomalies beyond the familiar Witten condition, while in others even that condition is absorbed by the global group structure [12]. The exact imported result is attached to the specified spin structure and matter assumptions.

18.3 Line operators and global completion

The line-operator spectrum is not a decorative addition. It determines which electric and magnetic charges are genuine, which one-form symmetries survive screening, and which background fields can

be introduced. The detailed classification is deferred to SYM-9, but SYM-3 records every anomaly statement that depends on it. An anomaly record with `global_form_dependence=unresolved` cannot be reported as globally complete.

The same discipline applies to theta periodicity and CP at special theta. A global-form change can alter the periodic identification and the action of discrete transformations even when the local topological density is unchanged. SYM-4 and SYM-9 must import the target-indexed record rather than one universal theta statement.

18.4 Bordism and embedding arguments

Embedding the Standard Model representation in an anomaly-free unified group can provide a useful consistency argument for a specific matter content, but it is not a substitute for a target-relative classification of arbitrary extensions. Bordism calculations make the tangential and global assumptions explicit. The proof record records whether the conclusion is representation-specific, group-family-wide or conditional on an embedding.

Global Standard Model record

The record contains: faithful group G_Γ ; spin/tangential structure; admitted bundles and large transformations; matter representation; local anomaly status; Witten parity; bordism/global anomaly result; line/operator category; monopole assumptions; theta/global-form sensitivity; and the exact SYM-9 handoff.

18.5 Exact boundary

SYM-3 establishes that global-form dependence is a mandatory field and imports the registered global results. It does not claim to enumerate every line operator, monopole sector or generalized symmetry for every Γ . That closure belongs to SYM-9. A new global anomaly found at the same fixed G_Γ , tangential structure and matter content reopens the SYM-3 record and every downstream import.

Electroweak $B + L$, Topology, and Operator-Order Ancestry

19.1 Classical and perturbative status

Baryon and lepton number are accidental global symmetries of the renormalizable minimal Standard Model Lagrangian in the appropriate classical and perturbative sense. They are not imposed as independent gauge principles. Their currents act on physical charged states and organize selection rules. The electroweak anomaly changes this quantum status without turning the currents into gauge redundancy.

For N_g generations, a unit electroweak topological transition changes baryon and lepton numbers by

$$\Delta B = N_g n_{\text{top}}, \quad \Delta L = N_g n_{\text{top}}, \quad \Delta(B + L) = 2N_g n_{\text{top}},$$

with sign and normalization set by the topological charge convention [31, 58]. The combination $B - L$ has different status and depends on the neutrino and gravitational/global target.

SYM3-T21 – Electroweak $B + L$ ancestry theorem. The renormalizable perturbative $B + L$ action, anomalous current relation, instanton/sphaleron selection rule, higher-dimension operator violations and neutrino-target changes are distinct atomic records with distinct causal ancestry. No record may be summarized as an unqualified exact conservation or generic breaking statement.

19.2 Perturbative versus nonperturbative

The local divergence equation is visible in perturbation theory, but actual charge violation between topological sectors is nonperturbative. At zero temperature instanton transitions are exponentially suppressed; at high temperature sphaleron transitions can become important. These dynamical rates are not determined by the structural anomaly classification. SYM-3 records the selection rule and target; numerical cosmological rates are outside its scope.

19.3 Neutrino and EFT variants

The minimal renormalizable Standard Model has no right-handed neutrino and no neutrino mass term. A dimension-five Weinberg operator violates lepton number by two units and changes flavor and discrete-remnant records. Dirac neutrino targets with right-handed fields have different anomaly

ledgers. Baryon-violating dimension-six operators introduce another ancestry. These operator-order effects must not be attributed to the electroweak anomaly.

The downstream SYM-7 record carries the lowest violating operator dimension, perturbative/non-perturbative status, neutrino target, anomaly class and preserved discrete remnants. A target update does not retroactively change the minimal-target theorem.

19.4 AASC role map

The perturbative accidental action is an effective physical action. The current and selection rule are derived witnesses. The electroweak topological anomaly is an obstruction/diagnostic for the continuous charge. The gauge and matter architecture explains why the renormalizable operator set has its form but does not promote B or L to primitive carrier status. Higher-dimension violations and neutrino variants are effective-route or target-update data.

Baryon/lepton falsifier

A row fails if it omits the quantum sector, operator dimension, neutrino target, topology or anomaly status. A proposed exact discrete remnant must be checked against both continuous anomaly data and the admitted higher-dimension operators. A sentence stating only “baryon number is conserved” or “lepton number is violated” is not complete at the declared target.

QCD Axial, Chiral, Vacuum, Wess–Zumino–Witten, and Defect Records

20.1 Five records that must not collapse

For N_f massless quark flavors, the classical kinetic and gauge terms admit chiral flavor actions. The following are distinct:

1. the classical continuous $U(1)_A$ action on massless quark fields;
2. the renormalized singlet axial current and its ABJ Ward identity;
3. the absence of an exact continuous quantum $U(1)_A$ conservation law;
4. surviving discrete or topology-sensitive remnants and their vacuum realization;
5. any separately constructed non-invertible defect action in a fixed extended-operator target.

Quark masses are explicit deformations. Vacuum selection is a realization status. The anomaly is quantum causal ancestry. A single phrase such as “chiral symmetry breaking” cannot substitute for these records.

Axial QCD target boundary

The closed $U(1)_A$ fiber contains the classical axial action, renormalized current, ABJ/Jacobian/index coordinates, integer instanton sectors, theta-vacuum realization, and the exact $\mathbb{Z}_6$ remnant for $N_f = 3$. The non-Abelian $SU(3)_L \times SU(3)_R$ Wess–Zumino–Witten match is proved in the separate matching fiber. A non-invertible defect completion remains an explicit SYM-9 frontier target.

20.2 Anomalous current and topology

The singlet axial current obeys a source-supported anomalous Ward identity whose normalization depends on representation and convention. The global transformation remains a meaningful classical physical action, but the quantum current is anomalous. Topological sectors relate axial rotations to theta and quark-mass phases. The strong-CP record must therefore synchronize the axial rotation, mass matrix, theta periodicity, and faithful global target rather than treating one formula as the entire anomaly object.

20.3 Vacuum and the $U(1)_A$ problem

The anomaly alters naive Goldstone counting and contributes to the special status of the η' . SYM-3 does not derive hadron masses. It records the distinction between classical action, anomalous current, nonperturbative topology, and vacuum realization, with exact limitations on each imported proposition.

20.4 Separate non-Abelian chiral matching target

The low-energy chiral theory contains a quantized Wess–Zumino–Witten term whose coefficient matches the microscopic $SU(3)_L \times SU(3)_R$ anomaly [15]. That calculation is executed in the separate fiber of Chapter 13; it is not a constituent of the closed $U(1)_A$ fiber. The separation prevents singlet-current, non-Abelian flavor, vacuum, and defect data from being conflated.

20.5 Conditional non-invertible defect record

In selected massless QED/QCD targets, topological defects can combine chiral transformations with topological field-theory data and obey non-group fusion [55]. This construction requires extended operators, fixed global form, massless limits, and monopole assumptions. It remains in the SYM-9 frontier fiber and cannot be used to claim an exact continuous quantum $U(1)_A$ symmetry or to discharge the closed QCD fiber’s role burdens.

20.6 Downstream transport

SYM-6 imports the classical massless action, ABJ class, vacuum status, WZW matching record, and mass-deformation ancestry. SYM-4 imports the theta/CP-facing specialization. SYM-9 owns the global-form and non-invertible defect completion. Changing light-flavor content, masses, theta target, line spectrum, or monopole category produces a explicit successor target.

Flavor Backgrounds, Rephasing, and Higher-Group Anomaly Interfaces

21.1 Flavor symmetry versus basis covariance

The zero-Yukawa limit of the minimal Standard Model has a large flavor symmetry acting on fermion species. Weak-basis transformations and rephasings in the full Yukawa theory are often presentation changes rather than physical symmetries. An anomaly calculation must therefore specify whether the transformation is an actual global action of a parameter-limit target or a basis covariance with spurions.

A trace of flavor generators is invariant under a witnessed basis change, but a background-field assignment can change if the spurion or global-form target changes. SYM-5 imports the anomaly record only after its basis quotient is fixed. Complex matrix entries are not physical CP or anomaly data by themselves.

21.2 Triangle data and higher flavor

When flavor backgrounds are coupled together with hypercharge, mixed triangle data can intertwine zero-form flavor transformations with magnetic hypercharge one-form structure. The resulting higher-group relation is described in the Standard Model analysis of Cordova and Koren [54]. The effect depends on the admitted magnetic background, global form, monopoles and Yukawa limit.

The extension class is structural data. The zero-form transformation, one-form action, mixed anomaly and background transformation law are separate atomic records. A spurion organization can display the breaking pattern without making numerical Yukawa entries primitive anomaly data.

21.3 Neutrino bifurcation

The flavor group, lepton number, PMNS rephasing freedom and anomaly conditions depend on whether neutrinos are massless, Dirac or Majorana. Majorana masses restrict rephasings and can break continuous lepton number to a discrete subgroup. Right-handed neutrinos add chiral fields and alter gauged $B - L$ conditions. The record uses target IDs rather than a nullable “neutrino included” flag.

21.4 Mixed anomaly allocation

A mixed flavor-hypercharge anomaly can be represented in different current bases, but its joint class and higher-group consequence are invariant under lawful counterterm transport. If hypercharge is dynamical, its gauge variation must cancel. The remaining background variation constrains the flavor symmetry. The target record explicitly records the Bardeen allocation used.

Flavor anomaly handoff

SYM-5 receives: exact flavor-limit target; basis quotient; acting background group; hypercharge/global-form branch; mixed coefficient and counterterm allocation; higher-group extension class; anomaly/matching status; neutrino variant; and source-supported proposition. A basis change cannot alter this invariant record without a target or source mismatch.

Discrete, CP/Theta, Trace-Anomaly, and RG Handoffs

22.1 Discrete anomalies

Discrete symmetries can have anomalies not captured by treating the group as an infinitesimal continuous symmetry. The target must specify the finite group, spin/tangential structure, background bundle and global classification method. Group-cohomology or bordism coordinates can apply in particular dimensions and symmetry types [44]; they are not universal substitutes for a target-specific computation.

C, P, T and CP are sector- and parameter-dependent physical transformations, while CPT is a theorem consequence under its premise record. A discrete anomaly record must preserve the active-action and theorem distinctions established in SYM-2. SYM-4 imports the record together with theta/global-form data.

22.2 Theta and CP

An anomalous axial rotation can shift the theta parameter and quark-mass phases. CP at special theta values depends on the global group, topological sectors and vacuum realization. The local density $F\tilde{F}$ is not the complete theta target. A change in periodicity or line spectrum is a global-form update.

The repair relation lattice prevents a field redefinition from being called physical elimination of theta unless all invariant combinations and global data are transported. The strong-CP problem and possible BSM resolutions are outside core SYM-3; the record supplies the anomaly and target relations needed by SYM-4.

22.3 Trace anomaly

Scale and conformal anomalies differ from internal gauge anomalies. The trace of the renormalized energy-momentum tensor contains beta-function and mass/operator terms, schematically

$$T^\mu_\mu = \sum_i \beta_i(g) \mathcal{O}_i + \sum_a (1 + \gamma_a) m_a \mathcal{O}_a + \text{improvement/contact terms},$$

with operator mixing and scheme conventions fixed [56, 59]. The trace identity is a diagnostic of quantum scale breaking. RG flow is theory-space transport, not automatically a global symmetry action on a fixed Hilbert space.

SYM-3 records the anomaly grammar and transport relation. SYM-10 completes the renormalized operator identity, scheme-independent content, fixed-point status and Standard Model running. No trace-anomaly coefficient is used here to infer numerical couplings or hierarchy solutions.

22.4 Anomaly transport under matching

Integrating out heavy fields can move anomaly information into Wess-Zumino or Chern-Simons-like terms in the EFT. The low-energy record must reproduce the gauge anomaly cancellation and any unbroken 't Hooft anomaly to the declared order. A change of renormalization scheme is presentation; crossing a physical threshold is transport between EFT targets.

22.5 Downstream update rules

SYM-4 may change a discrete/CP record only by declaring a sector, parameter, theta, vacuum or global-form update. SYM-10 may change a trace/RG record only by declaring scale interval, operator basis, threshold or fixed-point target. Neither module may infer anomaly status from a familiar name or reuse a local coefficient outside its source assumptions.

Part IV

Fiberwise Normal Forms and Arc Transport

Role-Blind Physical Anomaly Objects

23.1 Independent physical construction

The physical construction reads only the fixed target, typed theorem interfaces, mathematical data, and declared physical generation rules. It has no access to standing roles or intended assignments. Candidate records contain a mathematical type, domain, codomain, dependencies, target identity, computed payload, and derivation provenance. They contain no expected occupant or stored compatibility verdict.

The ten closed targets produce 590 physical candidates. They include chiral ledgers, anomaly-coefficient vectors, local descent data, determinant and mod-two invariants, current and charge relations, counterterm representatives, bulk and boundary classes, WZW levels and variations, symmetry maps, trace operators, basis transformations, and target-comparison records. This is a finite generation theorem relative to the declared constructors, not an open-world claim that no future same-target object can be discovered. Presentation-related objects remain distinct until an executable map establishes equivalence. Where only the complete declared invariant tuple agrees, the record is described as finite invariant agreement rather than universal formulation equivalence.

SYM3-T24 – Target-relative physical generation. For each closed target, every physical object entering the normal-form theorem is generated from the fixed target and its imported QFT interfaces by the declared finite construction grammar, independently of role identities and intended occupations. This is a grammar-relative coverage theorem, not an open-world saturation claim. A target-relevant object expressible in the declared grammar but absent from the generated universe is a candidate-completeness counterexample and reopens the local theorem.

Claim status: Registered-scope physical-generation theorem; no open-world completeness claim is made.

23.2 Surplus and nonoccupants

The physical universe is intentionally larger than the live occupation set. Objects may remain as presentation variants, witnesses, countermodels, incompatible alternatives, or target edges. Their nonoccupation is a conclusion of eligibility and compatibility, not a reason to suppress their generation.

Physical Equivalence and Presentation Maps

24.1 Why presentation equivalence is narrow

Two anomaly records are presentation-equivalent only when an explicit map preserves the target, physical quotient, invariant anomaly content, continuation, failure conditions, and reportable consequence. Equality of names, formulas, or role labels is not sufficient.

The executed map families include changes of anomaly-calculation presentation, consistent-to-covariant conversion, Bardeen counterterm redistribution, equivalent current representatives, species permutations in symmetric targets, WZW presentation changes, and operator-basis changes preserving the renormalized trace identity. By contrast, adding matter, adding a bulk, changing global form, changing WZW level, or changing the admitted operator category changes target or realization data.

Across the ten fibers, 8210 presentation relations are evaluated. Counterterm-affine shifts, charge-basis changes, finite scheme maps, and species permutations carry explicit forward and inverse data with target- and invariant-preservation checks. Independent computational methods whose records agree only on the complete declared target tuple are recorded as finite invariant agreement; that agreement is not promoted to a universal equivalence of continuum formulations.

SYM3-T25 – Executable presentation covariance. Every relation reported as an explicit presentation isomorphism has typed forward and inverse data, target preservation, invariant preservation, and composition checks. Relations supported only by equality of the complete declared target tuple are reported as finite invariant agreement. Lawfully mapped assignments remain within one normal-form component.

Claim status: Finite map theorem over the declared presentation families.

24.2 Counterterms and invariant content

A local counterterm may move anomaly density between currents or background factors while preserving the obstruction class. This is presentation or redistribution, not physical cancellation. Actual cancellation requires the total class in the declared target to vanish, or an integrated bulk–boundary system whose relative anomaly class vanishes.

Eligibility and Relational Compatibility

25.1 Eligibility

A physical anomaly object is eligible for a role position only when it satisfies the fixed target's complete placement contract and restores every theorem query lost under the role's denial. Each same-fiber pair is tested on eight independent coordinates: target identity, candidate kind, physical locus, mathematical domain, mathematical codomain, consequence restoration, continuation behavior, and failure behavior. The eligibility proof records the role and candidate contracts, every clause disposition, the restored consequences and continuations, the target relation, and the placement witness. This extensional criterion avoids an authored role–candidate name crosswalk.

All 4412 same-fiber role-position/candidate pairs are evaluated. Exactly 581 edges are positive, and no role position lacks an eligible candidate.

SYM3-T26 – Complete same-target eligibility. For every closed target, every role-position/candidate pair receives a positive or negative disposition from the complete eight-coordinate placement and restoration test. No pair is omitted because its assignment would be inconvenient, and no candidate is admitted merely because its display name resembles a role.

Claim status: Complete finite incidence theorem over the ten closed targets.

25.2 Compatibility

Pointwise eligibility does not establish that several eligible occupants coexist coherently. Compatibility evaluates complete maps against the physical relations of the fixed target: chirality fidelity, local and global consistency, current–charge–Ward synchronization, descent structure, counterterm-class preservation, bulk–boundary cancellation, orientation and quantization, ultraviolet–infrared pullback, WZW level and background map, and operator-basis transport.

The complete assignment spaces contain 91695 maps. Relational compatibility rejects 91597 and retains 98. Each rejection records the failed physical predicates and, where applicable, diagnostic notes.

SYM3-T27 – Relational compatibility. A complete pointwise-eligible assignment is lawful only if every required cross-object relation is satisfied in the fixed target. A named relation failure rejects the complete assignment without changing the independently derived role architecture or physical candidate universe.

Claim status: Complete finite compatibility theorem over all pointwise-eligible maps.

Lawful Automorphisms and the Fiberwise Normal Form

26.1 The classification groupoid

The normal-form object is not a single database row. Lawful relabelings may permute physically indistinguishable carrier positions or equivalent presentations. The finite theorem uses the lawful assignment action groupoid: its objects are compatible complete assignments and its morphisms are target-preserving automorphisms of the executed incidence, presentation, and relation structure.

Record identifiers, display names, row order, expected occupants, and software-specific metadata are excluded from the lawful signature. Physical target data, denial structure, mathematical object type, invariant content, eligibility, presentation maps, and compatibility relations are retained.

SYM3-T28 – Lawful automorphism theorem. The target-preserving automorphisms of the role–object incidence and relation structure act on the compatible assignments. Their stabilizers and assignment orbits preserve the complete physical anomaly tuple.

Claim status: Computed finite-group action for each closed fiber.

SYM3-T32 – Fiberwise anomaly normal form. For every Reference-complete closed target in the declared SYM-3 family, relative to the frozen burden grammar, candidate grammar, presentation witnesses, and compatibility relations, the lawful anomaly-normal-form action groupoid has exactly one connected component and one isomorphism class:

$$|\pi_0 \mathcal{N}(D_\lambda)| = 1.$$

The theorem permits nontrivial stabilizers and multiple compatible presentations inside that one class.

Claim status: Ten-fiber finite groupoid theorem.

26.2 Quantitative result

The ten groupoids contain 98 compatible objects. Their lawful automorphism groups range from order 2 to order 2592; the massless-QCD trace target has group order 12, the Standard Model local target order 4, and the symmetric vectorlike target order 2592 with nontrivial stabilizers. Every closed fiber nevertheless has one normal-form component.

26.3 Meaning of uniqueness

The theorem does not say that every formula, basis, counterterm representative, or labeling is literally identical. It says that after fixing the target, physical anomaly content, admissible presentation maps, and compatibility relations, no second governance-non-equivalent anomaly normal form survives in the declared closed target.

Downstream Anomaly Transport

27.1 Target-indexed anomaly interfaces

Each closed fiber exports a target-indexed anomaly interface containing the fixed target, anomaly kind, current or determinant locus, background dependence, global data, phase assumptions, scale range, matching obligations, normal-form identity, and exact limitations. Ten such interfaces support the declared downstream imports to SYM-4 through SYM-11. The exported record is a finite projection of the closed target and normal-form summary; it is not asserted to be a general functor on arbitrary future certificate categories.

SYM3-T30 – Downstream anomaly-interface identity. A downstream module importing a closed SYM-3 result receives the target-indexed anomaly record fixed by the completed fiber. This is an irreversibility-preserving record-transport rule: same-target continuation preserves its physical disposition and normal-form class, while a change of symmetry content, field content, background, global form, boundary or bulk data, phase, operator category, scale, or ultraviolet–infrared map creates an explicit successor record or target update. No general categorical functor is inferred from the record rule alone.

Claim status: AASC record-transport rule over the ten closed anomaly interfaces.

SYM3-T35 – Successor discipline. A later result may supersede an anomaly record only through an explicit successor relation that states which target coordinates, assumptions, conclusions, and limitations changed. Supersession does not retroactively alter the completed predecessor claim.

Claim status: Irreversibility-preserving transport rule.

27.2 Arc interfaces

The discrete-symmetry, flavor, chiral, accidental-symmetry, vacuum, global-form, and RG modules import one common anomaly grammar. SYM-9 owns the completion of global-form and generalized-operator inventories. SYM-11 owns final candidate completeness and whole-arc exhaustion.

Ten Closed Targets: Worked Normal-Form Dossiers

The ten dossiers below state the physical question, the natural alternatives retained in the candidate universe, the relations that eliminate incompatible combinations, and the lawful presentation freedom that remains. The complete role, candidate, eligibility, assignment, and groupoid records appear in the Technical Audit Companion.

Table 28.1: Ten closed anomaly fibers and their normal-form calculations

Fiber	Physical target	Roles	Candidates	Complete	Rejected	Compatible	$ \Gamma $	Classes
F-SM-LOCAL-1GEN	Standard Model local anomaly ledger	5	36	16384	16380	4	4	1
F-VA-BARDEEN-DESCENT	$U(1)_V \times U(1)_A$ descent and Bardeen structure	5	66	1600	1598	2	2	1
F-VECTORLIKE-3PAIR-SYMMETRIC	Symmetric vectorlike six-Weyl target	10	213	14616	14544	72	2592	1
F-SU2-WITTEN-GLOBAL	Ordinary $SU(2)$ global anomaly	5	16	64	62	2	2	1
F-SU2-NEW-SPIN-ANOMALY	New spin- $SU(2)$ anomaly	6	49	4096	4092	4	4	1
F-QCD-U1A-NF3	Massless three-flavor QCD $U(1)_A$	6	12	36	34	2	2	1
F-EW-BPLUSL-3GEN	Electroweak $B + L$	6	16	192	190	2	2	1
F-U1-WEYL-INFLOW-4D5D	Weyl anomaly with five-dimensional inflow	6	44	16875	16873	2	2	1
F-QCD3-WZW-MATCH	Three-flavor QCD gauged-WZW matching	7	102	32000	31998	2	2	1
F-QCD3-TRACE-MSBAR	QCD trace anomaly in a fixed $\overline{\text{MS}}$ target	7	36	5832	5826	6	12	1

28.1 One-generation Standard Model local anomalies

Every fermion is written as a left-handed Weyl field. The local coefficients are exact sums over representation multiplicities and hypercharges. The candidate universe contains alternative chirality conversions, contribution ledgers, aggregate sums, descent decisions, and the global $SU(2)$ parity test. Compatibility enforces faithful conversion of right-handed fields, equality of the aggregate with the selected species contributions, agreement between aggregate and descent status, and coexistence of local and global consistency. Of 16,384 complete maps, four survive; they form one class under a lawful group of order four. The multiplicity reflects presentation-related anomaly calculations, not distinct gauge theories.

28.2 $U(1)_V \times U(1)_A$ descent and Bardeen structure

The target fixes the vector and axial backgrounds, the admitted current convention, and the relevant descent relations. Candidates include local cocycle data, current representatives, counterterm conventions, and consistent/covariant conversions. Compatibility requires Wess–Zumino consistency, preservation of the admitted vector current, and agreement between the selected counterterm and current conversion. Two of 1,680 complete maps survive and are presentation-equivalent.

28.3 Symmetric vectorlike six-Weyl theory

Three charge-+1 and three charge-−1 left-handed Weyl fields define an anomaly-neutral target with exact species-permutation symmetry. The candidate universe contains complete and partial ledgers, contribution records, aggregate classes, and permutation-related presentations. Seventy-two of 14,616 complete maps survive. The lawful group has order 2592 and each compatible object has a nontrivial stabilizer; all objects lie in one normal-form component. This target demonstrates that uniqueness of the normal form does not require a unique stored assignment or a trivial automorphism group.

28.4 Ordinary $SU(2)$ global anomaly

The target fixes the four-dimensional spin structure, the $SU(2)$ representation content, and the mod-two obstruction test. Candidates include representation ledgers, doublet counts, mod-two classes, and global-consistency decisions. Compatibility requires that the selected obstruction be computed from the selected representation data. Two of sixty-four complete maps survive and form one class.

28.5 New spin- $SU(2)$ anomaly

The target distinguishes the newer spin- $SU(2)$ obstruction from the ordinary Witten anomaly. Candidates include tangential-structure data, representation criteria, five-dimensional characteristic data, and target-update alternatives. Compatibility enforces the selected spin structure, the relevant representation condition, and the declared global invariant. Four of 4,096 complete maps survive in one class.

28.6 Massless three-flavor QCD $U(1)_A$

The classical axial action, anomalous divergence, topological density, vacuum orbit, and surviving discrete remnant are separate physical records. Candidate alternatives differ in massless-limit assumptions, current normalization, topology, and vacuum realization. Compatibility requires a common target and the correct relation among action, divergence, topological charge, and remnant. Two of thirty-six complete maps survive and are lawfully equivalent.

28.7 Electroweak $B + L$

For three generations the target fixes the Standard Model chiral content, electroweak topology, and the baryon/lepton current conventions. Candidate records distinguish the perturbative global action, nonperturbative divergence, topological selection rule, and $B - L$ control. Compatibility rejects maps that conflate perturbative conservation with nonperturbative exactness or that fail the generation and topology relations. Two of 192 complete maps survive.

28.8 Four-dimensional Weyl anomaly with five-dimensional inflow

The bulk and boundary are declared as one oriented target. Candidate objects include the boundary anomaly class, bulk level, orientation, relative class, and cancellation decision. Compatibility requires the selected bulk term to use the declared orientation, satisfy the admitted quantization condition, and cancel the selected boundary class in the relative target. Two of 16,875 complete maps survive. A bulk introduced only after the boundary classification is a successor target, not a same-act repair.

28.9 Three-flavor QCD gauged-WZW matching

The target fixes the ultraviolet chiral anomaly, the infrared chiral manifold, WZW level, background symmetry map, and Bardeen convention. Candidates include quantized levels, background maps, counterterms, gauged variations, and matching decisions. Compatibility requires the correct level, the admitted background map, vector-current preservation, and equality of the ultraviolet class with the pulled-back infrared variation. Two of 32,000 complete maps survive. The matching condition constrains but does not uniquely select the infrared phase.

28.10 Massless-QCD trace anomaly

The target fixes N_c , N_f , the perturbative order, the $\overline{\text{MS}}$ operator basis, and the operator quotient. Candidates include beta-function coefficients, complete and incomplete operator bases, trace identities, scheme maps, and transported invariants. Compatibility requires the selected beta data and basis to support the complete trace identity, an invertible scheme map, and preservation of the physical trace class. Six of 5,832 complete maps survive; a lawful group of order twelve identifies them in one component.

28.11 Cross-target conclusion

The dossiers exercise different sources of competition: representation conversion, descent and counterterm choice, global topology, vacuum realization, bulk orientation, ultraviolet–infrared

maps, operator bases, and nontrivial species permutations. The common result is not obtained by imposing one universal anomaly formula. It follows from fixed target identity, independently derived standing obligations, role-blind physical alternatives, complete eligibility, and target-specific compatibility.

Falsification Conditions and Exact Scope

The closed-target theorem is reopened by any same-scope demonstration of:

1. a false imported QFT result or a false target specialization;
2. a target coordinate used by a claim but absent from the target declaration;
3. a missing same-target anomaly object or standing role;
4. a false presentation map or an overbroad quotient;
5. an omitted or false eligibility or compatibility relation;
6. an under- or over-constrained lawful automorphism group;
7. a second normal-form isomorphism class;
8. an illicit same-act repair, hidden selector, or target change presented as a same-target rival.

The result is target- and grammar-relative: it closes the explicitly fixed fibers under the declared burden, constructor, presentation, eligibility, and compatibility grammars. It does not claim unique infrared realization, derivation of the Standard Model from anomaly cancellation, complete classification of every ultraviolet completion, or completion of the faithful global-form and extended-operator inventory. Those boundaries are explicit dependencies of SYM-9 and SYM-11.

SYM3-T37 – Downstream no-second-classifier consequence. Given the separately established AASC apex hypotheses, the already kernel-bearing fixed target and its locally established singleton normal-form class admit no second same-domain standing classifier or governance-non-equivalent admissible interior.

Claim status: Downstream AASC consequence, applicable only after the local normal-form theorem.

Conclusion

SYM-3 establishes a common anomaly grammar for the remainder of the symmetry arc. It separates gauge inconsistency, anomalous global response, local descent, consistent and covariant representatives, global determinant and index obstructions, mixed anomalies, obstruction to gauging, inflow, matching, trace/RG structure, and generalized-anomaly interfaces. The Standard Model applications show how these distinctions coexist within fixed targets rather than collapsing into a single exact/broken or conserved/nonconserved label.

The ten closed fibers supply a substantive finite normal-form calculation. Independent denial analysis yields 63 role positions; the role-blind physical construction yields 590 candidates; all 4412 same-fiber pairs are evaluated; and relational compatibility reduces 91695 complete maps to 98 compatible objects. The compatible objects retain lawful automorphisms and stabilizers, yet each fiber has one isomorphism class. The result therefore distinguishes physical uniqueness from uniqueness of notation, presentation, or record identity.

Counterterm changes, presentation changes, physical cancellations, bulk extensions, infrared matching relations, and target updates are classified by their effect on the fixed theory. The resulting anomaly interfaces may be transported to later symmetry modules only while their target assumptions remain fixed. Ten closed interfaces support the anomaly needs of SYM-4 through SYM-10, while global-form and generalized-operator completion remains with SYM-9.

The result does not claim a unique infrared realization, derive the Standard Model from anomaly cancellation, or close the faithful global form and extended-operator inventory. Those remain downstream burdens of SYM-9 and SYM-11. Within its declared scope, SYM-3 provides the anomaly distinctions, target-relative normal forms, and transport rules required by every later module that invokes gauge consistency, global response, matching, vacuum realization, defects, or scale dependence.

Scientific endpoint

For every Reference-complete closed anomaly target in the declared SYM-3 family, non-degenerate determinacy implies necessary kernel activation. Independently derived target roles and QFT-established physical anomaly objects form a lawful normal-form groupoid with exactly one isomorphism class. Public anomaly coordinates and repair claims are admitted only when they preserve the fixed target and its invariant physical content. Frontier targets remain outside the closure theorem until their missing global-form or operator data are supplied. The resulting target-indexed anomaly interfaces constitute the SYM-3 anomaly authority for the downstream symmetry arc.

Part V

Scientific Appendices

Anomaly Conventions and Core Formulae

For a d -dimensional local anomaly with invariant polynomial I_{d+2} , choose a descent representative

$$I_{d+2} = dI_{d+1}^{(0)}, \quad sI_{d+1}^{(0)} + dI_d^{(1)} = 0.$$

The consistent anomaly is represented by the integrated descendant $I_d^{(1)}$. Equivalent representatives differ by BRST-exact and total-derivative terms. A Bardeen–Zumino current converts consistent to covariant form without changing the underlying obstruction class.

For a four-dimensional chiral $U(1)$ ledger, the perturbative local coefficients include

$$\sum_i d_i Y_i^3, \quad \sum_i d_i Y_i,$$

with the appropriate non-Abelian Dynkin-index factors for mixed anomalies. A nonzero aggregate coefficient is a gauge inconsistency when the corresponding symmetry is dynamical; for a background global symmetry it is an obstruction to gauging or a matching datum.

A global anomaly is encoded by a target-specific phase or sign of the fermion determinant or partition function under a large transformation. Its detection may require an eta invariant, an index, or a bordism invariant not visible in the local polynomial.

Exact Standard Model Anomaly Arithmetic

All fermions are written as left-handed Weyl fields. For one generation the minimal chiral multiplets are

$$Q : (\mathbf{3}, \mathbf{2})_{1/6},$$

$$u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-2/3}, \quad d^c : (\bar{\mathbf{3}}, \mathbf{1})_{1/3}, \quad L : (\mathbf{1}, \mathbf{2})_{-1/2}, \quad e^c : (\mathbf{1}, \mathbf{1})_1.$$

The aggregate $SU(3)^2 U(1)_Y$, $SU(2)^2 U(1)_Y$, $U(1)_Y^3$, and gravitational–hypercharge sums vanish generation by generation. The $SU(2)$ global parity condition is checked separately from the local polynomial. Full species-level tables and exact rational calculations are supplied in the Technical Audit Companion.

Closed Targets and Frontier Interfaces

The closed family contains ten target-fixed anomaly problems: the Standard Model local ledger; vector/axial descent with Bardeen structure; a symmetric vectorlike six-Weyl theory; the ordinary and newer $SU(2)$ global anomalies; massless three-flavor QCD axial structure; electroweak $B + L$; a fixed Weyl bulk–boundary inflow system; three-flavor QCD WZW matching; and a fixed trace-anomaly operator target.

The complete target-coordinate registry appears in the Technical Audit Companion. Table 28.1 gives the closed-fiber summary used by the scientific argument.

The frontier family contains faithful Standard Model global forms with genuine line and monopole sectors, higher-flavor/higher-group structure, and non-invertible defect completion. These targets remain outside the local closure quantifier until SYM-9 supplies their missing operator and global-form data.

Theorem Dependency Map

The theorem spine is asymmetric:

$$\begin{aligned}
 T01 &\Rightarrow T02 \Rightarrow (T03, T04, T05) \\
 &\Rightarrow (T06\text{--}T23) \\
 &\Rightarrow (T24\text{--}T32, T38) \\
 &\Rightarrow (T30, T33\text{--}T36, T39\text{--}T42) \\
 &\Rightarrow T37.
 \end{aligned}$$

T01 supplies the local physical antecedent for kernel activation. T02 supplies the necessary AASC implication. T06–T23 establish the QFT anomaly distinctions and target-specific physical results. T24–T32 and T38 establish independent physical generation, executable presentation equivalence, complete eligibility, relational compatibility, lawful automorphisms, and the ten-fiber normal-form theorem. T30 and T33–T36 govern relation closure, downstream transport, and successor discipline. T39–T42 state the exact naturality, independent reconstruction, and frontier-successor results. T37 is strictly downstream of the local normal-form theorem.

The complete theorem inventory and evidence paths appear in the Technical Audit Companion.

Exact Nonclaims

The result does not claim:

- derivation of Standard Model representations, hypercharges, couplings, masses, or mixing parameters from anomaly cancellation alone;
- a unique infrared realization of a fixed 't Hooft anomaly;
- open-world completeness over arbitrary ultraviolet completions;
- completion of faithful global-form, genuine-line, monopole, higher-group, or non-invertible inventories;
- that every anomalous global symmetry is inconsistent or meaningless;
- that a bulk added after classification repairs the original boundary target;
- that a local anomaly polynomial detects every global anomaly;
- that one normal-form isomorphism class means one formula, one basis, or one presentation;
- replacement of SYM-9 global-structure closure or SYM-11 whole-arc exhaustion.

Formalization Boundary

The finite formal target is the classification architecture: typed targets, physical objects, presentation maps, role obligations, eligibility, compatibility, lawful automorphisms, and the statement of the closure theorem. Imported continuum QFT theorems remain imported mathematical and physical results with explicit hypotheses. The formal machinery does not claim to reconstruct interacting path integrals, prove every anomaly theorem from first principles, or generate the AASC kernel.

Extended Capstone Theorem Package

This appendix states the theorem objects that complete the local normal-form, relation, transport, and successor-discipline spine. Their detailed finite records appear in the Technical Audit Companion. Each statement is scoped to the declared targets and executed grammars unless it is explicitly identified as an imported AASC theorem or a general protocol rule.

SYM3-T23 – QCD axial action, anomaly, vacuum, and exact-remnant theorem.

In the fixed massless three-flavor QCD target, the classical axial action, anomalous quantum current, topological-charge selection rule, vacuum orbit, and exact discrete remnant occupy distinct loci and have one compatible target-relative normal form.

SYM3-T29 – Registered benchmark and holdout discrimination. The declared calibration suite executes sixty-six adversarial checks and eleven frozen-grammar holdouts over the ten closed fibers. Each test retains its fixed target and reports the exact semantic disposition. This is adequacy for the registered suite, not a claim about every possible future counterexample.

SYM3-T31 – Closed-fiber migration completeness. Every anomaly fiber currently declared closed in the SYM-3 plan is represented in the ten-fiber proof architecture and satisfies the pre-role, role, physical-object, presentation, eligibility, compatibility, and normal-form obligations. The three SYM-9 frontier fibers are excluded from this claim.

SYM3-T33 – Registered coordinate and counterterm relation closure. Within every closed fiber, each coordinate, basis, counterterm, and presentation relation generated by the declared finite relation grammar receives one executed disposition. Explicit isomorphisms preserve target and invariant content; invariant-agreement records are not promoted beyond their stated target tuple.

SYM3-T34 – Registered repair, inflow, matching, and target-update closure. Every proposed cancellation or repair generated in the declared closed fibers is classified by the executed relation checks as same-target neutralization, representative redistribution, integrated inflow, matching transport, target update, scope enlargement, or an unresolved frontier. The theorem is exhaustive relative to the registered finite candidate and relation grammars.

SYM3-T36 – Conservative record equivalence and standing-changing reopening.

For the explicit evidence-only enrichment used in the audit companion, source and enriched

finite groupoid records have identical physical objects, lawful morphisms, and component structure under the identity map. A change to a target-defining or standing-bearing coordinate creates a successor target and requires a fresh classification.

SYM3-T38 – Normal-form groupoid theorem. For every Reference-complete closed anomaly fiber, the lawful groupoid of compatible complete role occupations has exactly one isomorphism class, while lawful automorphisms and stabilizers remain visible as morphisms.

SYM3-T39 – Finite naturality under the declared conservative enrichment. The explicit identity-on-physical-data map between each closed-fiber groupoid record and its evidence-enriched copy preserves the object set, lawful morphism set, stabilizers, and component partition. This is a computed equivalence for the declared enrichment, not a theorem about every possible certificate morphism.

SYM3-T40 – Declared downstream anomaly-interface maps. Each exported anomaly interface is the verified finite projection of a closed target, its physical anomaly disposition, normal-form class, assumptions, and limitations. The declared import records preserve those fields and compose with identity at their registered use sites. No broader categorical functoriality is claimed.

SYM3-T41 – Independent semantic endpoint reconstruction. A secondary implementation, isolated from the primary role, eligibility, compatibility, presentation, and groupoid builders, reconstructs all same-fiber eligibility dispositions, complete assignments, physical compatibility truth values, lawful assignment actions, stabilizers, and normal-form components for the ten closed fibers. The reconstruction reads the fixed targets, reviewed theorem interfaces, independently generated roles and physical candidates, dependencies, and executed presentation evidence; it does not read the primary eligibility or compatibility verdicts before producing its own result. Agreement is asserted up to the registered semantic invariants rather than serialization order.

SYM3-T42 – Frontier-successor monotonicity protocol. Completion of a SYM-9 frontier target must create an explicit successor target and successor anomaly interface. The protocol may enlarge the admitted closed family but cannot silently rewrite a completed SYM-3 target or its act-time theorem record. No concrete SYM-9 successor instance is claimed before SYM-9 fixes the successor target and map.

Part VI

Technical Audit Companion

Introduction to the Technical Audit Companion

The preceding scientific manuscript states the anomaly distinctions and the ten-fiber normal-form theorem in conventional academic form. This companion exposes the controlled objects behind those results: target declarations, theorem interfaces, denial records, physical candidates, presentation maps, eligibility proofs, compatibility dispositions, assignment groupoids, transport interfaces, and reproducibility evidence.

Dependency discipline

The audit does not treat AASC as an optional overlay. The logical implication

$$\text{NonDegDet}(D_{\lambda}^{\text{phys}}) \implies K_{D_{\lambda}} = (\text{Admissibility}, \text{Standing}, \text{Reference}, \text{Irreversibility})$$

precedes the target-specific QFT and finite normal-form calculations. The audit checks the local antecedent, the physical specialization, and the downstream construction as distinct obligations.

Ten-fiber normal-form reconstruction

The closed target family contains 10 physical fibers and 3 explicit SYM-9 frontiers. Independent denial analysis yields 63 role positions, while the role-blind physical construction yields 590 candidate objects. All 4412 same-fiber pairs are evaluated; 581 edges are positive. The finite assignment spaces contain 91695 complete maps, of which 91597 are rejected by physical compatibility and 98 survive. Each closed fiber has one normal-form isomorphism class. The combined evidence graph contains 184 nodes and 348 edges with no dangling theorem-bearing edge.

Controlled Research Object and Exact Scope

The theorem-bearing research object consists of the indexed target family, typed QFT interfaces, pre-role physical acts, executed denial transitions, role-blind physical candidates, presentation maps, complete same-fiber eligibility, relational compatibility, lawful assignment groupoids, and downstream anomaly interfaces. The scientific manuscript and this companion are coordinated presentations of that object.

The closed family contains 10 targets and the frontier family contains 3 targets owned by SYM-9. The finite calculation contains 63 role positions, 590 physical candidates, 91695 complete maps, and 98 compatible objects. Each closed fiber has one normal-form isomorphism class.

Finite scientific identity

The reconstructed ten-fiber certificate has identity `88d27ad19c082c93`. The combined evidence graph has identity `8431e3df190d2eb51ec2ea48b2572ea8240a01c5017c86e776adccec5eb64965`. These identifiers allow two builds to be compared; the scientific justification remains in the physical calculations and theorem relations documented below.

Literature Basis and Imported Results

The closed fibers use freely accessible primary papers and technical reviews containing the required derivations. Each imported result is represented by a typed interface that states its physical conclusion, hypotheses, target specialization, and source identity. The cited literature remains the authority for the continuum theorem; the local project supplies the exact target match and finite specialization.

Historical papers may be cited for attribution even when a later open source supplies the operative derivation. The complete proof-core source and theorem-interface tables appear in Chapter 5, followed by the broader literature basis used by the scientific exposition.

Evidence and Dependency Graph

The combined evidence graph connects open literature to typed theorem interfaces, fixed targets, pre-role acts, executed denial transitions, physical candidates, eligibility proofs, compatibility relations, groupoid calculations, manuscript theorems, and downstream anomaly interfaces.

The graph contains 184 nodes and 348 edges. Every theorem-bearing edge has existing endpoints. A graph path exposes dependency but does not replace the corresponding physical proof.

Graph closure

The combined graph has no dangling theorem-bearing edge. Its identity is 8431e3df190d2eb51ec2ea48b2572ea8240a01c5017c86e776adccec5eb64965. The underlying machine-readable graph is stored in `data/v11/derived/combined_evidence_graph_v11.json`.

Adversarial Reconstruction Standard

The proof architecture is assessed against the following independent requirements:

1. the local non-degeneracy acts precede role construction;
2. target identity is unchanged within a same-target comparison;
3. roles arise from executed denial consequences;
4. physical candidates contain no expected-role or intended-assignment fields;
5. presentation equivalence is witnessed by executable maps;
6. every same-fiber role/candidate pair is evaluated;
7. compatibility rejects complete but physically incoherent maps;
8. lawful automorphisms and normal-form components are calculated;
9. frontier targets remain outside the closure quantifier;
10. holdouts and a separate implementation reproduce the declared endpoint semantics.

Sixty-three adversarial tests and eleven frozen-grammar holdouts exercise these requirements. Passing the suite does not make the theorem immune to criticism. A false imported theorem, a missing target coordinate, an omitted same-target object or role, a false relation, an overbroad presentation quotient, or a second isomorphism class remains a direct falsifier.

Imported-Result Register

Ordinary citations in the scientific manuscript are the primary reading surface. The following tables expose the exact open sources and typed theorem interfaces used by the ten-fiber normal-form proof.

Table 6.1: Open primary sources used by the ten-fiber proof core

Source	Authors	Title	arXiv	Rev.
SRC-ACC-2208.01441v2	Taushif Ahmed; Long Chen; Michał Czakon	A note on quark and gluon energy-momentum tensors	2208.01441v2	
SRC-BBH-hep-th-0002245v3	Glenn Barnich; Friedemann Brandt; Marc Henneaux	Local BRST cohomology in gauge theories	hep-th/0002245v3	
SRC-BILAL-0802.0634v1	Adel Bilal	Lectures on Anomalies	0802.0634v1	
SRC-FREED-HOPKINS-1604.06527v6	Daniel S. Freed; Michael J. Hopkins	Reflection positivity and invertible topological phases	1604.06527v6	
SRC-HATTA-1810.05116v1	Yoshitaka Hatta; Abha Rajan; Kazuhiro Tanaka	Quark and gluon contributions to the QCD trace anomaly	1810.05116v1	
SRC-HHH-0712.1230v1	Jeffrey A. Harvey; Christopher T. Hill; Richard J. Hill	Standard Model Gauging of the WZW Term: Anomalies, Global Currents and pseudo-Chern-Simons Interactions	0712.1230v1	
SRC-HILL-hep-th-0603060v2	Christopher T. Hill	Exact Equivalence of the D=4 Gauged Wess-Zumino-Witten Term and the D=5 Yang-Mills Chern-Simons Term	hep-th/0603060v2	
SRC-LANDSTEINER-1610.04413v4	Karl Landsteiner	Notes on Anomaly Induced Transport	1610.04413v4	
SRC-PPS-2010.02062v2	George Panagopoulos; Haralambos Panagopoulos; Gregoris Spanoudes	Two-loop renormalization and mixing of gluon and quark energy-momentum tensor operators	2010.02062v2	
SRC-RS-hep-ph-9603208v2	V. A. Rubakov; M. E. Shaposhnikov	Electroweak Baryon Number Non-Conservation in the Early Universe and in High Energy Collisions	hep-ph/9603208v2	
SRC-TANIZAKI-1807.07666v3	Yuya Tanizaki	Anomaly constraint on massless QCD and the role of Skyrmions in chiral symmetry breaking	1807.07666v3	
SRC-WWW-1810.00844v4	Juven Wang; Xiao-Gang Wen; Edward Witten	A New SU(2) Anomaly	1810.00844v4	
SRC-YONEKURA-1607.01873v1	Kazuya Yonekura	Dai-Freed theorem and topological phases of matter	1607.01873v1	

Table 6.2: Typed theorem interfaces used by the proof core

Interface	Statement	Hypotheses	Sources
THM-ABELIAN-DESCENT-BARDEEN	Solving the vector Ward constraints selects the Bardeen counterterm class; the remaining axial anomaly has the vector-preserving 3:1 coefficient ratio in the one-Dirac target.	four-dimensional left-handed Weyl basis; commuting $U(1)_V$ and $U(1)_A$ backgrounds; local polynomial anomaly; common overall factor suppressed	SRC-BILAL-0802.0634v1; SRC-BBH-hep-th-0002245v3; SRC-LANDSTEINER-1610.04413v4
THM-ABELIAN-TRACE-PERMUTATION	The symmetric vectorlike target has vanishing local $U(1)$ anomalies and lawful $S_3 \times S_3$ species permutations.	left-handed Weyl fermions; fixed $U(1)$ charges; no flavor-breaking target data for the symmetric fiber	0802.0634v1
authority_allow list THM-EW-BPLUSL	B+L changes by $2N_g Q$ while B-L is preserved.	three Standard Model generations; unit SU(2) _L topological charge; standard baryon and lepton normalization	SRC-RS-hep-ph-9603208v2; SRC-BILAL-0802.0634v1
THM-4D-LOCAL-ANOMALY-TRACE	The aggregate local anomaly vector is the exact sum of the species contributions.	four-dimensional chiral Weyl fermions; fixed local gauge representations; left-handed basis	SRC-BILAL-0802.0634v1

continued

Interface	Statement	Hypotheses	Sources
THM-QCD3-TRACE-MSBAR	The renormalized trace is represented in the fixed operator quotient and is invariant under the declared invertible finite basis transport.	massless $SU(N_c)$ QCD; fixed $\overline{MS}$ scheme; renormalized operator basis; two-loop beta coefficients	SRC-HATTA-1810.05116v1; SRC-PPS-2010.02062v2; SRC-ACC-2208.01441v2
THM-QCD3-GAUGED-WZW	The gauged WZW functional matches the UV chiral anomaly when $k=N_c$, the background map is fixed, and the vector-preserving counterterm is included.	massless $N_f=3$ QCD; $SU(3)_L$ and $SU(3)_R$ backgrounds; gauged WZW functional; vector-preserving counterterm convention	SRC-HHH-0712.1230v1; SRC-HILL-hep-th-0603060v2
THM-QCD-U1A-NF3	The continuous classical axial action has an anomalous current; an exact $Z_2(2N_f)$ remnant remains and the condensate preserves Z_2 .	massless N_f -flavor QCD; integer topological charge; axial normalization $q_L=+1$ $q_R=-1$	SRC-BILAL-0802.0634v1; SRC-TANIZAKI-1807.07666v3
THM-SU2-NEW-SPIN	The newer anomaly is detected by the w_2w_3 bulk invariant and is distinct from the ordinary Witten anomaly.	spin- $SU(2)$ tangential structure; no chosen spin structure; left-handed isospin representations	SRC-WWW-1810.00844v4
THM-SU2-ORDINARY-GLOBAL	Odd ordinary mod-two index gives determinant sign -1 and a global gauge obstruction.	four-dimensional spin $SU(2)$ gauge theory; left-handed Weyl representations; nontrivial π_4 transformation	SRC-WWW-1810.00844v4
THM-U1-WEYL-INFLOW	A quantized bulk class cancels the full boundary class only when the oriented relative class vanishes.	charge-one 4d Weyl boundary; fixed oriented 5d bulk; basis cubic and mixed gauge-gravitational; lattice scales 6 and 24	SRC-YONEKURA-1607.01873v1; SRC-FREED-HOPKINS-1604.06527v6

Research-Object Schema

The controlled object families are:

Family	Required content
Target fiber	Field content, interactions, backgrounds, global form, boundary, phase, operator category, scale, quantum regime, and source interfaces.
Imported result	Source identity, hypotheses, conclusion, locator, limitations, and typed application constructor.
Physical fact	Target-matched result produced by an imported theorem or exact local calculation.
Standing burden	Denied function, required consequence, locus, continuation and failure behavior, and denial loss.
Physical object	Kind, locus, acted-upon object, invariant content, provided consequence, source fact, and presentation data.
Eligibility proof	Clause-level comparison of one role instance and one presentation residue.
Compatibility record	Named cross-object physical relations and their calculated outcomes.
Normal-form groupoid	Compatible assignments, lawful automorphisms, stabilizers, morphisms, and connected components.
Downstream interface	Target identity, normal-form identity, allowed successor relation, and recipient SYM modules.

No primitive physical candidate may contain an expected role, intended occupant, stored compatibility result, or canonical/defective label.

Controlled Types and Allowed Values

The top-level candidate kinds are *Action*, *Structure*, *Diagnostic*, and *Transport*. Loci distinguish target, field space, state space, local operators, extended operators, boundary, bulk–boundary system, background space, and theory space. Physical status is recorded independently from standing role.

An incomplete target, unresolved imported result, absent presentation map, unknown required relation, or unsupported frontier operator category is reported as an open burden rather than receiving a default classification.

Indexed Target Registry

A fiber changes whenever a coordinate capable of changing the physical theorem object changes. Closed fibers enter the normal-form quantifier; frontier fibers remain exact interfaces to SYM-9. The complete indexed target table appears in Scientific Appendix C; the companion retains the underlying target records in the project data.

Pre-Role Determinacy and Standing Roles

The pre-role process reads the fixed target and QFT calculation interfaces before role, candidate, eligibility, assignment, or groupoid construction. The summary below records the accepted and rejected complete acts for each target.

Table 10.1: Pre-role same-target physical acts

Fiber	Accepted	Rejected	Process boundary	Digest
F-SM-LOCAL-1GEN	1	3	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	1c8b45e90822
F-VA-BARDEEN-DESCENT	1	9	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	83b12513094e
F-VECTORLIKE-3PAIR-SYMMETRIC	1	202	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	fa39a7b13a36
F-SU2-WITTEN-GLOBAL	2	6	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	30d942b65fa4
F-SU2-NEW-SPIN-ANOMALY	2	6	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	e80336554a66
F-QCD-U1A-NF3	2	16	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	79909abf01df
F-EW-BPLUSL-3GEN	2	6	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	262c6e36bc58
F-U1-WEYL-INFLOW-4D5D	2	13	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	622cb8687eda
F-QCD3-WZW-MATCH	2	8	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	3c068d70fa3d
F-QCD3-TRACE-MSBAR	6	12	src.common.io, src.common.combinatorics, src.common.qft_math, src.common.expanded_qft	a2d21096b037

Standing roles are then obtained from executed denial transitions. The complete registry contains all 63 role positions.

Table 10.2: Extensional standing roles derived from executed denial behavior

Fiber	Role	Denied interface(s)	Lost theorem queries	Available
F-SM-LOCAL-1GEN	ROL-b864d5d63d7d007a	aggregate sum	aggregate local class, quantum descent	3

continued

Fiber	Role	Denied interface(s)	Lost theorem queries	Available
F-SM-LOCAL-1GEN	ROL-d3c5a317f55048bd	local index evaluation	aggregate local class, quantum descent, species contributions	2
F-SM-LOCAL-1GEN	ROL-b9861e2172c5e9f4	quantum descent decision	quantum descent	4
F-SM-LOCAL-1GEN	ROL-8096027bb6607478	chirality conversion, physical field content	aggregate local class, ledger complete, quantum descent, species contributions, su2 global parity	0
F-SM-LOCAL-1GEN	ROL-60bb6a700e64b716	su2 mod2 test	su2 global parity	4
F-VA-BARDEEN-DESCENT	ROL-4f19964c039ea826	left handed conversion, physical charge content	axial ward identity, consistent ward polynomial, left charge ledger, presentation relation, six form charge tensor, vector gauge resolution	0
F-VA-BARDEEN-DESCENT	ROL-66994469625483b6	presentation basis change	presentation relation	5
F-VA-BARDEEN-DESCENT	ROL-c2d75404d4d3a2f5	local counterterm variation, vector ward requirement	axial ward identity, presentation relation, vector gauge resolution	3
F-VA-BARDEEN-DESCENT	ROL-c4977c70da69ba65	symmetric charge tensor	axial ward identity, consistent ward polynomial, presentation relation, six form charge tensor, vector gauge resolution	1
F-VA-BARDEEN-DESCENT	ROL-59fcc1531a566dc8	abelian descent	axial ward identity, consistent ward polynomial, presentation relation, vector gauge resolution	2
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-b9f0f806e187597b	species content	aggregate abelian anomaly, carrier response::charge +1, carrier response::charge -1, lawful species automorphisms, physical presentation quotient	0
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-569d8c0dcef4a664	carrier p3	carrier response::charge +1	8
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-974f17ca8dda2510	observable algebra, physical automorphism discovery	lawful species automorphisms, physical presentation quotient	7
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-815f00b938d145c9	carrier p2	carrier response::charge +1	8
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-1e50af0b481a316d	presentation quotient test	physical presentation quotient	8
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-0efe5b85e701f050	abelian trace evaluation, aggregate sum	aggregate abelian anomaly	8
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-00ccee602cd2106a	carrier m3	carrier response::charge -1	8
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-32edc680f799373b	carrier m1	carrier response::charge -1	8
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-64914f77284d22be	carrier m2	carrier response::charge -1	8
F-VECTORLIKE-3PAIR-SYMMETRIC	ROL-66001a7496e40743	carrier p1	carrier response::charge +1	8
F-SU2-WITTEN-GLOBAL	ROL-82162df77f7fe037	representation content	representation ledger	4
F-SU2-WITTEN-GLOBAL	ROL-d8db41562313a464	local polynomial test	local polynomial	4
F-SU2-WITTEN-GLOBAL	ROL-7deef2d9d4da9e21	ordinary mod2 index	ordinary mod2	4
F-SU2-WITTEN-GLOBAL	ROL-44385dd583389447	large transform phase	determinant phase	4
F-SU2-WITTEN-GLOBAL	ROL-0bfccbe41ea4a085	global consistency decision	global consistency	4
F-SU2-NEW-SPIN-ANOMALY	ROL-5097093a65819d23	ordinary mod2 index	ordinary mod2	5
F-SU2-NEW-SPIN-ANOMALY	ROL-42a5036d142a4143	new mod2 index	new mod2	5
F-SU2-NEW-SPIN-ANOMALY	ROL-1883e74f27eac517	tangential structure	tangential packet	5
F-SU2-NEW-SPIN-ANOMALY	ROL-8b4ee73f1d4a1f1b	w2w3 bulk invariant	w2w3 bulk	5
F-SU2-NEW-SPIN-ANOMALY	ROL-38c50138244dde1b	representation content	representation ledger	5
F-SU2-NEW-SPIN-ANOMALY	ROL-877ecc7b69a602be	global consistency decision	global consistency	5
F-QCD-U1A-NF3	ROL-6682e485451030d5	massless quark content	massless content	5
F-QCD-U1A-NF3	ROL-fd923b434206c166	classical axial action	classical axial action	5
F-QCD-U1A-NF3	ROL-bc805c2415bf3238	vacuum realization	vacuum realization	5
F-QCD-U1A-NF3	ROL-355e7d53fc223e27	exact discrete remnant	discrete remnant	5
F-QCD-U1A-NF3	ROL-4bd77c9fcf68c0b6	axial current ward identity	axial ward identity	5
F-QCD-U1A-NF3	ROL-855604443be32edf	measure index anomaly	measure index anomaly	5
F-EW-BPLUSL-3GEN	ROL-4ff6879f59f7d3b4	nonperturbative status	nonperturbative status	5
F-EW-BPLUSL-3GEN	ROL-276f06fd78bab88e	electroweak index	electroweak index	5
F-EW-BPLUSL-3GEN	ROL-9c5338e1e3423d72	generation charge content	generation content	5
F-EW-BPLUSL-3GEN	ROL-6059f8f303875616	baryon selection rule	baryon rule	5
F-EW-BPLUSL-3GEN	ROL-468d2cd785385718	combined charge rule	combined rule	5
F-EW-BPLUSL-3GEN	ROL-6d3d9df6bd0d3b0a	lepton selection rule	lepton rule	5

continued

Fiber	Role	Denied interface(s)	Lost theorem queries	Available
F-U1-WEYL-INFLOW-4D5D	ROL-876fb11bab63448c	relative gluing	relative gluing	5
F-U1-WEYL-INFLOW-4D5D	ROL-68a74f7f93258f19	boundary charge content	boundary content	5
F-U1-WEYL-INFLOW-4D5D	ROL-6dbbc8a3a2174b0e	boundary anomaly class	boundary class	5
F-U1-WEYL-INFLOW-4D5D	ROL-f3daf1c85fceaed6	integral quantization	integral quantization	5
F-U1-WEYL-INFLOW-4D5D	ROL-816de4e2a1620dea	inflow cancellation	inflow cancellation	5
F-U1-WEYL-INFLOW-4D5D	ROL-0a952db07620367c	bulk anomaly class	bulk class	5
F-QCD3-WZW-MATCH	ROL-c6b67fa6d8704232	wzw level	wzw level	6
F-QCD3-WZW-MATCH	ROL-f25a4ebd6138a5d4	background symmetry map	background map	6
F-QCD3-WZW-MATCH	ROL-1a6e5617402db2c8	uv quark anomaly	uv anomaly	6
F-QCD3-WZW-MATCH	ROL-f4c09b97cb b91a7c	bardeen counterterm	counterterm	6
F-QCD3-WZW-MATCH	ROL-4a102d2352c29da7	ir chiral target	ir target	6
F-QCD3-WZW-MATCH	ROL-17e65367b58e4e86	uv ir matching	uv ir matching	6
F-QCD3-WZW-MATCH	ROL-6e3bd4139fc86ec5	gauged wzw variation	gauged variation	6
F-QCD3-TRACE-MSBAR	ROL-c35a991d757d0c6c	qcd parameter content	qcd parameters	6
F-QCD3-TRACE-MSBAR	ROL-132e14c85584b330	renormalized operator basis	operator basis	6
F-QCD3-TRACE-MSBAR	ROL-af88f42e4b9e1b73	eom total derivative quotient	operator quotient	6
F-QCD3-TRACE-MSBAR	ROL-94c5bff56de0c099	trace operator identity	trace identity	6
F-QCD3-TRACE-MSBAR	ROL-6164d36e802d5c65	finite scheme map	finite scheme map	6
F-QCD3-TRACE-MSBAR	ROL-44c5f31403b9c472	beta gamma coefficients	beta gamma	6
F-QCD3-TRACE-MSBAR	ROL-6a794ec114d5554e	scheme invariant trace	scheme invariant trace	6

Role-Blind Physical Object Generation and Presentation

Physical candidates are constructed from fixed targets, mathematical data, and typed QFT interfaces without access to the role architecture. The generated universe contains 590 candidates. The scientific volume summarizes the physical-object universe in Chapter 23.

Table 11.1: Physical candidate counts by mathematical type

Fiber	Mathematical type	Count
F-SM-LOCAL-1GEN	GaugeDescentDecision	8
F-SM-LOCAL-1GEN	GlobalMod2Index	8
F-SM-LOCAL-1GEN	LocalAnomalyVector	8
F-SM-LOCAL-1GEN	LocalContributionTable	8
F-SM-LOCAL-1GEN	WeylLedger	4
F-VA-BARDEEN-DESCENT	AbelianChargeLedger	1
F-VA-BARDEEN-DESCENT	AxialWardIdentity	20
F-VA-BARDEEN-DESCENT	ConsistentWardPolynomial	2
F-VA-BARDEEN-DESCENT	CountertermWardBundle	20
F-VA-BARDEEN-DESCENT	PresentationIsomorphism	21
F-VA-BARDEEN-DESCENT	SymmetricChargeTensor	2
F-VECTORLIKE-3PAIR-SYMMETRIC	AbelianAnomalyVector	2
F-VECTORLIKE-3PAIR-SYMMETRIC	PermutationGroup	1
F-VECTORLIKE-3PAIR-SYMMETRIC	PhysicalSpeciesPacket	1
F-VECTORLIKE-3PAIR-SYMMETRIC	SpeciesQuotient	203
F-VECTORLIKE-3PAIR-SYMMETRIC	SpeciesResponse	6
F-SU2-WITTEN-GLOBAL	DeterminantPhase	4
F-SU2-WITTEN-GLOBAL	GlobalConsistencyDecision	8
F-SU2-WITTEN-GLOBAL	LocalPolynomialNull	1
F-SU2-WITTEN-GLOBAL	OrdinaryMod2Index	2
F-SU2-WITTEN-GLOBAL	SU2RepresentationPacket	1
F-SU2-NEW-SPIN-ANOMALY	GlobalConsistencyDecision	32
F-SU2-NEW-SPIN-ANOMALY	NewMod2Index	4
F-SU2-NEW-SPIN-ANOMALY	OrdinaryMod2Index	2
F-SU2-NEW-SPIN-ANOMALY	SU2RepresentationPacket	1
F-SU2-NEW-SPIN-ANOMALY	TangentialStructurePacket	2
F-SU2-NEW-SPIN-ANOMALY	W2W3BulkInvariant	8
F-QCD-U1A-NF3	AxialAction	1
F-QCD-U1A-NF3	AxialAnomalyCoefficient	2
F-QCD-U1A-NF3	AxialWardIdentityQCD	2
F-QCD-U1A-NF3	DiscreteRemnant	3
F-QCD-U1A-NF3	QCDMasslessPacket	1
F-QCD-U1A-NF3	VacuumRealization	3
F-EW-BPLUSL-3GEN	BaryonSelectionRule	4
F-EW-BPLUSL-3GEN	CombinedChargeRule	2
F-EW-BPLUSL-3GEN	ElectroweakIndex	2
F-EW-BPLUSL-3GEN	GenerationChargePacket	1
F-EW-BPLUSL-3GEN	LeptonSelectionRule	4
F-EW-BPLUSL-3GEN	NonperturbativeStatus	3
F-U1-WEYL-INFLOW-4D5D	BoundaryAnomalyClass	3
F-U1-WEYL-INFLOW-4D5D	BoundaryChargePacket	1
F-U1-WEYL-INFLOW-4D5D	BulkAnomalyClass	5
F-U1-WEYL-INFLOW-4D5D	InflowDecision	15
F-U1-WEYL-INFLOW-4D5D	QuantizationDecision	5
F-U1-WEYL-INFLOW-4D5D	RelativeAnomalyClass	15
F-QCD3-WZW-MATCH	AnomalyMatchingDecision	80
F-QCD3-WZW-MATCH	BackgroundSymmetryMap	2
F-QCD3-WZW-MATCH	BardeenCounterterm	2
F-QCD3-WZW-MATCH	GaugedWZWVariation	10
F-QCD3-WZW-MATCH	IRChiralTarget	1
F-QCD3-WZW-MATCH	UVAAnomalyClass	2

continued

Fiber	Mathematical type	Count
F-QCD3-WZW-MATCH	WZWLevel	5
F-QCD3-TRACE-MSBAR	BetaGammaData	2
F-QCD3-TRACE-MSBAR	FiniteSchemeMap	3
F-QCD3-TRACE-MSBAR	OperatorQuotient	3
F-QCD3-TRACE-MSBAR	QCDParameterPacket	1
F-QCD3-TRACE-MSBAR	RenormalizedOperatorBasis	3
F-QCD3-TRACE-MSBAR	SchemeInvariantTrace	18
F-QCD3-TRACE-MSBAR	TraceOperatorIdentity	6

Presentation maps are evaluated separately and are admitted only when their forward, inverse, invariant, target, and composition checks pass. The executable presentation-map summary appears in Chapter 24.

Source and Theorem-Interface Records

The complete reviewed source and theorem-interface tables appear in Audit Chapter 5. This appendix records the schema-level rule: a load-bearing application must cite one fixed source identity and one typed theorem interface, must satisfy every declared hypothesis, and must inherit the stated limitation without strengthening it.

Principal Physical Formulas and Conventions

The closed-fiber calculations use exact rational or symbolic data at their declared scope. Representative formulas include

$$I_6 = \frac{1}{6} \operatorname{tr}_R \left(\frac{F}{2\pi} \right)^3 - \frac{1}{24} p_1(T) \operatorname{tr}_R \left(\frac{F}{2\pi} \right), \quad I_6 = dI_5^{(0)}, \quad sI_5^{(0)} + dI_4^{(1)} = 0,$$

with target-specific normalizations recorded in the corresponding fiber. For the renormalized trace target,

$$T^\mu{}_\mu = \frac{\beta(g)}{2g} [F^2] + \text{EOM} + \partial_\mu J^\mu + \text{contact terms},$$

where the operator basis, perturbative order, and scheme are fixed. For matching targets, the anomaly condition is expressed as a pullback equality under the declared UV–IR symmetry map rather than as equality of untyped coefficients.

Exact Standard Model Local-Anomaly Ledger

The one-generation calculation uses a left-handed Weyl convention and exact rational arithmetic. The aggregate row vanishes in every admitted perturbative local-anomaly coordinate.

Table 14.1: One-generation Standard Model anomaly coefficients

Field	SU3 cub.	SU3–U1	SU2–U1	U1 cub.	grav.–U1
Q_L	2	1/6	1/4	1/36	1
u_R	-1	-1/3	0	-8/9	-2
d_R	-1	1/6	0	1/9	1
L_L	0	0	-1/4	-1/4	-1
e_R	0	0	0	1	1
Aggregate	0	0	0	0	0

Adversarial and Frozen-Grammar Tests

The proof-core suite contains sixty-three adversarial checks spanning process isolation, target fidelity, answer-blind records, eligibility completeness, compatibility, presentation maps, lawful automorphisms, and determinism.

Table 15.1: Adversarial proof-core tests

Test	Result	Evidence
pre_role_import_isolation	PASS	['src.roles', 'src.candidates', 'src.comparator', 'src.compatibility', 'src.groupoid']
role_process_has_no_candidate_import	PASS	None
candidate_process_has_no_role_import	PASS	None
F-SM-LOCAL-1GEN:pre_role_bivalent	PASS	{'accepted': 1, 'rejected': 3}
F-SM-LOCAL-1GEN:pre_role_target_partition	PASS	{'same_target': 4, 'target_updates': 0}
F-VA-BARDEEN-DESCENT:pre_role_bivalent	PASS	{'accepted': 1, 'rejected': 9}
F-VA-BARDEEN-DESCENT:pre_role_target_partition	PASS	{'same_target': 10, 'target_updates': 0}
F-VECTORLIKE-3PAIR-SYMMETRIC:pre_role_bivalent	PASS	{'accepted': 1, 'rejected': 202}
F-VECTORLIKE-3PAIR-SYMMETRIC:pre_role_target_partition	PASS	{'same_target': 203, 'target_updates': 0}
F-SU2-WITTEN-GLOBAL:pre_role_bivalent	PASS	{'accepted': 2, 'rejected': 6}
F-SU2-WITTEN-GLOBAL:pre_role_target_partition	PASS	{'same_target': 8, 'target_updates': 0}
F-SU2-NEW-SPIN-ANOMALY:pre_role_bivalent	PASS	{'accepted': 2, 'rejected': 6}
F-SU2-NEW-SPIN-ANOMALY:pre_role_target_partition	PASS	{'same_target': 8, 'target_updates': 8}
F-QCD-U1A-NF3:pre_role_bivalent	PASS	{'accepted': 2, 'rejected': 16}
F-QCD-U1A-NF3:pre_role_target_partition	PASS	{'same_target': 18, 'target_updates': 0}
F-EW-BPLUSL-3GEN:pre_role_bivalent	PASS	{'accepted': 2, 'rejected': 6}
F-EW-BPLUSL-3GEN:pre_role_target_partition	PASS	{'same_target': 8, 'target_updates': 0}
F-U1-WEYL-INFLOW-4D5D:pre_role_bivalent	PASS	{'accepted': 2, 'rejected': 13}
F-U1-WEYL-INFLOW-4D5D:pre_role_target_partition	PASS	{'same_target': 15, 'target_updates': 0}
F-QCD3-WZW-MATCH:pre_role_bivalent	PASS	{'accepted': 2, 'rejected': 8}
F-QCD3-WZW-MATCH:pre_role_target_partition	PASS	{'same_target': 10, 'target_updates': 30}
F-QCD3-TRACE-MSBAR:pre_role_bivalent	PASS	{'accepted': 6, 'rejected': 12}
F-QCD3-TRACE-MSBAR:pre_role_target_partition	PASS	{'same_target': 18, 'target_updates': 0}
candidate_records_contain_no_answer_fields	PASS	[]
role_candidate_human_vocabularies_disjoint	PASS	[]
F-SM-LOCAL-1GEN:eligibility_complete	PASS	{'RP0S-0ec57031e9554df8': 8, 'RP0S-1d82ea8a290f5e1f': 8, 'RP0S-d0222553beb11ab1': 8, 'RP0S-e16cf16420fcd0a4': 8, 'RP0S-ebbe507d51b5c3ee': 4}
F-VA-BARDEEN-DESCENT:eligibility_complete	PASS	{'RP0S-0a73c90213d2d08b': 20, 'RP0S-1e419ce6a0d85bcb': 1, 'RP0S-8c8d104e37decf64': 2, 'RP0S-9bdd48f424e67405': 2, 'RP0S-a7a455d0493f4f15': 20}
F-VECTORLIKE-3PAIR-SYMMETRIC:eligibility_complete	PASS	{'RP0S-013e235f96322924': 1, 'RP0S-544ed184a47aba2f': 3, 'RP0S-59a48a706585a1da': 1, 'RP0S-62584e61f37182af': 3, 'RP0S-656486937c1d2897': 2, 'RP0S-76b53be02e2b7761': 3, 'RP0S-779b3
F-SU2-WITTEN-GLOBAL:eligibility_complete	PASS	{'RP0S-0f236a8efae83c65': 4, 'RP0S-38ab277746ac8f5d': 1, 'RP0S-3bc6fe5dd99e1e54': 8, 'RP0S-8a1c0a8b54214034': 1, 'RP0S-a2fcd8599017386': 2}

continued

Test	Result	Evidence
F-SU2-NEW-SPIN-ANOMALY:eligibility__complete	PASS	{'RP0S-4f12442678a99f50': 32, 'RP0S-5700c8a295cec3f1': 2, 'RP0S-5f8501a99258f10d': 2, 'RP0S-62ea927b58516587': 4, 'RP0S-b4bffafbbdaed511': 1, 'RP0S-f20da4575c521256': 8}
F-QCD-U1A-NF3:eligibility__complete	PASS	{'RP0S-50059c07d49ce620': 3, 'RP0S-5f9f4bbf1d164d05': 1, 'RP0S-63efe44d67fcdcbff': 2, 'RP0S-7254da3dcf45411c': 2, 'RP0S-7ac711b3f5954dfe': 3, 'RP0S-f5db893466497808': 1}
F-EW-BPLUSL-3GEN:eligibility__complete	PASS	{'RP0S-14d2a9284b57c7f0': 2, 'RP0S-1a6f2a8161e3154f': 4, 'RP0S-d500f10c0b119dea': 1, 'RP0S-e06983f185130b7a': 2, 'RP0S-e2db39919f7444b6': 4, 'RP0S-f2f82afdcaf734ef': 3}
F-U1-WEYL-INFLOW-4D5D:eligibility__complete	PASS	{'RP0S-3d06feec5589820b': 5, 'RP0S-651e97d96131fca5': 5, 'RP0S-6776ff26318a44a6': 15, 'RP0S-99c1d10dad c71072': 15, 'RP0S-cb2f301eafe5bb67': 3, 'RP0S-e643b27595588887': 1}
F-QCD3-WZW-MATCH:eligibility__complete	PASS	{'RP0S-00b74faaf3dc7b39': 80, 'RP0S-011aac614df1705': 10, 'RP0S-8f167bd45b517e64': 2, 'RP0S-b4d5f6bcf1a10d53': 2, 'RP0S-c21a1696fa0c1542': 1, 'RP0S-e4b0c9cf7df4daf3': 2, 'RP0S-faf
F-QCD3-TRACE-MSBAR:eligibility__complete	PASS	{'RP0S-004b66e1a4476828': 3, 'RP0S-0c3de586995821c5': 3, 'RP0S-44b0d6d08d7e3314': 6, 'RP0S-8cd99aa4e77629ff': 2, 'RP0S-e1347b4b7c72a5a0': 3, 'RP0S-f5ba1edb9f9a109e': 1, 'RP0S-fc76d
F-SM-LOCAL-1GEN:compatibility__nontrivial	PASS	{'compatible': 4, 'complete': 16384, 'rejected': 16380}
F-VA-BARDEEN-DESCENT:compatibility__nontrivial	PASS	{'compatible': 2, 'complete': 1600, 'rejected': 1598}
F-VECTORLIKE-3PAIR-SYMMETRIC:compatibility__nontrivial	PASS	{'compatible': 72, 'complete': 14616, 'rejected': 14544}
F-SU2-WITTEN-GLOBAL:compatibility__nontrivial	PASS	{'compatible': 2, 'complete': 64, 'rejected': 62}
F-SU2-NEW-SPIN-ANOMALY:compatibility__nontrivial	PASS	{'compatible': 4, 'complete': 4096, 'rejected': 4092}
F-QCD-U1A-NF3:compatibility__nontrivial	PASS	{'compatible': 2, 'complete': 36, 'rejected': 34}
F-EW-BPLUSL-3GEN:compatibility__nontrivial	PASS	{'compatible': 2, 'complete': 192, 'rejected': 190}
F-U1-WEYL-INFLOW-4D5D:compatibility__nontrivial	PASS	{'compatible': 2, 'complete': 16875, 'rejected': 16873}
F-QCD3-WZW-MATCH:compatibility__nontrivial	PASS	{'compatible': 2, 'complete': 32000, 'rejected': 31998}
F-QCD3-TRACE-MSBAR:compatibility__nontrivial	PASS	{'compatible': 6, 'complete': 5832, 'rejected': 5826}
F-SM-LOCAL-1GEN:normal_form_single_component	PASS	{'components': 1, 'group_order': 4, 'objects': 4}
F-VA-BARDEEN-DESCENT:normal_form_single_component	PASS	{'components': 1, 'group_order': 2, 'objects': 2}
F-VECTORLIKE-3PAIR-SYMMETRIC:normal_form_single_component	PASS	{'components': 1, 'group_order': 2592, 'objects': 72}
F-SU2-WITTEN-GLOBAL:normal_form_single_component	PASS	{'components': 1, 'group_order': 2, 'objects': 2}
F-SU2-NEW-SPIN-ANOMALY:normal_form_single_component	PASS	{'components': 1, 'group_order': 4, 'objects': 4}
F-QCD-U1A-NF3:normal_form_single_component	PASS	{'components': 1, 'group_order': 2, 'objects': 2}
F-EW-BPLUSL-3GEN:normal_form_single_component	PASS	{'components': 1, 'group_order': 2, 'objects': 2}
F-U1-WEYL-INFLOW-4D5D:normal_form_single_component	PASS	{'components': 1, 'group_order': 2, 'objects': 2}
F-QCD3-WZW-MATCH:normal_form_single_component	PASS	{'components': 1, 'group_order': 2, 'objects': 2}
F-QCD3-TRACE-MSBAR:normal_form_single_component	PASS	{'components': 1, 'group_order': 12, 'objects': 6}
source_formula_mutation_changes_physics	PASS	{'nondegenerate': True}
bardeen_formula_mutation_changes_selected_counterterm	PASS	[{'counterterm_coefficients': [6, 0]}]
F-SM-LOCAL-1GEN:presentation_maps_do_real_work	PASS	{'after': 4, 'before': 1}
F-VA-BARDEEN-DESCENT:presentation_maps_do_real_work	PASS	{'after': 2, 'before': 1}
presentation_inverse_corruption_detected	PASS	['PMAP-d5da71e5fddcd31']
partial_species_quotient_rejected	PASS	[[['m1'], ['m2'], ['m3'], ['p1'], ['p2'], ['p3']]]
opaque_idenfifer_blinding_invariant	PASS	{'blinded': [36, 4, 1, 4]}
frozen_clause6_complete_denial_coordinates	PASS	[]
frozen_clause8_complete_eligibility_coordinates	PASS	[]
eligibility_coordinate_mutations_fail	PASS	{'codomain': True, 'domain': True, 'locus': True}
primary_build_deterministic	PASS	88d27ad19c082c930fcd580186bdf244ce1fa9a869c722c8faeecd70a3275e4

Eleven committed holdouts are evaluated only after the proof grammar is frozen.

Table 15.2: Frozen-grammar holdout results

Case	Fiber	Result	Expected	Actual
H-ALPHA		PASS		
H-BETA		PASS		
H-GAMMA		PASS		
H-DELTA		PASS		
H-EPSILON		PASS		
H-ZETA		PASS		
H-ETA		PASS		
H-THETA		PASS		
H-IOTA		PASS		
H-KAPPA		PASS		
H-LAMBDA		PASS		

Complete Same-Fiber Eligibility

Every role position is compared with every candidate in its fixed fiber. Cross-fiber pairs are excluded by target identity and are not counted as physical competition. The fiberwise eligibility totals appear in Chapter 25.

The complete positive incidence relation contains 581 edges. All 4412 positive and negative dispositions remain available in the machine-readable per-fiber ledgers under `engine/ten_fiber_core/generated/eligibility/`.

Table 16.1: All positive role-position/candidate eligibility edges

Fiber	Role position	Candidate	Type	Restored queries	Witness
F-SM-LOCAL-1GEN	RPOS-e16cf16420fcf0a4	ART-c5b0b7a334c217d4	LocalAnomalyVector	aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-e16cf16420fcf0a4	ART-4ccd69e807007d6e	LocalAnomalyVector	aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-e16cf16420fcf0a4	ART-52296a8a5f18db26	LocalAnomalyVector	aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-e16cf16420fcf0a4	ART-30c430fd94ed6da7	LocalAnomalyVector	aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-e16cf16420fcf0a4	ART-d38ec89568b7ba2a	LocalAnomalyVector	aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-e16cf16420fcf0a4	ART-fa771b8394b758fd	LocalAnomalyVector	aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-e16cf16420fcf0a4	ART-1d84c9a883399ecf	LocalAnomalyVector	aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-e16cf16420fcf0a4	ART-dba1e55cf7f960d4	LocalAnomalyVector	aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-1d82ea8a290f5e1f	ART-fc3f7feef7b197f4	LocalContributionTable	species contributions, aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-1d82ea8a290f5e1f	ART-55e9ae960a1865b8	LocalContributionTable	species contributions, aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-1d82ea8a290f5e1f	ART-28947be651d5fc92	LocalContributionTable	species contributions, aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-1d82ea8a290f5e1f	ART-9f54d19f32045b8c	LocalContributionTable	species contributions, aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-1d82ea8a290f5e1f	ART-1f9eb2bfda7b566c	LocalContributionTable	species contributions, aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-1d82ea8a290f5e1f	ART-68c3435f41f06705	LocalContributionTable	species contributions, aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-1d82ea8a290f5e1f	ART-439ad10b4236d91f	LocalContributionTable	species contributions, aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-1d82ea8a290f5e1f	ART-1194f9f2b54f8224	LocalContributionTable	species contributions, aggregate local class, quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-d0222553beb11ab1	ART-3deaa997992ebc8d	GaugeDescentDecision	quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-d0222553beb11ab1	ART-eb87a903068236c8	GaugeDescentDecision	quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-d0222553beb11ab1	ART-4af0698a68423a9b	GaugeDescentDecision	quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-d0222553beb11ab1	ART-5eea31af50c82c88	GaugeDescentDecision	quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-d0222553beb11ab1	ART-5104afc248b91d13	GaugeDescentDecision	quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-d0222553beb11ab1	ART-e8e095445e069e06	GaugeDescentDecision	quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-d0222553beb11ab1	ART-24b3a44be91b5113	GaugeDescentDecision	quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-d0222553beb11ab1	ART-0e1263a7212d788d	GaugeDescentDecision	quantum descent	identity
F-SM-LOCAL-1GEN	RPOS-ebbe507d51b5c3ee	ART-ba34646744d24227	WeylLedger	ledger complete, species contributions, aggregate local class, quantum descent, su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-ebbe507d51b5c3ee	ART-5cd405b4aa4b44e1	WeylLedger	ledger complete, species contributions, aggregate local class, quantum descent, su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-ebbe507d51b5c3ee	ART-f6c4bbcd08299c0e	WeylLedger	ledger complete, species contributions, aggregate local class, quantum descent, su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-ebbe507d51b5c3ee	ART-63334eb042521a66	WeylLedger	ledger complete, species contributions, aggregate local class, quantum descent, su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-0ec57031e9554df8	ART-b0b72c525a66effd	GlobalMod2Index	su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-0ec57031e9554df8	ART-511e4e7028b18e8c	GlobalMod2Index	su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-0ec57031e9554df8	ART-5a7188893b1ee596	GlobalMod2Index	su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-0ec57031e9554df8	ART-298219b305b6b0ac	GlobalMod2Index	su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-0ec57031e9554df8	ART-a8bf4cf696c678c1	GlobalMod2Index	su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-0ec57031e9554df8	ART-41a972b615fa2b2c	GlobalMod2Index	su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-0ec57031e9554df8	ART-c674d2dfc6dac0a1	GlobalMod2Index	su2 global parity	identity
F-SM-LOCAL-1GEN	RPOS-0ec57031e9554df8	ART-1474e3a4e97f2be6	GlobalMod2Index	su2 global parity	identity
F-VA-BARDEEN-DESCENT	RPOS-1e419ce6a0d85bcb	ART-e678fc2ae88b4fb5	AbelianChargeLedger	left charge ledger, six form charge tensor, consistent ward polynomial, vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-c2d856a0fdea8705	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-034bbc4b6aaef7f9c	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-87a6d01b0dcc280	PresentationIsomorphism	presentation relation	identity

continued

Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-596f3d533b72d060	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-10e8793bc74f0c7d	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-193eeba8f61461ec	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-a0038c7c2a95e44d	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-ad615be09c5c29f0	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-e23cc846a44e8894	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-a333902599a7ec35	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-275a63f6df21eb0f	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-41128f2462f1aa56	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-7125669847b3d0aa	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-c25e0cf14c765a9a	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-c7e4c9462578a530	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-a60117fbf863e9b1	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-f553500bfb8b3b7a	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-3b423017332f7635	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-6c2eb36912cf2889	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-0a73c90213d2d08b	ART-4c466a41a80fa4c1	PresentationIsomorphism	presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-8d1ae583b82d18dd	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-8b2fee121fdebab8	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-01dbc0cfa902cdb	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-7144d66b9e479562	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-98ff91b68e8f8aa5	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-1b38a2f3c59767b2	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-9ed0d2da33553263	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-6fa090b327db03db	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-098852354c2ce47f	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-408c9bca92d84ae9	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-8a7e18c0328ffc06	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-13baed56b0a4b29e	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-eadbe8ce2234ff4a	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-4bd881f5973a9fd2	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-f9448282197be31e	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-f5e30bb1391c02e1	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-2b14a16971fb5cb9	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-6550cbc919343417	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-2991ad78a33380ae	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-a7a455d0493f4f15	ART-acbeca4ec01cf3d0	CountertermWardBundle	vector gauge resolution, axial ward identity, presentation relation	identity

continued

Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VA-BARDEEN-DESCENT	RPOS-8c8d104e37decf64	ART-65584f52f393bd35	SymmetricChargeTensor	six form charge tensor, consistent ward polynomial, vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-8c8d104e37decf64	ART-38443d146ae57976	SymmetricChargeTensor	six form charge tensor, consistent ward polynomial, vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-9bdd48f424e67405	ART-e796949f10aa28b9	ConsistentWardPolynomial	consistent ward polynomial, vector gauge resolution, axial ward identity, presentation relation	identity
F-VA-BARDEEN-DESCENT	RPOS-9bdd48f424e67405	ART-7060a06c062ef2b7	ConsistentWardPolynomial	consistent ward polynomial, vector gauge resolution, axial ward identity, presentation relation	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-013e235f96322924	ART-07d74c454be87b52	PhysicalSpeciesPacket	carrier response::p1, carrier response::p2, carrier response::p3, carrier response::m1, carrier response::m2, carrier response::m3, aggregate abelian anomaly, lawful species automorphisms, physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-62584e61f37182af	ART-b7a93596fe51b144	SpeciesResponse	carrier response::p3	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-62584e61f37182af	ART-9e21c67ed17240b3	SpeciesResponse	carrier response::p3	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-62584e61f37182af	ART-f598c35e1ade5fe1	SpeciesResponse	carrier response::p3	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-59a48a706585a1da	ART-1afd90b90aceeff9	PermutationGroup	lawful species automorphisms, physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-544ed184a47aba2f	ART-b7a93596fe51b144	SpeciesResponse	carrier response::p2	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-544ed184a47aba2f	ART-9e21c67ed17240b3	SpeciesResponse	carrier response::p2	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-544ed184a47aba2f	ART-f598c35e1ade5fe1	SpeciesResponse	carrier response::p2	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-a05b48b7516ab2b0	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-8465a7a9aa4dd5df	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-7ccb5d254550d3e2	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-2ad382059fb34d45	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-0974284b0e4aeefa	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-3f6f8a4140d51872	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-143f2cb11526c679	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-3cff6b69ef10b334	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-459cac7fc87124dc	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-3973463c2afb0e5f	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-c7ba000fb6835a19	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-c75756ef12b41bb1	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-5699a57ace916a95	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-aa21594f880e29af	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-8ddabebc3a229d5a	SpeciesQuotient	physical presentation quotient	identity

continued

Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-49aef7da9d113ad8	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-109b6cd4aa2ac9e4	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-39153da165cd177d	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d13bfff26bc2d33f	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-a6f5622ae254f833	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-498891b6fbec969f	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-5edf1d5d30961ed0	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-03fbdb2a51d9af87	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-48e454f165d8c77e	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b90ffea2abed5206	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-a15775b3d0f37724	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-1bdbf9f099cf55c6	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-aefab529e4e8e959	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-0a61cd363fc5bfaf	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-66507d1b0ebef7b2	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-1cdd0dc73eba7b18	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-3a904a7eba78cd6b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-514ca8aabade66f5	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f542c986a7a417d7	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-1278992ec5bf2223	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-2b1b2d10ec3f95d0	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b340b282564346f0	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-ba4eb8bd087a8fbb	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-c50da946cd5b2a34	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-94d8e3f60e4f4418	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-dddd9248206c99e9	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-9671f633fa2c0400	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d7bde42e9dfeb150	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d999a4514ad75851	SpeciesQuotient	physical presentation quotient	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-6dc857db29af0f7a	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-9741d09432eaa712	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-5c1e1124ef5b89cc	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f7b248cfa54ee5e8	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-6e562e516d4230c9	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f3e411a1cf58d3c8	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-ba2d0bb2eabb2db3	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-908c56aecc41212b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-1b194b6d583b7699	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-1e5b7a35bbc5f7de	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-08275b6ed01af407	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-10888b4dc291d412	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-c87a1b1fb72bb1cc	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-0cbc7f7764e3fe5e	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-4944a74a297cf178	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-45e44e538060b02b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-a8946ade0149bb78	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-15fc7917841d1a31	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-89ff74c91766ae29	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-eb8542eaa7fc95b2	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-0e5a1098a740dd2c	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-789b0df6a766848c	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-254ef09176ddb0d1	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-daea97b76ef86ac3	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f7daeb0a09095bcf	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d51bea985468608e	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d893039de22483b7	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-92f2d93dcb243113	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-4ea759b235724f16	SpeciesQuotient	physical presentation quotient	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-9b92cb4b0b267044	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-530f20084742109b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-d9798a61b4e150bb	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-ba623f90a62d73f5	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-b4ff77b29297a502	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-ef9e4b6ccad4e645	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-541ee02cecf252fb	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-b678615ceacfe1f4	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-b54bc9e9aa1f0803	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-eea9f370e7423374	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-3aebd514a208c4e9	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-2f5420a21f9e238d	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-3b13eb68fcd2f132	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-d424f635fe6e1cf9	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-e760ab97413afc2b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-058b12153f271cd0	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-22824819385cf565	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-a83364223947dbad	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-de8486c86eebf842	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-7d86cf623b31f4d5	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-12ec779b29ca976b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-38d6d100a72460a7	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-ddb45b90a74891bb	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-f3e945b8acd54770	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-716be0702dd5f26a	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-1c5543ee26b59c22	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-1350743bf9d4a6e9	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-ece9f8615f9c3e3b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RP0S-fd72b9c194e79874	ART-ca188087412188b3	SpeciesQuotient	physical presentation quotient	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-9722d5202aeef4b3	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-86424e9f588e6afb	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-4b5535502cafd005	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-dd1fbf79068bebea	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-bdf7ee0631666ceb	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-fad0c2f44bd15c8c	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-5444748df6657877	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-def37dba6205ef24	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-948610527747a0ea	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-235f922e562928c7	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-bcca173f71136825	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f6ec3ec22618fd8d	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-2fab452f943ad544	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b31c810dde2a6ac3	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-2ba814523d3a624d	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-40e11f17d6f0591f	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-866cc7be1ac5b71a	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-4613eefba7b1e3f2	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-419f4f9946fa72cf	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-09b240f9851d28eb	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-dae565aabe6cf9a4	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-6dc4feaa6db73ef7	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-775efc3be08e80fa	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-1b643d123e66cdf2	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-02291cc5d4d4bf20	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-814032142010fbee	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b1cc4262ca515d89	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-a9b9009c76b23115	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-04ab88fac5ebd94b	SpeciesQuotient	physical presentation quotient	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-8ae96178a90798d0	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f1ac776de26e8ff1	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-0df9190f9e22f25f	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-fbc15a52a9b9e3fa	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-5d61230e9d4322cd	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-5557a374413b7d26	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b81681007fa88a5a	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-2c6f19f685c8f95e	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-95dbe1cede73179c	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b57eb5442375bf91	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f46b2c3513268d0e	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-1bb4c0672cdeea55	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b312d5a9918975aa	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-106ee1fd7575e314	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-8bd42c4eebf04130	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-991116f755b7a468	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b25969f1abf15987	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-11aa4d1474489664	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-765bff527cd23356	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-7751e94ef800cfc	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-150a077bff07a28b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d39f6b2459d13f36	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f189730420439535	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-418ff811b4b2b455	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-6f092f9d7523b8b8	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-cc531d89da023e15	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-ba9575054ee64427	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-367523485d9cd57f	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-30cd415197d68f6f	SpeciesQuotient	physical presentation quotient	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-594bffafc0d4b7f8	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-ddf0fba6af5685ce	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f08277152acf9506	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b20fedea18f557d2	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d45f7f8e35dcb72c	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-41b4d165c6391759	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-02686fc25919d3e5	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-634038581dc30a75	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-1456e15f6af6e025	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-67cd9e1961344d4e	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-978f24eabf35447b	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-54c52c329855d798	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-62817cdf74364fc8	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d9437738355cbf0c	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-f499bc174fa2bc8e	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-e041c2c968f472ba	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-a1a292e12d16cdd4	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-59b1ec5491734a1c	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-0d538196aa533da2	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-edaebf14d7de00a3	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-e94187871f7e139d	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-df6f2e9fa2ceb8c0	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-d4491e0498e63511	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-290d5426042f4703	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-86a549c87e35e552	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-155884b111c141fd	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-9e2b99529ee9e865	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-4b190a8123f8987c	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-cf1bf77fbc68637e	SpeciesQuotient	physical presentation quotient	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-5fc409fad7b17fbc	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-47bdd8a8eb57d8b7	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-7b87c97abaf961a	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-79ca876a140a6073	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-b857815b6446fef	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-26f53c127ff284bc	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-ba962afbcbd709f6	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-e5323f2fea7c00b0	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-cf7770bfeb543e46	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-fd3c59d1130ff0d4	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-78ef05e17e127d07	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-cdee04b467c1b906	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-c217c6326d34e438	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-fd72b9c194e79874	ART-a5f105e73a5cb739	SpeciesQuotient	physical presentation quotient	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-656486937c1d2897	ART-deacbd760ac6ea61	AbelianAnomalyVector	aggregate abelian anomaly	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-656486937c1d2897	ART-70ed4e3635947b22	AbelianAnomalyVector	aggregate abelian anomaly	identity
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-782bdec835282e56	ART-22c08b1bf0105513	SpeciesResponse	carrier response::m3	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-782bdec835282e56	ART-d29cf0cbb875b5d7	SpeciesResponse	carrier response::m3	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-782bdec835282e56	ART-430d6a4c5cf00016	SpeciesResponse	carrier response::m3	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-76b53be02e2b7761	ART-22c08b1bf0105513	SpeciesResponse	carrier response::m1	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-76b53be02e2b7761	ART-d29cf0cbb875b5d7	SpeciesResponse	carrier response::m1	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-76b53be02e2b7761	ART-430d6a4c5cf00016	SpeciesResponse	carrier response::m1	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-9eb4d30473c0b5ae	ART-22c08b1bf0105513	SpeciesResponse	carrier response::m2	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-9eb4d30473c0b5ae	ART-d29cf0cbb875b5d7	SpeciesResponse	carrier response::m2	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-9eb4d30473c0b5ae	ART-430d6a4c5cf00016	SpeciesResponse	carrier response::m2	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-779b3d33e31f5018	ART-b7a93596fe51b144	SpeciesResponse	carrier response::p1	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-779b3d33e31f5018	ART-9e21c67ed17240b3	SpeciesResponse	carrier response::p1	charge-sector-placement
F-VECTORLIKE-3PAIR-SYMMETRIC	RPOS-779b3d33e31f5018	ART-f598c35e1ade5fe1	SpeciesResponse	carrier response::p1	charge-sector-placement
F-SU2-WITTEN-GLOBAL	RPOS-38ab277746ac8f5d	ART-fca71e7245ba13fc	SU2RepresentationPacket	representation ledger	identity
F-SU2-WITTEN-GLOBAL	RPOS-8a1c0a8b54214034	ART-7cab77ec2b6bb967	LocalPolynomialNull	local polynomial	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-SU2-WITTEN-GLOBAL	RPOS-a2fced8599017386	ART-5363c3aa5860ed2b	OrdinaryMod2Index	ordinary mod2	identity
F-SU2-WITTEN-GLOBAL	RPOS-a2fced8599017386	ART-ea6fc0405a43da11	OrdinaryMod2Index	ordinary mod2	identity
F-SU2-WITTEN-GLOBAL	RPOS-0f236a8efae83c65	ART-f65f04bafb9b91e1	DeterminantPhase	determinant phase	identity
F-SU2-WITTEN-GLOBAL	RPOS-0f236a8efae83c65	ART-af80974a5f8529bb	DeterminantPhase	determinant phase	identity
F-SU2-WITTEN-GLOBAL	RPOS-0f236a8efae83c65	ART-4c4a145c48e691c1	DeterminantPhase	determinant phase	identity
F-SU2-WITTEN-GLOBAL	RPOS-0f236a8efae83c65	ART-152a000619e3dd51	DeterminantPhase	determinant phase	identity
F-SU2-WITTEN-GLOBAL	RPOS-3bc6fe5dd99e1e54	ART-38995a692337ae28	GlobalConsistencyDecision	global consistency	identity
F-SU2-WITTEN-GLOBAL	RPOS-3bc6fe5dd99e1e54	ART-c9a8195125e700fc	GlobalConsistencyDecision	global consistency	identity
F-SU2-WITTEN-GLOBAL	RPOS-3bc6fe5dd99e1e54	ART-e3d139f5096f91	GlobalConsistencyDecision	global consistency	identity
F-SU2-WITTEN-GLOBAL	RPOS-3bc6fe5dd99e1e54	ART-f7bd0d230c8da1d0	GlobalConsistencyDecision	global consistency	identity
F-SU2-WITTEN-GLOBAL	RPOS-3bc6fe5dd99e1e54	ART-238138cf241f4f87	GlobalConsistencyDecision	global consistency	identity
F-SU2-WITTEN-GLOBAL	RPOS-3bc6fe5dd99e1e54	ART-ddd2869600cad8d6	GlobalConsistencyDecision	global consistency	identity
F-SU2-WITTEN-GLOBAL	RPOS-3bc6fe5dd99e1e54	ART-6aafcbba5ff3df77	GlobalConsistencyDecision	global consistency	identity
F-SU2-WITTEN-GLOBAL	RPOS-3bc6fe5dd99e1e54	ART-06ee194521fec71f	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-5700c8a295cec3f1	ART-75039314df31a2c8	OrdinaryMod2Index	ordinary mod2	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-5700c8a295cec3f1	ART-5e659cdbc9047c79	OrdinaryMod2Index	ordinary mod2	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-62ea927b58516587	ART-07f22935afa7841c	NewMod2Index	new mod2	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-62ea927b58516587	ART-fcbfab833721d623	NewMod2Index	new mod2	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-62ea927b58516587	ART-9d047d1778dfe170	NewMod2Index	new mod2	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-62ea927b58516587	ART-2bcdaf2bf4af5222	NewMod2Index	new mod2	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-5f8501a99258f10d	ART-9b5ac91618e7b17f	TangentialStructurePacket	tangential packet	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-5f8501a99258f10d	ART-c59c51778895ebd0	TangentialStructurePacket	tangential packet	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-f20da4575c521256	ART-5c688f8828000818	W2W3BulkInvariant	w2w3 bulk	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-f20da4575c521256	ART-09a3aa9ee19f6fed	W2W3BulkInvariant	w2w3 bulk	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-f20da4575c521256	ART-3a0eb3e2ef1f87ce	W2W3BulkInvariant	w2w3 bulk	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-f20da4575c521256	ART-872e9859f25b1e71	W2W3BulkInvariant	w2w3 bulk	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-f20da4575c521256	ART-f9838d40e3c84a2e	W2W3BulkInvariant	w2w3 bulk	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-f20da4575c521256	ART-5f800842b0ea22de	W2W3BulkInvariant	w2w3 bulk	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-f20da4575c521256	ART-fb6e648f129dd57a	W2W3BulkInvariant	w2w3 bulk	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-f20da4575c521256	ART-12b107a77d916321	W2W3BulkInvariant	w2w3 bulk	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-b4bffafbbdaed511	ART-5c99939e93e517f4	SU2RepresentationPacket	representation ledger	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-4f12442678a99f50	ART-01a1500df063d141	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-4f12442678a99f50	ART-7966f4f2cd58dafb	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-4f12442678a99f50	ART-85e9177b9bd29cbb	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMA LY	RPOS-4f12442678a99f50	ART-095bd3dff3f46851	GlobalConsistencyDecision	global consistency	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-bfa8d1341908d8e1	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-b6a8e95d144943b0	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-691f6494489f8e88	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-79dfca318fb9ac9e	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-5070c69cfa858c04	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-b80f50bfdbf05d32	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-53faaa8868c94f7b	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-80d97d251f4d13ec	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-4a02f07f82868f45	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-f08278789087fbf8	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-4aedabaea6e9d267	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-a335eadd9eaa00ea	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-895456a15b4b89c0	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-672b1d873ad1aac3	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-5b5ae928b879b8ce	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-00c9700202c17adf	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-10f3a47fa7bb59c8	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-836caaea1146f034	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-b81b91b777e1515c	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-ad49875c2a027f96	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-6ac9c826a471daaa	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-406e572b0e4dbb1f	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-b7de887cd0e2eed0	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-5865feee0356562a	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-7788878f344d2b6e	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-a421df79e24254fa	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-6d08e554c94724a9	GlobalConsistencyDecision	global consistency	identity
F-SU2-NEW-SPIN-ANOMALY	RPOS-4f12442678a99f50	ART-5c4ee6a619668153	GlobalConsistencyDecision	global consistency	identity
F-QCD-U1A-NF3	RPOS-5f9f4bbf1d164d05	ART-803217a883ef3779	QCDMasslessPacket	massless content	identity
F-QCD-U1A-NF3	RPOS-f5db893466497808	ART-59873a08e1f068bf	AxialAction	classical axial action	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-QCD-U1A-NF3	RPOS-7ac711b3f5954dfe	ART-87734a08d04a5b97	VacuumRealization	vacuum realization	identity
F-QCD-U1A-NF3	RPOS-7ac711b3f5954dfe	ART-a4e838ed5992ea0	VacuumRealization	vacuum realization	identity
F-QCD-U1A-NF3	RPOS-7ac711b3f5954dfe	ART-541813bed8183f8b	VacuumRealization	vacuum realization	identity
F-QCD-U1A-NF3	RPOS-50059c07d49ce620	ART-174278e40c845d73	DiscreteRemnant	discrete remnant	identity
F-QCD-U1A-NF3	RPOS-50059c07d49ce620	ART-5beaf309700d8d51	DiscreteRemnant	discrete remnant	identity
F-QCD-U1A-NF3	RPOS-50059c07d49ce620	ART-2e06f6cff7482835	DiscreteRemnant	discrete remnant	identity
F-QCD-U1A-NF3	RPOS-63efe44d67fcd bff	ART-13fd3c74f55c7960	AxialWardIdentityQCD	axial ward identity	identity
F-QCD-U1A-NF3	RPOS-63efe44d67fcd bff	ART-9c87eab86521806b	AxialWardIdentityQCD	axial ward identity	identity
F-QCD-U1A-NF3	RPOS-7254da3dcf45411c	ART-3f4abeb05e1943a8	AxialAnomalyCoefficient	measure index anomaly	identity
F-QCD-U1A-NF3	RPOS-7254da3dcf45411c	ART-88189619dadcd3e7	AxialAnomalyCoefficient	measure index anomaly	identity
F-EW-BPLUSL-3GEN	RPOS-f2f82afdcaf734ef	ART-e2940815386a2a9b	NonperturbativeStatus	nonperturbative status	identity
F-EW-BPLUSL-3GEN	RPOS-f2f82afdcaf734ef	ART-94c0c97aa95cbc8b	NonperturbativeStatus	nonperturbative status	identity
F-EW-BPLUSL-3GEN	RPOS-f2f82afdcaf734ef	ART-70f43b9e95e46792	NonperturbativeStatus	nonperturbative status	identity
F-EW-BPLUSL-3GEN	RPOS-e06983f185130b7a	ART-28bb7b42c9e4d3e5	ElectroweakIndex	electroweak index	identity
F-EW-BPLUSL-3GEN	RPOS-e06983f185130b7a	ART-db3808af502d785f	ElectroweakIndex	electroweak index	identity
F-EW-BPLUSL-3GEN	RPOS-d500f10c0b119dea	ART-a09374e54f4c843e	GenerationChargePacket	generation content	identity
F-EW-BPLUSL-3GEN	RPOS-1a6f2a8161e3154f	ART-246945c8092ab462	BaryonSelectionRule	baryon rule	identity
F-EW-BPLUSL-3GEN	RPOS-1a6f2a8161e3154f	ART-70fee137a0e84678	BaryonSelectionRule	baryon rule	identity
F-EW-BPLUSL-3GEN	RPOS-1a6f2a8161e3154f	ART-f473c515d8047f8d	BaryonSelectionRule	baryon rule	identity
F-EW-BPLUSL-3GEN	RPOS-1a6f2a8161e3154f	ART-9612762b9f71775d	BaryonSelectionRule	baryon rule	identity
F-EW-BPLUSL-3GEN	RPOS-14d2a9284b57c7f0	ART-0d9bc45f66b5fb15	CombinedChargeRule	combined rule	identity
F-EW-BPLUSL-3GEN	RPOS-14d2a9284b57c7f0	ART-02656f42cef43ebf	CombinedChargeRule	combined rule	identity
F-EW-BPLUSL-3GEN	RPOS-e2db39919f7444b6	ART-29e2333f17fa73c1	LeptonSelectionRule	lepton rule	identity
F-EW-BPLUSL-3GEN	RPOS-e2db39919f7444b6	ART-e44e2311afe0276f	LeptonSelectionRule	lepton rule	identity
F-EW-BPLUSL-3GEN	RPOS-e2db39919f7444b6	ART-a006f0dd36000a69	LeptonSelectionRule	lepton rule	identity
F-EW-BPLUSL-3GEN	RPOS-e2db39919f7444b6	ART-be5a8629103f3b50	LeptonSelectionRule	lepton rule	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-8e87c18e6c79e9c2	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-5cc1901ee7d0c384	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-91b6a4858e020b7	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-d9fd567ad611ed79	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-1fec9d1d930e97c2	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-61fc3d86994914c0	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-1e0a4c3cb468e4d2	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-486a00b5b50a0a64	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-b77a92bee8d8ea90	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-7fb0386208cb4eb9	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-c9dc56bd3a01beb7	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-2a8b76ca053c9769	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-11e02fe98b6b3bdc	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-9e68c65b0634f7b8	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-99c1d10dadcf71072	ART-a118a58c786c346c	RelativeAnomalyClass	relative gluing	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-e643b27595588887	ART-968d5e7734899773	BoundaryChargePacket	boundary content	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-cb2f301eafe5bb67	ART-6c784f8097f662c1	BoundaryAnomalyClass	boundary class	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-cb2f301eafe5bb67	ART-6a557417823e9c16	BoundaryAnomalyClass	boundary class	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-cb2f301eafe5bb67	ART-5caedadba1bba35f	BoundaryAnomalyClass	boundary class	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-3d06feec5589820b	ART-b2051b1612705285	QuantizationDecision	integral quantization	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-3d06feec5589820b	ART-615a5221209f03b9	QuantizationDecision	integral quantization	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-3d06feec5589820b	ART-1f600998b64f6f24	QuantizationDecision	integral quantization	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-3d06feec5589820b	ART-96517b42bddc8c06	QuantizationDecision	integral quantization	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-3d06feec5589820b	ART-5b114eef6bd23773	QuantizationDecision	integral quantization	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-8881e707989acea2	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-1b9a22ffdc1c8c29a	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-759e5108e3bec507	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-c79f6fb00b41fa7a	InflowDecision	inflow cancellation	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-8cc144a38c898552	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-679ab90cc9bed39a	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-aea21ed77b30d8ab	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-7ed25b476a3173ee	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-802a837766c6944e	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-70fe98030d770012	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-b23049023be04938	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-e684e3185e4621d2	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-ee6dbc0febcb04d1c	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-a2ce9047eee97677	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-6776ff26318a44a6	ART-be3f091d96e6f363	InflowDecision	inflow cancellation	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-651e97d96131fca5	ART-41eb0943375c78fa	BulkAnomalyClass	bulk class	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-651e97d96131fca5	ART-90aa9bf6c0c69a38	BulkAnomalyClass	bulk class	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-651e97d96131fca5	ART-98b69e02628e4584	BulkAnomalyClass	bulk class	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-651e97d96131fca5	ART-97c7fdeb4fb30915	BulkAnomalyClass	bulk class	identity
F-U1-WEYL-INFLOW-4D5D	RPOS-651e97d96131fca5	ART-51935d6007d2dd9c	BulkAnomalyClass	bulk class	identity
F-QCD3-WZW-MATCH	RPOS-fafb5b8eaab0c7ea	ART-0d513adfed636aca	WZWLevel	wzw level	identity
F-QCD3-WZW-MATCH	RPOS-fafb5b8eaab0c7ea	ART-8c622c31be999217	WZWLevel	wzw level	identity
F-QCD3-WZW-MATCH	RPOS-fafb5b8eaab0c7ea	ART-25e9bfb697bb2fe0	WZWLevel	wzw level	identity
F-QCD3-WZW-MATCH	RPOS-fafb5b8eaab0c7ea	ART-a5b551ece777bf0f	WZWLevel	wzw level	identity
F-QCD3-WZW-MATCH	RPOS-fafb5b8eaab0c7ea	ART-a92683500cc6c483	WZWLevel	wzw level	identity
F-QCD3-WZW-MATCH	RPOS-b4d5f6bcf1a10d53	ART-eb6a3681b0587f61	BackgroundSymmetryMap	background map	identity
F-QCD3-WZW-MATCH	RPOS-b4d5f6bcf1a10d53	ART-aa72e4420aa51da6	BackgroundSymmetryMap	background map	identity
F-QCD3-WZW-MATCH	RPOS-e4b0c9cf7df4daf3	ART-6e814df426075e88	UVAnomalyClass	uv anomaly	identity
F-QCD3-WZW-MATCH	RPOS-e4b0c9cf7df4daf3	ART-57d08b36f45e999f	UVAnomalyClass	uv anomaly	identity
F-QCD3-WZW-MATCH	RPOS-8f167bd45b5176e4	ART-ec714a4f6e11cc8	BardeenCounterterm	counterterm	identity
F-QCD3-WZW-MATCH	RPOS-8f167bd45b5176e4	ART-de8a73dafca92d7c	BardeenCounterterm	counterterm	identity
F-QCD3-WZW-MATCH	RPOS-c21a1696fa0c1542	ART-77181f1c7cc44952	IRChiralTarget	ir target	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-0a684d835d82eaf8	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-2fa055df53012470	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-3793e96317a25fe1	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-937604eada33cde5	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-4db133cf8ca3a189	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-6c2f52f29c957f32	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-0abd287a42c7074b	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-a8fa99fcd13d47bf	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-616921be3c1c57338	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-d2db4e80780ce76e	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-2e52d2eee40ad00f	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-0b298e34ca8c6bd9	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-5ae9f7ebbd6e3509	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-d67f15c14f7bfa2c	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-20fd3a93b6f26c59	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-a90d3a5b24870887	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-f8da2c02a5ebf8e6	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-332407c5918299b6	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-d90ae6f2989464a6	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-a415b0d9ce3e4b08	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-647903e59fc23804	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-8243d660cfe6aa0c	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-d52336b28030d191	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-72676458af813e40	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-dbdaf6d5374ea8d1	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-b4b8648f9d1923eb	AnomalyMatchingDecision	uv ir matching	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-da192266e001b8bf	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-b953b06873f34dd7	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-a5d8df7a106637e7	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-47a807cf2dd5d537	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-cbb49a68dcbfcf57	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-521d30b8ff161b1c	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-e946c5903b0941a6	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-011bc34bf4e4b993	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-62364378530439d1	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-f0b75571a5062d12	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-edcb1e41a46ac39e	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-fe53d4dd8e7c23e3	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-e4c116deaeab89b5	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-82b6eac381ec4d81	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-516f891a9ff9c6ae0	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-e5fd9b7ff25091e8	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-f010b22cb6e2b46b	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-da83960dc5f3cc32	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-442e02bd5565bb15	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-7800ace2347f6b86	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-c9b959ff7dd1052c9	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-4c6664a15c98cbd7	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-55995b0a46b11b63	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-35ca2917071e9645	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-e24ea304b27c7652	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-c2c89bf82a4e59a0	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-f0b4bae35c7ffffb3	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-0d90c61839aec580	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-5d7b58a4db5bcc09	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-6102c6b9b4827d69	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-24d8cab59a07be39	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-cb4bfd4429a49cdc	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-dd54881ee2b118e4	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-b8d5f496c1a65edb	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-006af9d06adf2f5a	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-30c01d8e8a39199a	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-4ee06f239b406980	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-bcd2f68708b6e1c7	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-6320a63b13d1705a	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-f4753968b6863c47	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-fcbdb92337a52f05	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-501b5bc8280e844f	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-170abefcf2bd1528	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-caa5292d8d2558c8	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-87776daf101286bc	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-13ad017a5e043661	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-611f1846bdec33e3	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-d31f36cf3503e919	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-d0902528d4101971	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-1307cc985f73bbbd	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-7fd40a2f15118152	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-d7e4bcef46e32ff2	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-f51b1a070f096817	AnomalyMatchingDecision	uv ir matching	identity
F-QCD3-WZW-MATCH	RPOS-00b74faaf3dc7b39	ART-0986bdcef77aa52c	AnomalyMatchingDecision	uv ir matching	identity

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Fiber	Role position	Candidate	Type	Restored queries	Witness
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-c9d72b941686cb79	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-6913a700c26d44f1	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-27be1da0de8807b3	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-c7b34da5fcc1924e	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-b73f71ea73c46a36	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-83da0eb9950b631c	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-bed0f5480f90fb78	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-0299363730ca127f	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-727bdcdec0251437	GaugedWZWVariation	gauged variation	identity
F-QCD3-WZW-MATCH	RPOS-011aaac614df1705	ART-2643be306353d7f9	GaugedWZWVariation	gauged variation	identity
F-QCD3-TRACE-MSBAR	RPOS-f5ba1edb9f9a109e	ART-67ff1b730559eda3	QCDParameterPacket	qcd parameters	identity
F-QCD3-TRACE-MSBAR	RPOS-0c3de586995821c5	ART-a83adf8f6009bbaf	RenormalizedOperatorBasis	operator basis	identity
F-QCD3-TRACE-MSBAR	RPOS-0c3de586995821c5	ART-3ca93aea0d211684	RenormalizedOperatorBasis	operator basis	identity
F-QCD3-TRACE-MSBAR	RPOS-0c3de586995821c5	ART-b70bb13b23be767d	RenormalizedOperatorBasis	operator basis	identity
F-QCD3-TRACE-MSBAR	RPOS-e1347b4b7c72a5a0	ART-88affbf9e36e88db	OperatorQuotient	operator quotient	identity
F-QCD3-TRACE-MSBAR	RPOS-e1347b4b7c72a5a0	ART-3f93f3236ca82812	OperatorQuotient	operator quotient	identity
F-QCD3-TRACE-MSBAR	RPOS-e1347b4b7c72a5a0	ART-772a05fbfb6d6f73	OperatorQuotient	operator quotient	identity
F-QCD3-TRACE-MSBAR	RPOS-44b0d6d08d7e3314	ART-db8eaa90a80bb999	TraceOperatorIdentity	trace identity	identity
F-QCD3-TRACE-MSBAR	RPOS-44b0d6d08d7e3314	ART-83a6c2b434fce77a	TraceOperatorIdentity	trace identity	identity
F-QCD3-TRACE-MSBAR	RPOS-44b0d6d08d7e3314	ART-72802b7e85a35e44	TraceOperatorIdentity	trace identity	identity
F-QCD3-TRACE-MSBAR	RPOS-44b0d6d08d7e3314	ART-ec05fef040809fa3	TraceOperatorIdentity	trace identity	identity
F-QCD3-TRACE-MSBAR	RPOS-44b0d6d08d7e3314	ART-9968a8aa991d9fde	TraceOperatorIdentity	trace identity	identity
F-QCD3-TRACE-MSBAR	RPOS-44b0d6d08d7e3314	ART-c767a157987f6aec	TraceOperatorIdentity	trace identity	identity
F-QCD3-TRACE-MSBAR	RPOS-004b66e1a4476828	ART-150a839df9e8c95d	FiniteSchemeMap	finite scheme map	identity
F-QCD3-TRACE-MSBAR	RPOS-004b66e1a4476828	ART-15d59c01f069e882	FiniteSchemeMap	finite scheme map	identity
F-QCD3-TRACE-MSBAR	RPOS-004b66e1a4476828	ART-96895ee6570ba365	FiniteSchemeMap	finite scheme map	identity
F-QCD3-TRACE-MSBAR	RPOS-8cd99aa4e77629ff	ART-05393346524cfa8f	BetaGammaData	beta gamma	identity
F-QCD3-TRACE-MSBAR	RPOS-8cd99aa4e77629ff	ART-0a8615b1e75c1469	BetaGammaData	beta gamma	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-ba468a4c2860310e	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-ff9f94b483d8bd24	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-90165710a90b671b	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-1fd9aebddfbff0fe	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-7db744814045b1a5	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-ea882cc22432d2f1	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-4543eef84cb97c47	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-ee6c6d82bb2c4ac5	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-0a4cd3d9ed4223fd	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-a2bd8c886f4909c2	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-00d869a13a1a5130	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-5fefb14104f67e4e	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-7342553047e48aa2	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-42faeb34096c696b	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-704eef903b83fcfb	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-ce4f868b08d4c685	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-73009f580918ba34	SchemeInvariantTrace	scheme invariant trace	identity
F-QCD3-TRACE-MSBAR	RPOS-fc76d888efda546a	ART-b36fe1e40383be47	SchemeInvariantTrace	scheme invariant trace	identity

Relational Compatibility

Pointwise eligibility is necessary but not sufficient. Every complete map is evaluated against the physical relations of its target. The complete/rejected/compatible totals appear in Chapter 25.

The aggregate failure counts identify which relations perform the eliminative work:

Table 17.1: Compatibility-failure counts by physical relation

Fiber	Failed relation	Count
F-SM-LOCAL-1GEN	ledger faithful to physical chirality	12288
F-SM-LOCAL-1GEN	local and global consistency	12288
F-SM-LOCAL-1GEN	aggregate uses selected contributions	14336
F-SM-LOCAL-1GEN	descent uses selected aggregate	14336
F-SM-LOCAL-1GEN	aggregate equals contribution sum	12288
F-SM-LOCAL-1GEN	descent matches aggregate	6144
F-SM-LOCAL-1GEN	contribution uses selected ledger	12288
F-SM-LOCAL-1GEN	parity uses selected ledger	12288
F-VA-BARDEEN-DESCENT	axial ratio three to one	1120
F-VA-BARDEEN-DESCENT	gauged vector current preserved	1440
F-VA-BARDEEN-DESCENT	presentation coefficients match	1440
F-VA-BARDEEN-DESCENT	presentation map targets selected bundle	1560
F-VA-BARDEEN-DESCENT	counterterm uses selected anomaly	800
F-VA-BARDEEN-DESCENT	anomaly uses selected tensor	800
F-VECTORLIKE-3PAIR-SYMMETRIC	quotient equals automorphism orbits	14544
F-SU2-WITTEN-GLOBAL	decision matches index	32
F-SU2-WITTEN-GLOBAL	decision uses selected phase	48
F-SU2-WITTEN-GLOBAL	phase matches mod2	32
F-SU2-WITTEN-GLOBAL	phase uses selected index	32
F-SU2-NEW-SPIN-ANOMALY	bulk evaluates generator	2048
F-SU2-NEW-SPIN-ANOMALY	bulk phase matches index	3072
F-SU2-NEW-SPIN-ANOMALY	decision matches combined obstruction	2048
F-SU2-NEW-SPIN-ANOMALY	decision uses selected indices and bulk	4032
F-SU2-NEW-SPIN-ANOMALY	bulk uses selected new index	3584
F-SU2-NEW-SPIN-ANOMALY	new index matches target	2048
F-SU2-NEW-SPIN-ANOMALY	new index uses selected geometry	2048
F-SU2-NEW-SPIN-ANOMALY	tangential structure matches target	2048
F-QCD-U1A-NF3	exact remnant correct	24
F-QCD-U1A-NF3	vacuum stabilizer correct	24
F-QCD-U1A-NF3	vacuum subgroup of remnant	8
F-QCD-U1A-NF3	ward uses selected anomaly	18
F-EW-BPLUSL-3GEN	selection rules correct	144
F-EW-BPLUSL-3GEN	lepton rule uses selected index	96
F-EW-BPLUSL-3GEN	baryon rule uses selected index	96
F-EW-BPLUSL-3GEN	quantum status correct	128
F-EW-BPLUSL-3GEN	combined rule method synchronized	96
F-U1-WEYL-INFLOW-4D5D	decision uses selected relative	15750
F-U1-WEYL-INFLOW-4D5D	full relative class cancelled	16200
F-U1-WEYL-INFLOW-4D5D	relative class recomputed	14400
F-U1-WEYL-INFLOW-4D5D	relative uses selected boundary and bulk	15750
F-U1-WEYL-INFLOW-4D5D	quantization uses selected bulk	13500
F-U1-WEYL-INFLOW-4D5D	complete boundary class	5625
F-QCD3-WZW-MATCH	level quantized and correct	25600
F-QCD3-WZW-MATCH	uv ir pullback matches	31980
F-QCD3-WZW-MATCH	matching uses selected objects	31600
F-QCD3-WZW-MATCH	variation uses selected level counterterm target	28800
F-QCD3-WZW-MATCH	background map matches	16000
F-QCD3-WZW-MATCH	counterterm preserves vector current	16000
F-QCD3-TRACE-MSBAR	basis complete	3888
F-QCD3-TRACE-MSBAR	quotient uses selected basis	5184
F-QCD3-TRACE-MSBAR	scheme transport preserves trace	3888
F-QCD3-TRACE-MSBAR	trace identity complete	3888
F-QCD3-TRACE-MSBAR	invariant uses selected trace and map	5508

F-QCD3-TRACE-MSBAR

trace uses selected beta and basis

4860

Representative rejected maps are shown below; the full 91597-record ledger remains in the per-fiber compatibility files.

Table 17.2: Representative complete assignments rejected by relational compatibility

Fiber	Assignment	Failed relations	Physical notes
F-SM-LOCAL-1GEN	ASN-9a4e14b6552cd5e7	ledger faithful to physical chirality; local and global consistency	u_R: representation not conjugated; u_R: U(1) charge not conjugated
F-SM-LOCAL-1GEN	ASN-4a4ba870adcaaff7	aggregate uses selected contributions; descent uses selected aggregate; ledger faithful to physical chirality; local and global consistency	u_R: representation not conjugated; u_R: U(1) charge not conjugated
F-VA-BARDEEN-DESCENT	ASN-dcdd7f9d5640f4ac	axial ratio three to one; gauged vector current preserved	
F-VA-BARDEEN-DESCENT	ASN-8860f14766a966ee	axial ratio three to one; gauged vector current preserved; presentation coefficients match; presentation map targets selected bundle	
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-c79b3c9901f16349	quotient equals automorphism orbits	
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-d1cf28439ae1337d	quotient equals automorphism orbits	
F-SU2-WITTEN-GLOBAL	ASN-d4592c94baadb08a	decision matches index	
F-SU2-WITTEN-GLOBAL	ASN-a629fba683dbbcd6	decision uses selected phase	
F-SU2-NEW-SPIN-ANOMALY	ASN-584c3d3446a921e4	bulk evaluates generator; bulk phase matches index	
F-SU2-NEW-SPIN-ANOMALY	ASN-5692903323679d27	bulk evaluates generator; bulk phase matches index; decision matches combined obstruction	
F-QCD-U1A-NF3	ASN-676c57c01f5a84d0	exact remnant correct; vacuum stabilizer correct	
F-QCD-U1A-NF3	ASN-8e63dc2468131032	exact remnant correct	
F-EW-BPLUSL-3GEN	ASN-17d3ed5fa1d3ab3f	selection rules correct	
F-EW-BPLUSL-3GEN	ASN-1eb5f476079f2206	selection rules correct	
F-U1-WEYL-INFLOW-4D5D	ASN-eeb3d0a3c86adede	decision uses selected relative; full relative class cancelled; relative class recomputed; relative uses selected boundary and bulk	
F-U1-WEYL-INFLOW-4D5D	ASN-bcbc5d709691d9d6	decision uses selected relative; full relative class cancelled; relative class recomputed; relative uses selected boundary and bulk	
F-QCD3-WZW-MATCH	ASN-b6c4fc0a66d274cd	level quantized and correct; uv ir pullback matches	
F-QCD3-WZW-MATCH	ASN-b552268a2398aa7c	level quantized and correct; matching uses selected objects; uv ir pullback matches	
F-QCD3-TRACE-MSBAR	ASN-697e8a85bb591bda	basis complete; quotient uses selected basis; scheme transport preserves trace; trace identity complete	
F-QCD3-TRACE-MSBAR	ASN-b68a86614cb42d6c	basis complete; invariant uses selected trace and map; quotient uses selected basis; scheme transport preserves trace; trace identity complete	

Compatible Assignments and Normal-Form Groupoids

The ninety-eight compatible complete assignments are listed below.

Table 18.1: All compatible complete assignments

Fiber	Assignment	Occupied roles	Digest
F-SM-LOCAL-1GEN	ASN-63bac83f9c40491a	5	f9c9f5dbbab9
F-SM-LOCAL-1GEN	ASN-58c6177971116f83	5	86219beb4130
F-SM-LOCAL-1GEN	ASN-39251653d156bd8a	5	71ca935e675d
F-SM-LOCAL-1GEN	ASN-2e356c33756b4d36	5	3e43dc2da947
F-VA-BARDEEN-DESCENT	ASN-1730fbce3718e6d0	5	8245b16a9f88
F-VA-BARDEEN-DESCENT	ASN-198f22aa761efdcd	5	c3ccd63f66c7
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-9989ef262acfcbd5	10	91ce63eaf8d2
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-0db4a2234d25242a	10	c6c9e7b1f620
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-93ed9d87d8503252	10	3dc8c46574fa
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-fbdbda91237d73c9	10	b13c47ae3552
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-a164ea3acf47ba94	10	ef5796e5a344
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-aad83995f31d65cb	10	6ab56d06b1a9
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-49009de8a872f747	10	64cf5985f9ee
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-4f4650585f824977	10	4886d4b29fd4
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-2bc6d48fdee5f160	10	7f818f27ea7f
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-8206bfc72e44b084	10	620b8256785e
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-907e8d90e997df6b	10	885d1f09e2c5
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-3b06688a7189d2da	10	9ae8d01f08ba
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-98f40295c92e3bb6	10	580a25e0ff2a
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-645de44a3f03324a	10	a628db1c9f65
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-7e525f6b5825adeb	10	2c79193b24e1
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-24d1ed8dd2b806bd	10	81485426b038
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-05f54d1e9110813e	10	af54497a6d78
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-f4b064e0d953ddb3	10	4678f02c678e
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-f2bb5433801ef60b	10	54cf5c206099
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-042807c528fe0d74	10	9cbd61597d79
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-25904f1ab3c3de48	10	dcc9d259e940
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-7badb35e0959ba6c	10	ace5138f8184
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-cb27a6864bcf2e77	10	c67167d42680
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-07089580e8495aa1	10	21af790a5c8f
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-418b024faa3d78c8	10	537d3a81b5f4
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-283a08812145d036	10	261bf1473dac
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-da07d118a57c6186	10	3c18d9c4fc8a
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-d7d400f7a7254f1b	10	ea41c187248f
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-5d721adef006ddcf	10	f96e7e4762a6
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-4f0b6673a3f8a4d8	10	b427519afa42
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-9624e58591346435	10	a04d2e1925f0
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-8009178b4e692f99	10	e89364674d9f
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-ad5c6c4e5f708ab3	10	6e34a1720e48
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-35c87ac90406a138	10	bffc38bfbbd4
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-a52e7c2024212150	10	9876d812735e
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-cefc7d4a1db4576c	10	6bf04a14465f
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-401de368215a20be	10	99bbf4ccb2ae
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-082a5dcc536a5828	10	760fc27283d1
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-2c2bc34906658c0b	10	a1f18d4fd653
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-9b58e5c13ea13f52	10	2a75c8464e49
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-b06326a3849c6945	10	d19c55eb0953
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-459bbb6cf4c0d32c	10	4787a120b6ac
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-0183fd690c8f87c8	10	c65a8b82c2b7
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-721952cb80228c7b	10	c243f6fcd551
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-03d12281cf8bbefe	10	6a8a1e1af726
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-fa453275549915a6	10	db0c3a3ad524
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-3dc8c24cc7b5f04	10	fc2bcba0721f
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-d74e5fc2f5571c3a	10	018250612c7f

continued

Fiber	Assignment	Occupied roles	Digest
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-f3e4894bb638cbb5	10	af34fb3982d5
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-c021f74d16013b1e	10	01b6c5dfcefb
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-a61daee57cb21784	10	4c1705235975
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-d2142f07a91158dd	10	c594214b9700
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-450d75c42b1273cb	10	916173e7956a
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-4cb29b0f99b688f6	10	73012ac6664a
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-6f8277ba2b5c7f3c	10	f11279c98961
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-eb5264c40c0b8cd1	10	c46dae2f5808
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-85acc338c88f0192	10	ab55be0c3c9a
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-466984fcb5244337	10	a34db970591b
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-1d310c7e61ef20af	10	43b1ad3fbced
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-034b65694d2b354b	10	657238fd6628
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-5a498af333225bf0	10	587557de7caf
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-0e0f1e39fb2b8449	10	5f91bd527afc
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-b20231675d414445	10	41ad1986f38d
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-d733c41b7770fc11	10	745106ce7720
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-6eb138f12989ad08	10	1a3afb2b5842
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-b84289c009061aaa	10	77e52904e19e
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-5d0a6edfe9882e89	10	0c3457954c8b
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-8262c9ccba1f63ce	10	97b9f7abfbfc4
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-743c2c8db33c3995	10	d96b7236cccc
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-437f76f0497b3613	10	75f4dcf6ca72
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-a85664652c07ef76	10	6eb5cb66f2bf
F-VECTORLIKE-3PAIR-SYMMETRIC	ASN-1a352817b589ec25	10	cb71e534a44d
F-SU2-WITTEN-GLOBAL	ASN-08a83faba752e11b	5	d6e3a5d3d270
F-SU2-WITTEN-GLOBAL	ASN-f032ed2b6b584269	5	4db033672c20
F-SU2-NEW-SPIN-ANOMALY	ASN-9f4d1ea8a68192c8	6	548f32bb8145
F-SU2-NEW-SPIN-ANOMALY	ASN-b50cbe247899ae42	6	1a47aba88c22
F-SU2-NEW-SPIN-ANOMALY	ASN-2a450369bc5c4699	6	0b17ff60f87a
F-SU2-NEW-SPIN-ANOMALY	ASN-2591f3ba75c9d50e	6	055531939af5
F-QCD-U1A-NF3	ASN-daaca94719d9609c	6	89feec333198
F-QCD-U1A-NF3	ASN-aeb10da3db4dbdfa	6	bfc2018b92d3
F-EW-BPLUSL-3GEN	ASN-64041d92812132ed	6	5bc317f05a6c
F-EW-BPLUSL-3GEN	ASN-1a99098293b97ffc	6	c6e460e43544
F-U1-WEYL-INFLOW-4D5D	ASN-89a3d42a5498f5e2	6	18d6529edf4b
F-U1-WEYL-INFLOW-4D5D	ASN-e04941fbf809b165	6	e18f3801ce43
F-QCD3-WZW-MATCH	ASN-58fe8d6311294daa	7	e8d68e993896
F-QCD3-WZW-MATCH	ASN-74bf0bb0eabc4f18	7	adc1873902f2
F-QCD3-TRACE-MSBAR	ASN-64a7ed8caae98bee	7	58d7e4f31030
F-QCD3-TRACE-MSBAR	ASN-292ea1bb2530abd9	7	18d99e9e1007
F-QCD3-TRACE-MSBAR	ASN-b0fe86f7e7cf0f85	7	63413cbac7be
F-QCD3-TRACE-MSBAR	ASN-e60ea679ab338e0e	7	0433b7249174
F-QCD3-TRACE-MSBAR	ASN-8581b755b0c8fa7b	7	0252b825e5fe
F-QCD3-TRACE-MSBAR	ASN-85881821ab9cfd23	7	349474d6705c

The groupoid calculation retains lawful automorphisms and stabilizers while reducing every closed fiber to one isomorphism class. The normal-form groupoid summary appears in Chapter 26.

Physical Relation Records

The relation families represented in the closed targets include target fidelity, chirality conversion, contribution and aggregate consistency, current–charge–Ward synchronization, descent, counterterm preservation, ordinary and newer global-obstruction tests, bulk–boundary cancellation, orientation and quantization, ultraviolet–infrared pullback, WZW level and background-map consistency, and trace-operator basis transport.

Every proposed cancellation or repair generated by the declared ten-fiber candidate grammars receives one target-relative disposition. Presentation changes preserve the declared anomaly tuple; same-target neutralization vanishes inside the fixed theory; inflow is licensed only in an integrated bulk–boundary target; matching relates fixed ultraviolet and infrared data; target updates create successor records. The table below summarizes the executed relations. Completeness is relative to the declared targets, candidate generators, and relation grammar.

Table 19.1: Executed relation families and presentation dispositions

Fiber	Executed physical relations (selected counts)	Explicit isomor- phisms	Invariant agree- ments
F-EW-BPLUSL-3GEN	baryon rule uses selected index: 96; combined rule method synchronized: 96; lepton rule uses selected index: 96; quantum status correct: 128; selection rules correct: 144	0	12
F-QCD-U1A-NF3	exact remnant correct: 24; vacuum stabilizer correct: 24; vacuum subgroup of remnant: 8; ward uses selected anomaly: 18	0	4
F-QCD3-TRACE-MSBAR	basis complete: 3888; invariant uses selected trace and map: 5508; quotient uses selected basis: 5184; scheme-transport: 1; scheme transport preserves trace: 3888; trace identity complete: 3888; trace uses selected beta and basis: 4860	3	176
F-QCD3-WZW-MATCH	background map matches: 16000; counterterm preserves vector current: 16000; level quantized and correct: 25600; 0 matching-transport: 1; matching uses selected objects: 31600; target-update: 30; uv ir pullback matches: 31980; variation uses selected level counterterm target: 28800	0	82
F-SM-LOCAL-1GEN	aggregate equals contribution sum: 12288; aggregate uses selected contributions: 14336; contribution uses selected ledger: 12288; descent matches aggregate: 6144; descent uses selected aggregate: 14336; ledger faithful to physical chirality: 12288; local and global consistency: 12288; parity uses selected ledger: 12288	0	80
F-SU2-NEW-SPIN-ANOMALY	bulk evaluates generator: 2048; bulk phase matches index: 3072; bulk uses selected new index: 3584; decision matches combined obstruction: 2048; decision uses selected indices and bulk: 4032; new index matches target: 2048; new index uses selected geometry: 2048; tangential structure matches target: 2048	0	174
F-SU2-WITTEN-GLOBAL	decision matches index: 32; decision uses selected phase: 48; phase matches mod2: 32; phase uses selected index: 32	0	30
F-U1-WEYL-INFLOW-4D5D	complete boundary class: 5625; decision uses selected relative: 15750; full relative class cancelled: 16200; integrated-0 inflow: 1; quantization uses selected bulk: 13500; relative class recomputed: 14400; relative uses selected boundary and bulk: 15750	0	38
F-VA-BARDEEN-DESCENT	anomaly uses selected tensor: 800; axial ratio three to one: 1120; counterterm-redistribution: 1; counterterm uses selected anomaly: 800; gauged vector current preserved: 1440; presentation coefficients match: 1440; presentation map targets selected bundle: 1560	21	64
F-VECTORLIKE-3PAIR-SYMMETRIC	quotient equals automorphism orbits: 14544; same-target-neutralization: 1	7524	2

Downstream Interfaces and Successor Discipline

The ten closed targets export finite anomaly-interface records containing target identity, physical anomaly disposition, normal-form class, assumptions, and limitations. The declared downstream import records preserve these fields. They implement the AASC record-transport rule; they are not represented as a general categorical functor on arbitrary future projects.

Table 20.1: Downstream anomaly-interface record projections

Record	Fiber	Consumers	Classes	Projection Digest
ANOM-F-SM-LOCAL-1GEN	F-SM-LOCAL-1GEN	SYM-4, SYM-5, SYM-6, SYM-7, SYM-11	yes	484cf3716095
ANOM-F-VA-BARDEEN-DESCENT	F-VA-BARDEEN-DESCENT	SYM-5, SYM-6, SYM-7, SYM-9, SYM-11	yes	34966273579f
ANOM-F-VECTORLIKE-3PAIR-SYMMETRIC	F-VECTORLIKE-3PAIR-SYMMETRIC	SYM-5, SYM-7, SYM-11	1	yes1061e65c0981
ANOM-F-SU2-WITTEN-GLOBAL	F-SU2-WITTEN-GLOBAL	SYM-4, SYM-5, SYM-9, SYM-11	1	yes1ebc6e592796
ANOM-F-SU2-NEW-SPIN-ANOMALY	F-SU2-NEW-SPIN-ANOMALY	SYM-4, SYM-9, SYM-11	1	yes99ab38f1189e
ANOM-F-QCD-U1A-NF3	F-QCD-U1A-NF3	SYM-6, SYM-7, SYM-8, SYM-9, SYM-11	yes	848c5b884b5f
ANOM-F-EW-BPLUSL-3GEN	F-EW-BPLUSL-3GEN	SYM-4, SYM-7, SYM-8, SYM-10, SYM-11	1	yes5c4d5f6d389a
ANOM-F-U1-WEYL-INFLOW-4D5D	F-U1-WEYL-INFLOW-4D5D	SYM-4, SYM-9, SYM-11	1	yesab8a254ff2ec
ANOM-F-QCD3-WZW-MATCH	F-QCD3-WZW-MATCH	SYM-5, SYM-6, SYM-8, SYM-9, SYM-11	yes	f06e9f29b25d
ANOM-F-QCD3-TRACE-MSBAR	F-QCD3-TRACE-MSBAR	SYM-6, SYM-7, SYM-10, SYM-11	1	yescdc2458bc011

The audit companion also records one explicit evidence-only enrichment for each closed fiber. The physical objects, lawful morphisms, stabilizers, and component partition are unchanged under the identity-on-physical-data map. This verifies the declared finite conservative equivalence only.

Table 20.2: Finite equivalence under the declared evidence-only enrichment

Fiber	Objects	Arrows	Stabilizers	Components	Isomorphism	Digest
F-SM-LOCAL-1GEN	4	4	yes	yes	yes	30e63b759a3e
F-VA-BARDEEN-DESCENT	2	2	yes	yes	yes	95f6eff0eea0
F-VECTORLIKE-3PAIR-SYMMETRIC	72	2592	yes	yes	yes	736c1c1e9849
F-SU2-WITTEN-GLOBAL	2	2	yes	yes	yes	e0a4c7db9070
F-SU2-NEW-SPIN-ANOMALY	4	4	yes	yes	yes	4293fb9ef268
F-QCD-U1A-NF3	2	2	yes	yes	yes	4f602b951ff5
F-EW-BPLUSL-3GEN	2	2	yes	yes	yes	ddcf0bd6a153
F-U1-WEYL-INFLOW-4D5D	2	2	yes	yes	yes	570e1ddb841c
F-QCD3-WZW-MATCH	2	2	yes	yes	yes	106bf270824e
F-QCD3-TRACE-MSBAR	6	12	yes	yes	yes	ef09e202ffce

The remaining global-form and generalized-operator targets are controlled successor interfaces

owned by SYM-9.

Table 20.3: Explicit SYM-9 frontier targets and successor protocol

Frontier	Description	Owner	Successor protocol
F-SM-GLOBALFORM-LINES	SYM-9 frontier target	SYM-9	a fixed successor target and explicit relation must be supplied; no predecessor record is silently rewritten
F-SM-HIGHER-FLAVOR	SYM-9 frontier target	SYM-9	a fixed successor target and explicit relation must be supplied; no predecessor record is silently rewritten
F-SM-NONINVERTIBLE-DEFECT	SYM-9 frontier target	SYM-9	a fixed successor target and explicit relation must be supplied; no predecessor record is silently rewritten

Theorem Inventory and Closure Conditions

The scientific and audit surfaces share the following canonical theorem inventory.

Table 21.1: Canonical SYM-3 theorem inventory

ID	Statement	Scope
SYM3-T01	Pre-role same-target non-degenerate determinacy.	closed-fiber or protocol theorem
SYM3-T02	Necessary kernel activation.	closed-fiber or protocol theorem
SYM3-T03	Indexed target fixation and target-update discipline.	closed-fiber or protocol theorem
SYM3-T04	Anomaly-kind noncollapse.	closed-fiber or protocol theorem
SYM3-T05	Extensional occupant-independent role necessity.	closed-fiber or protocol theorem
SYM3-T06	Gauge anomaly as failure of quantum descent.	closed-fiber or protocol theorem
SYM3-T07	Anomalous global-current response.	closed-fiber or protocol theorem
SYM3-T08	't Hooft anomaly as obstruction to gauging and matching condition.	closed-fiber or protocol theorem
SYM3-T09	Wess-Zumino consistency and relative BRST classification.	closed-fiber or protocol theorem
SYM3-T10	Consistent/covariant/Bardeen-Zumino relation.	closed-fiber or protocol theorem
SYM3-T11	Global-anomaly target theorem.	closed-fiber or protocol theorem
SYM3-T12	Mixed-anomaly target theorem.	closed-fiber or protocol theorem
SYM3-T13	Integral relative inflow theorem.	closed-fiber or protocol theorem
SYM3-T14	Bulk-boundary target and orientation theorem.	closed-fiber or protocol theorem
SYM3-T15	Gauged-WZW anomaly-matching theorem.	closed-fiber or protocol theorem
SYM3-T16	Infrared realization nonuniqueness boundary.	closed-fiber or protocol theorem
SYM3-T17	Generalized-anomaly interface theorem.	SYM-9 frontier interface
SYM3-T18	Global-form and defect frontier theorem.	SYM-9 frontier interface
SYM3-T19	Exact Standard Model local anomaly-neutrality theorem.	closed-fiber or protocol theorem
SYM3-T20	Ordinary and newer SU(2) global-anomaly separation.	closed-fiber or protocol theorem
SYM3-T21	Electroweak B+L topology and B-L control.	closed-fiber or protocol theorem

continued

ID	Statement	Scope
SYM3-T22	Renormalized trace and scheme-transport theorem.	closed-fiber or protocol theorem
SYM3-T23	QCD axial action, anomaly, vacuum, and exact-remnant theorem.	closed-fiber or protocol theorem
SYM3-T24	Role-blind physical-object saturation relative to the declared constructors.	closed-fiber or protocol theorem
SYM3-T25	Executable presentation covariance.	closed-fiber or protocol theorem
SYM3-T26	Complete same-fiber eligibility.	closed-fiber or protocol theorem
SYM3-T27	Relational compatibility.	closed-fiber or protocol theorem
SYM3-T28	Lawful automorphism computation.	closed-fiber or protocol theorem
SYM3-T29	Registered benchmark and holdout discrimination.	closed-fiber or protocol theorem
SYM3-T30	Downstream anomaly-interface identity.	closed-fiber or protocol theorem
SYM3-T31	Closed-fiber migration completeness.	closed-fiber or protocol theorem
SYM3-T32	Fiberwise anomaly normal-form theorem.	closed-fiber or protocol theorem
SYM3-T33	Coordinate and counterterm relation closure.	closed-fiber or protocol theorem
SYM3-T34	Repair, inflow, matching, and target-update closure.	closed-fiber or protocol theorem
SYM3-T35	Downstream transport rule.	closed-fiber or protocol theorem
SYM3-T36	Registered conservative enrichment and standing-changing reopening.	closed-fiber or protocol theorem
SYM3-T37	Downstream no-second-classifier consequence.	downstream AASC consequence
SYM3-T38	Normal-form groupoid theorem.	closed-fiber or protocol theorem
SYM3-T39	Finite naturality under the declared conservative enrichment.	closed-fiber or protocol theorem
SYM3-T40	Declared downstream anomaly-interface maps.	closed-fiber or protocol theorem
SYM3-T41	Independent semantic endpoint reconstruction.	closed-fiber or protocol theorem
SYM3-T42	Frontier-successor protocol and no-silent-rewrite rule.	SYM-9 frontier interface

The ten-fiber proof core supplies the following local executable theorem objects:

Table 21.2: Ten-fiber proof-core theorem inventory

ID	Title	Statement	Class
PC-T01	Pre-role same-target non-degeneracy	For each of the ten declared closed fibers, the pre-role process produces at least one accepted and one rejected complete physical classification act while preserving the immutable target packet.	local executable theorem
PC-T02	Necessary kernel activation	Each established non-degenerate determinate physical fiber necessarily instantiates Admissibility, Standing, Reference, and Irreversibility.	imported AASC necessity with local antecedent
PC-T03	Independent burden and physical-object construction	Target burdens and physical candidate objects are constructed in isolated processes and first meet in the eligibility calculation.	process-isolation theorem
PC-T04	Extensional eligibility	Every same-fiber role-position/candidate pair is evaluated by denial, candidate injection, and restoration of the lost theorem queries.	finite executable theorem
PC-T05	Relational compatibility	Complete pointwise-eligible assignments are accepted only when their finite executable physical dependency, target-fidelity, Ward, quotient, matching, inflow, and consistency relations are jointly satisfied.	theorem
PC-T06	Executable presentation covariance	The registered presentation maps have executable forward, inverse, and composition checks, and lawfully related assignments remain in one normal-form component.	finite map theorem
PC-T07	Ten-fiber singleton normal form	For each of the ten declared closed anomaly fibers, the lawful assignment groupoid has exactly one connected component, with nontrivial lawful morphisms retained.	fiberwise finite groupoid theorem
PC-T08	Independent semantic reconstruction	A Go implementation reading primitive target inputs only reconstructs the selected physical invariants and declared fiber-level endpoint semantics for all ten closed fibers.	independent implementation theorem

continued

ID	Title	Statement	Class
PC-T09	Frozen-grammar internal holdout adequacy	Eleven committed holdout cases are evaluated after the grammar digest is frozen and receive the committed anomaly or symmetry dispositions.	blinded internal stress theorem
PC-T10	Closed-fiber migration completeness	Every currently declared closed SYM-3 anomaly fiber is represented in the clean-room proof core and passes the registered normal-form obligations.	registered-scope migration theorem
PC-T11	SYM-9 frontier separation	Faithful-global-form and line-operator completion, higher-flavor completion, and non-invertible-defect completion remain explicit SYM-theorem 9 frontier targets and are excluded from the closed-fiber quantifier.	arc-boundary
PC-T12	Exact proof-core boundary	The result establishes ten-fiber proof-core migration; it does not establish the revised full SYM-3 archival manuscript, SYM-9 frontier closure, or SYM-11 atlas exhaustion.	scope theorem

A closed-fiber result requires a same-target pre-role witness, Reference-complete target, imported physical theorem interfaces, independent role and candidate construction, executable presentation maps, complete same-fiber eligibility, relational compatibility, a computed normal-form groupoid, and a closed evidence path.

Exact Nonclaims

The project does not claim that anomaly cancellation uniquely derives the Standard Model, fixes numerical couplings, or determines one infrared realization. It does not claim open-world completeness beyond the declared target and constructor grammars. It does not close global-form, line-operator, monopole, higher-group, or non-invertible inventories assigned to SYM-9. It does not substitute finite software checks for imported continuum theorems. It does not infer the AASC kernel from software output.

Formalization Boundary

The executable layer formalizes finite target records, role and residue generation, presentation maps, eligibility, compatibility, and assignment groupoids. Continuum QFT theorems enter through reviewed typed interfaces. The finite system checks target matching and downstream consequences; it does not rederive path integrals, determinant-line theory, bordism classifications, or renormalized operator theorems from first principles.

Source Identity and Literature Review Record

The reviewed source registry fixes identities, revisions, locators, hypotheses, conclusions, and limitations. Its signature protects the reviewed record against unnoticed mutation. Source bytes are not redistributed and are not represented as locally retained unless explicitly stated.

The operative source and theorem-interface tables appear in Chapters 2 and 5. The evidence graph records how each imported result enters a target-specific calculation.

Independent Reconstruction and Reproducibility

The primary implementation constructs the ten finite targets, roles, physical residues, assignments, and normal-form groupoids. A separately maintained verifier reconstructs eligibility, complete assignments, physical compatibility truth, lawful assignment actions, stabilizers, and component counts from the fixed targets, theorem interfaces, role and candidate records, dependencies, and executed presentation evidence. The reconstruction does not read the primary eligibility or compatibility verdicts before producing its own result. Agreement is evaluated on the registered semantic invariants rather than implementation-specific identifiers or serialization order.

The project includes the complete source, inputs, generated ledgers, adversarial tests, holdouts, evidence graph, and deterministic build scripts. The ten-fiber certificate and extension identities begin `88d27ad19c082c93` and `2556c629670967e2`, respectively; their complete SHA-256 values appear in the release inventory and machine-readable summaries.

Open-Literature Authority Policy

No paywalled paper is required to reproduce a load-bearing closed-fiber claim. Freely accessible primary papers or derivational technical reviews supply the operative interfaces. Historical inaccessible papers may remain in the bibliography for attribution but do not close an evidence burden.

A closed fiber may use a composite open-source chain only when its component hypotheses are compatible and the bridge is explicit. If adequate open support is unavailable, that fiber remains outside the closure quantifier.

Falsification Conditions and Reopening Rules

The normal-form result is reopened by any of the following same-target findings:

- a false imported theorem or false target specialization;
- an omitted target-relevant anomaly object or standing burden;
- a lawful physical presentation not represented by the presentation category;
- a false positive or false negative eligibility edge;
- an omitted compatibility relation or incorrectly evaluated relation;
- an overbroad automorphism quotient;
- a second normal-form isomorphism class;
- a target change, added sector, hidden selector, or retrospective repair presented as a same-target refinement.

The mutation suite and independent reconstructions are regression evidence, not an exhaustion of criticism. A genuine counterexample must be recorded and must trigger an explicit successor rather than defensive relabeling.

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